

Individual-rollout structural validation

29 fold-switching proteins × 1,000 iid MarFold contact maps

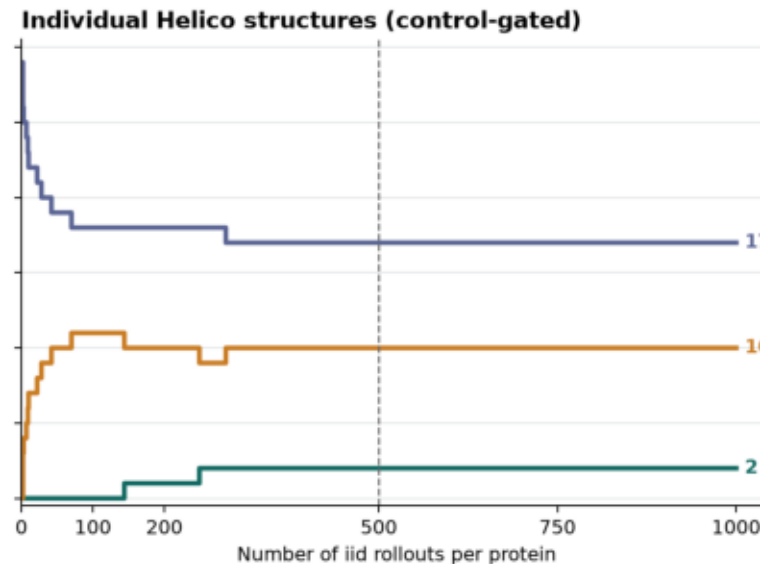
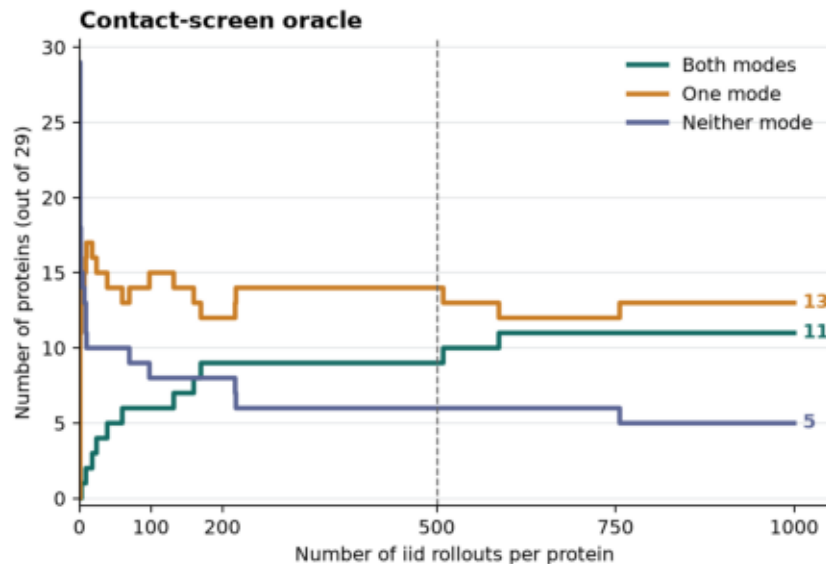
Each map was folded independently with Helico; no consensus contacts were used.

Exploratory post-hoc structural hit: common-position Kabsch GDT-TS $\geq \max(0.35, 90\% \text{ of the corresponding true-contact control})$ and switch
Every page reports raw Kabsch GDT-TS and standard CA RMSD. All PDBs and score tables are retained so this rule and the presentation can

Each protein has two pages: contact-map evidence and structural evidence. Candidate selection is an oracle analysis against the references.

Aggregate result and interpretation

Does iid sampling reveal both experimentally observed folds?



Post-hoc structural hit: common-position Kabsch GDT-TS $\geq \max(0.35, 90\% \text{ of the true-contact control})$ and switching-region advantage ≥ 0.10 .
Each Helico prediction uses one individual MarFold map; both controls pass for 17/29 proteins; dashed line: 500 draws.

At 1,000 rollouts

Both folds: 2/29

One fold: 10/29

Neither fold: 17/29

Exact-sequence subset

Both: 2/17

One: 6/17

Neither: 9/17

Validator gate

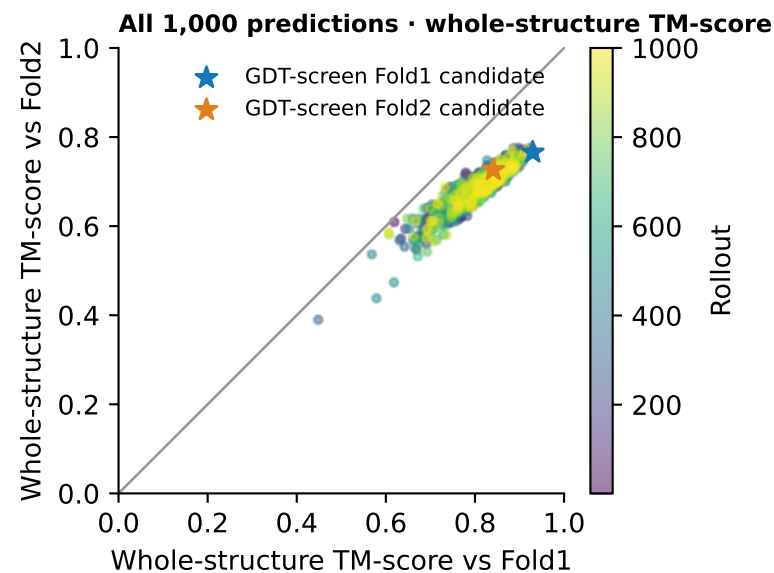
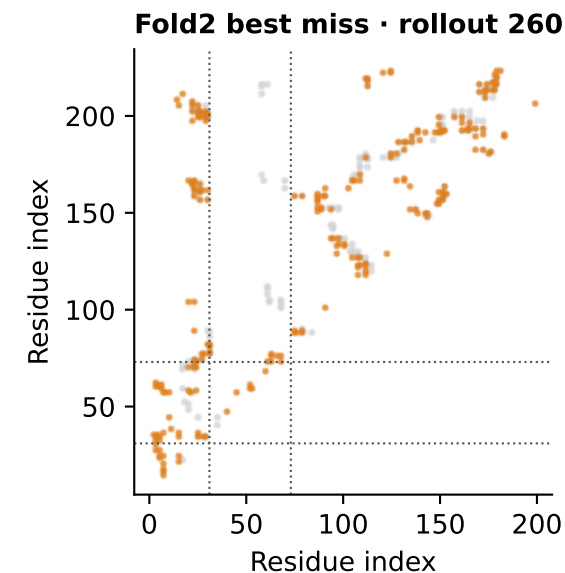
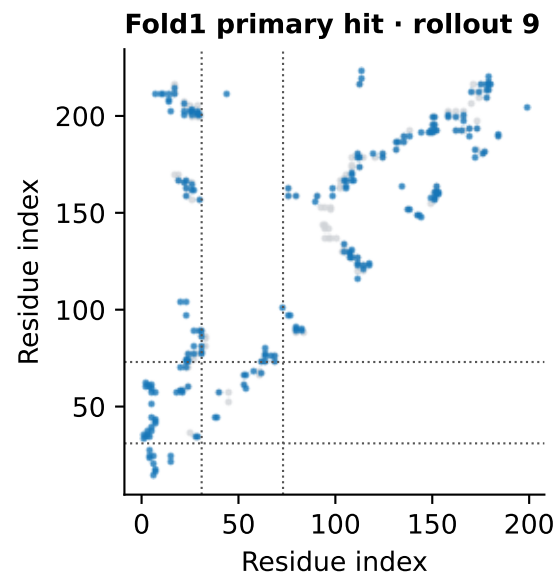
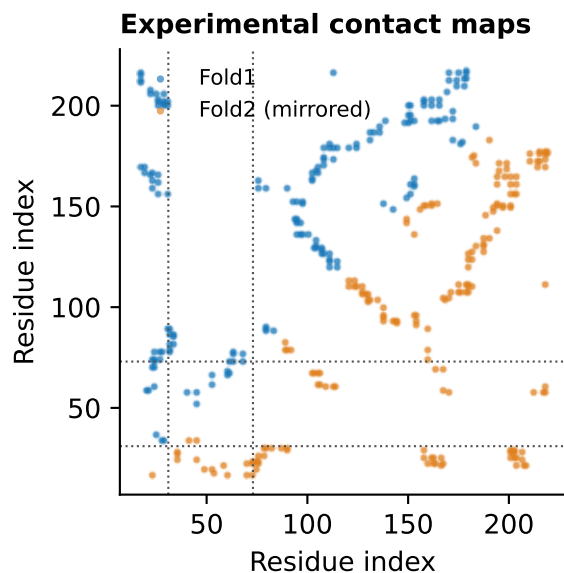
Both true-contact controls pass: 17/29

Dual among those: 2/17

A missing structural mode is evidence about the search only when Helico's true-contact control for that mode succeeds. Candidate selection on later pages is reference-aware and serves as an oracle analysis.

01. 1k0na_1rk4b — contact maps and structural scores

L=229 · Fold1=1k0nA · Fold2=1rk4B · 1,000 rollouts · near-match sequence · state=1 modes · reference TM-score=0.813



Post-hoc structural screen
Fold1 hits: 3
Fold2 hits: 0
Contact n structure: 3 / 0
First Fold1: 9.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

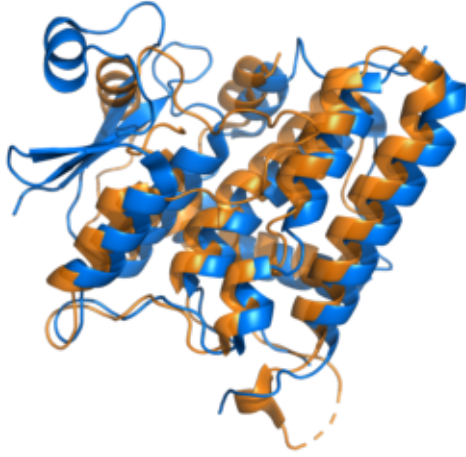
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.609 / 0.000
Fold1 TM-score F1/F2: 0.930 / 0.766
Fold1 GDT: 0.868 vs 0.431 (need 0.835)
Fold1 region: 0.939 vs 0.116
Fold1 CA RMSD: 1.88 Å
Fold1 mean pLDDT: 84.6
Fold2 contact recall F1/F2: 0.565 / 0.065
Fold2 TM-score F1/F2: 0.841 / 0.727
Fold2 GDT: 0.474 vs 0.610 (need 0.822)
Fold2 region: 0.159 vs 0.561
Fold2 CA RMSD: 8.02 Å
Fold2 mean pLDDT: 77.9

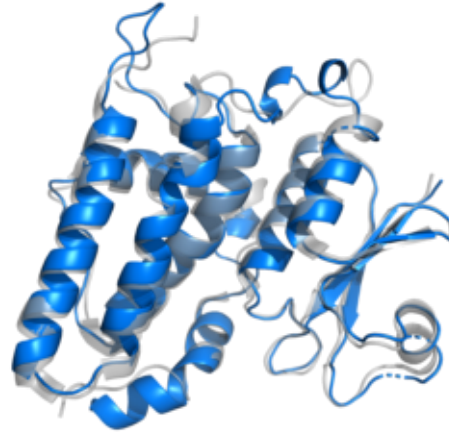
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

01. 1k0na_1rk4b — experimental and Helico structures

Experimental folds (aligned)



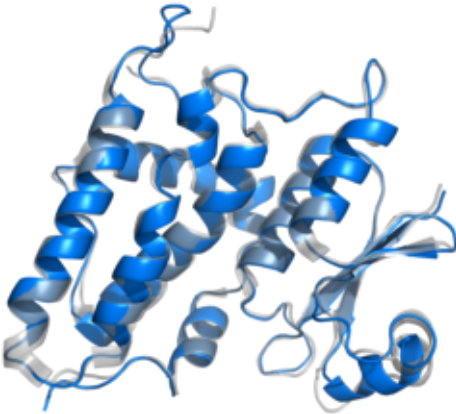
Primary Hit · rollout 9 vs Fold1



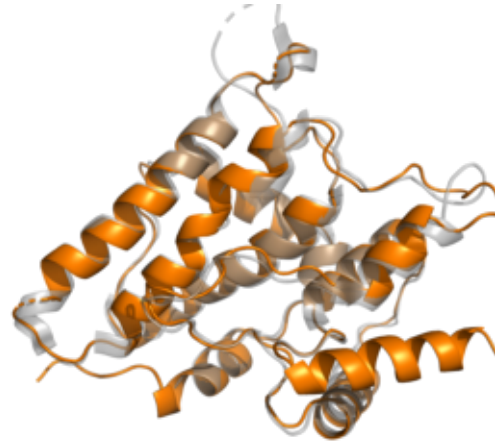
Best Miss · rollout 260 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.928 / 0.491
region: 0.744 / 0.122

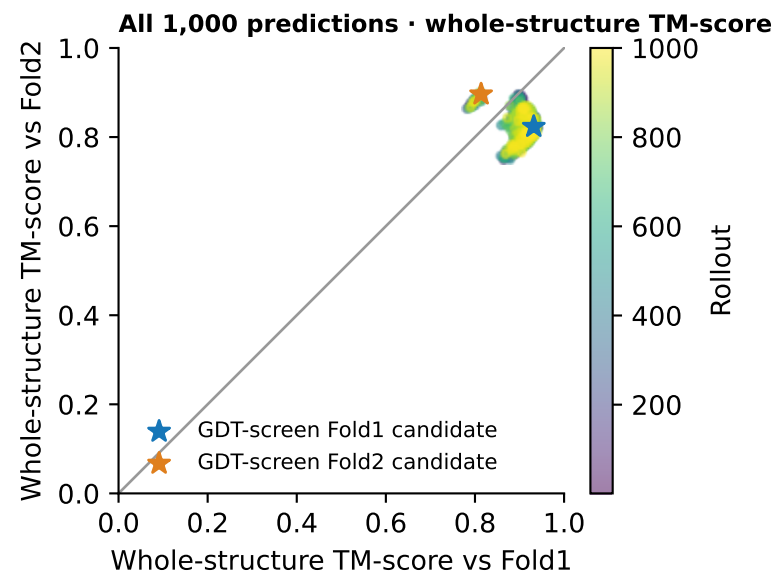
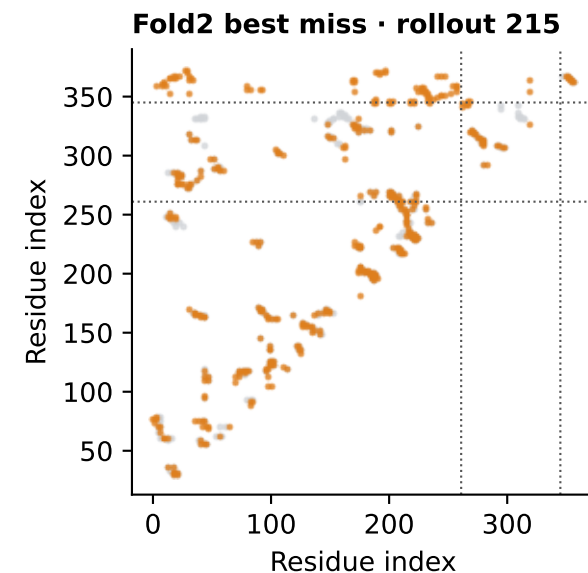
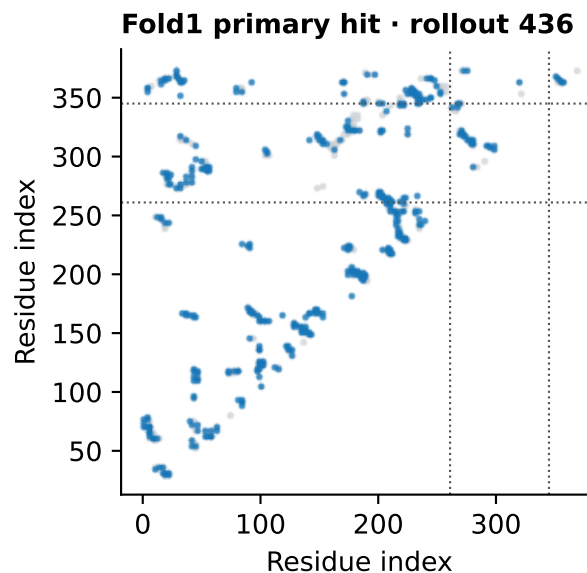
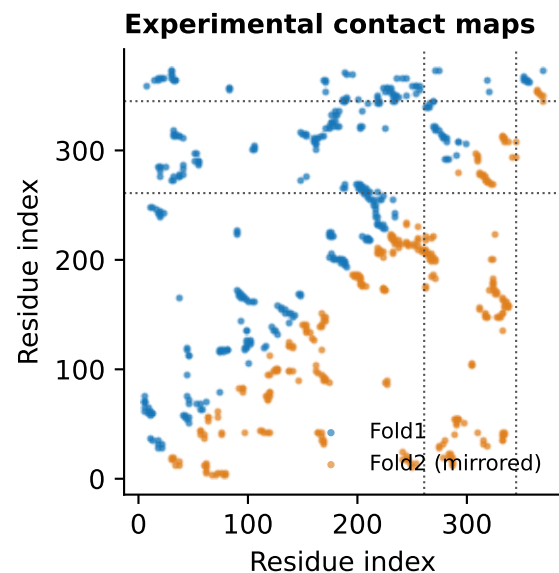
True Fold2-contact control
GDT Fold1/Fold2: 0.458 / 0.913
region: 0.104 / 0.860

No-contact baseline
GDT Fold1/Fold2: 0.046 / 0.035
region: 0.049 / 0.030

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

02. 1kcta_3t1pa — contact maps and structural scores

L=377 · Fold1=1kctA · Fold2=3t1pA · 1,000 rollouts · near-match sequence · state=1 modes · reference TM-score=0.812



Post-hoc structural screen
Fold1 hits: 373
Fold2 hits: 0
Contact n structure: 24 / 0
First Fold1: 2.0
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

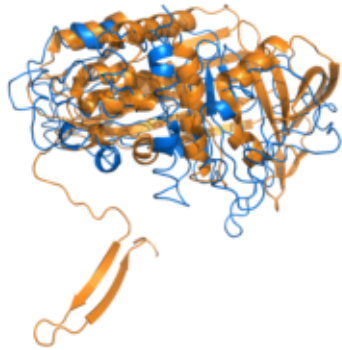
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.222 / 0.281
Fold1 TM-score F1/F2: 0.932 / 0.823
Fold1 GDT: 0.786 vs 0.220 (need 0.718)
Fold1 region: 0.645 vs 0.198
Fold1 CA RMSD: 2.88 Å
Fold1 mean pLDDT: 87.2
Fold2 contact recall F1/F2: 0.222 / 0.266
Fold2 TM-score F1/F2: 0.814 / 0.896
Fold2 GDT: 0.462 vs 0.115 (need 0.350)
Fold2 region: 0.420 vs 0.068
Fold2 CA RMSD: 8.21 Å
Fold2 mean pLDDT: 88.2

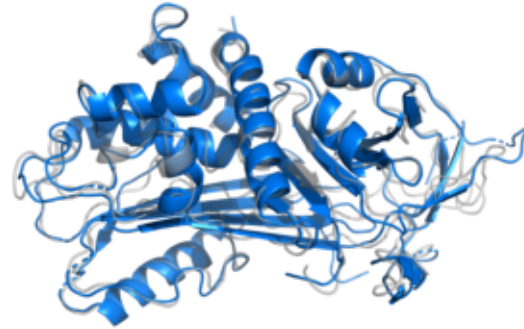
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

02. 1kcta_3t1pa — experimental and Helico structures

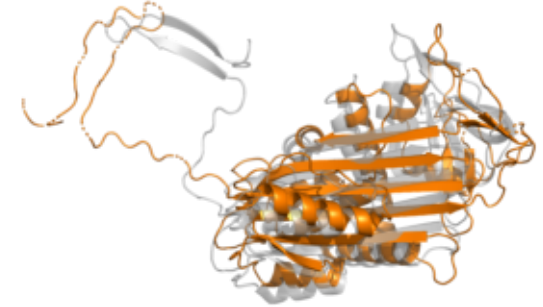
Experimental folds (aligned)



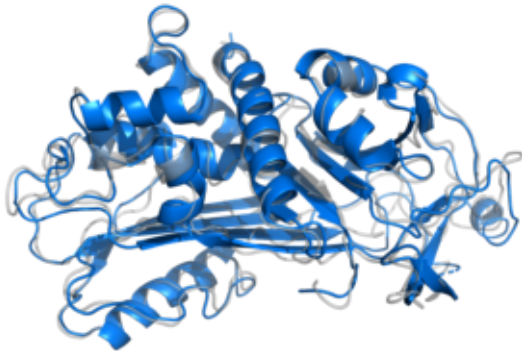
Primary Hit · rollout 436 vs Fold1



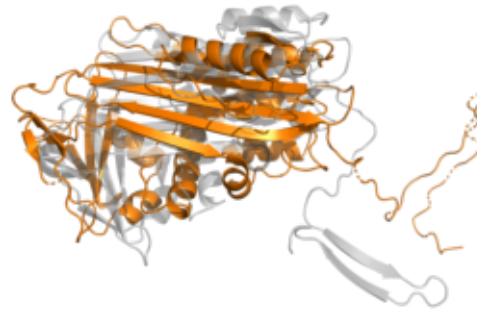
Best Miss · rollout 215 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.798 / 0.214
region: 0.756 / 0.188

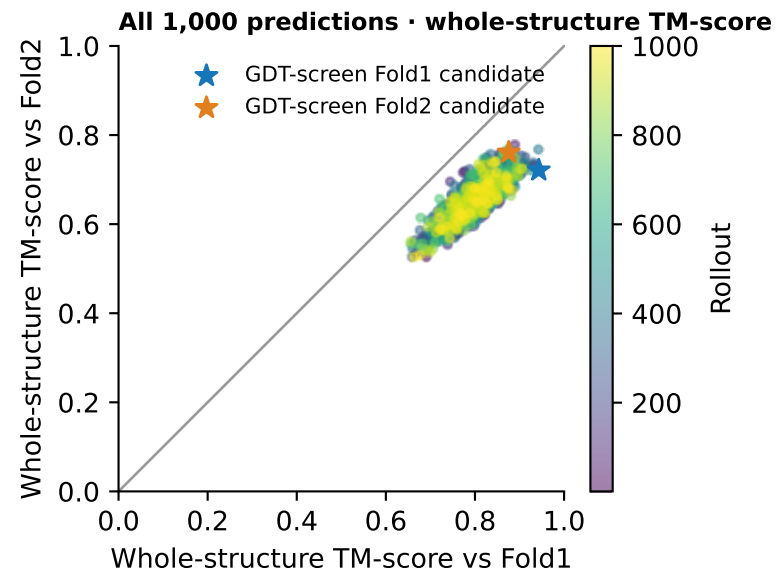
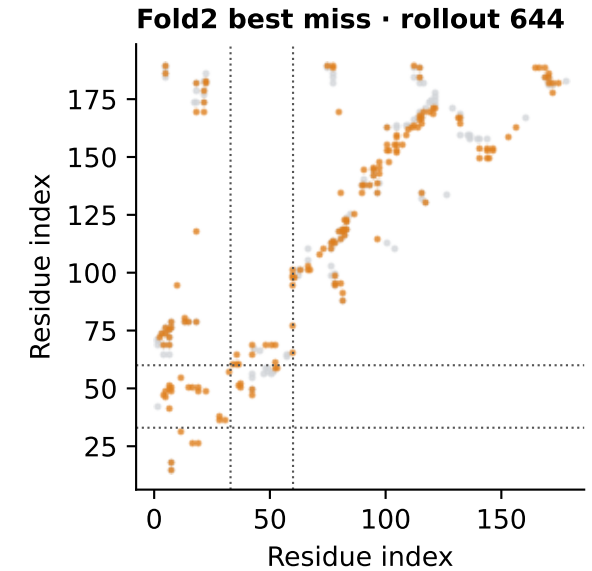
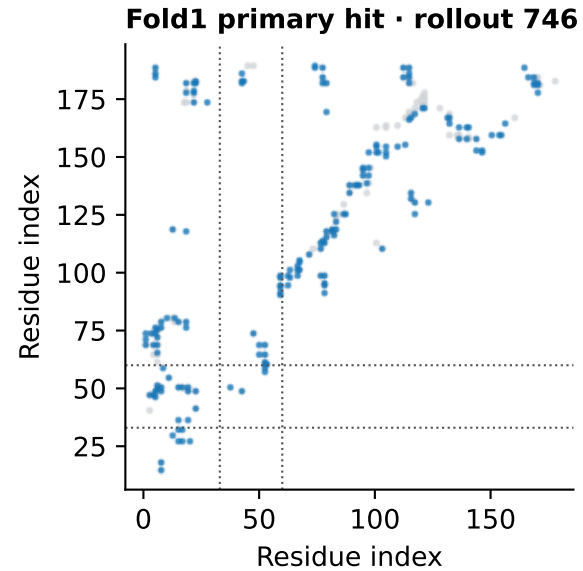
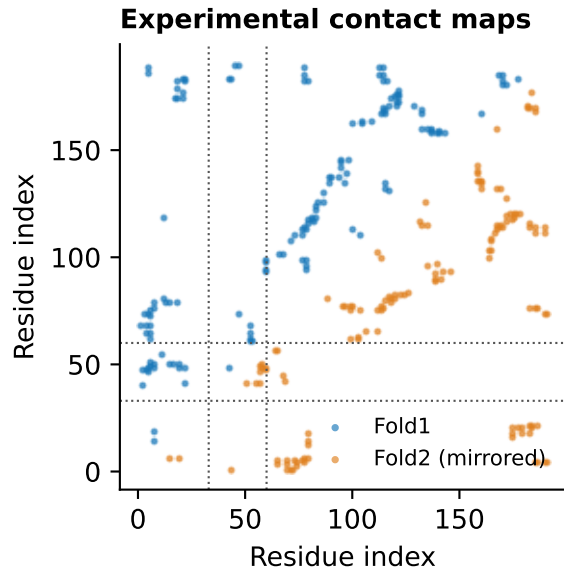
True Fold2-contact control
GDT Fold1/Fold2: 0.120 / 0.215
region: 0.086 / 0.151

No-contact baseline
GDT Fold1/Fold2: 0.012 / 0.010
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

03. 1nrjb_2gedb — contact maps and structural scores

L=191 · Fold1=1nrjB · Fold2=2gedB · 1,000 rollouts · exact sequence · state=1 modes · reference TM-score=0.745



Post-hoc structural screen
Fold1 hits: 6
Fold2 hits: 0
Contact n structure: 6 / 0
First Fold1: 42.0
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.903 / 0.000
Fold1 TM-score F1/F2: 0.943 / 0.721
Fold1 GDT: 0.915 vs 0.400 (need 0.865)
Fold1 region: 1.000 vs 0.025
Fold1 CA RMSD: 1.27 Å
Fold1 mean pLDDT: 83.8
Fold2 contact recall F1/F2: 0.581 / 0.071
Fold2 TM-score F1/F2: 0.876 / 0.761
Fold2 GDT: 0.460 vs 0.785 (need 0.433)
Fold2 region: 0.037 vs 0.625
Fold2 CA RMSD: 7.10 Å
Fold2 mean pLDDT: 77.4

Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

03. 1nrjb_2gedb — experimental and Helico structures

Experimental folds (aligned)



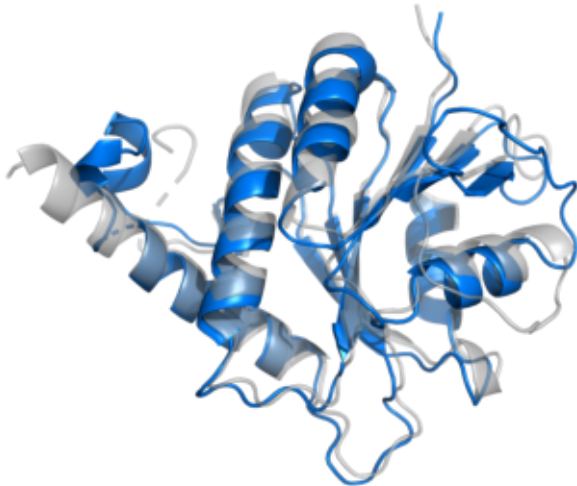
Primary Hit • rollout 746 vs Fold1



Best Miss • rollout 644 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.962 / 0.429
region: 0.887 / 0.013

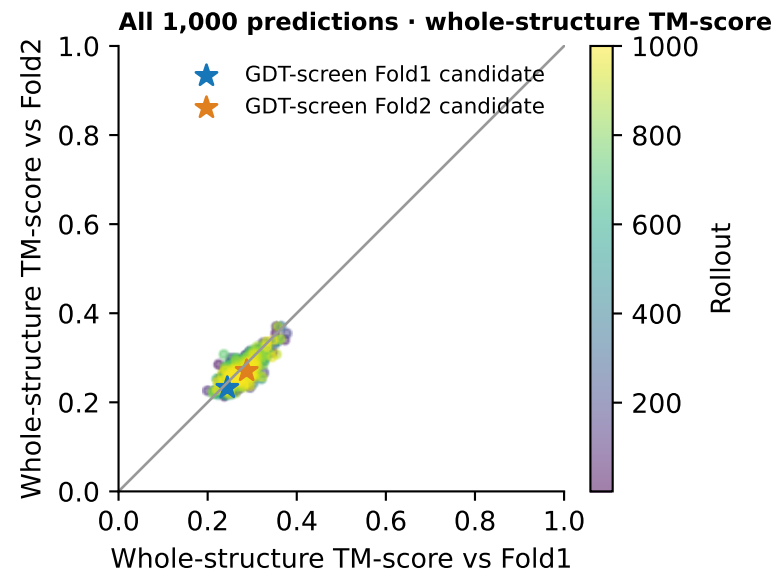
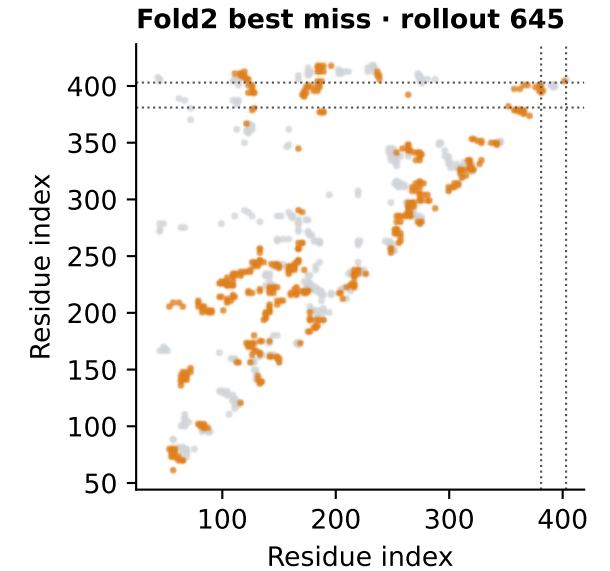
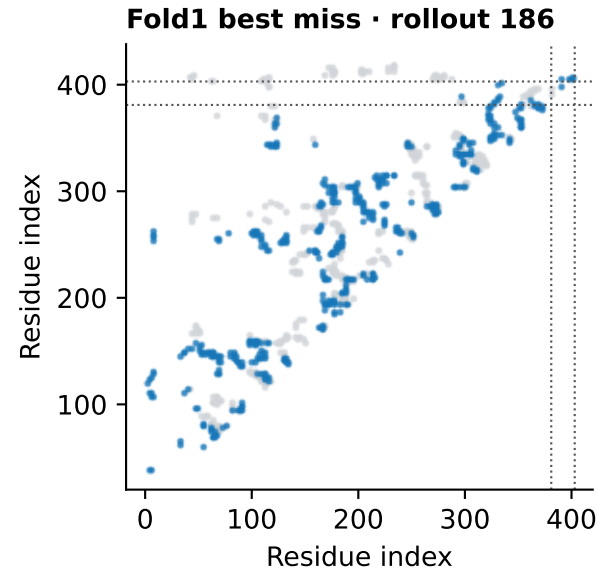
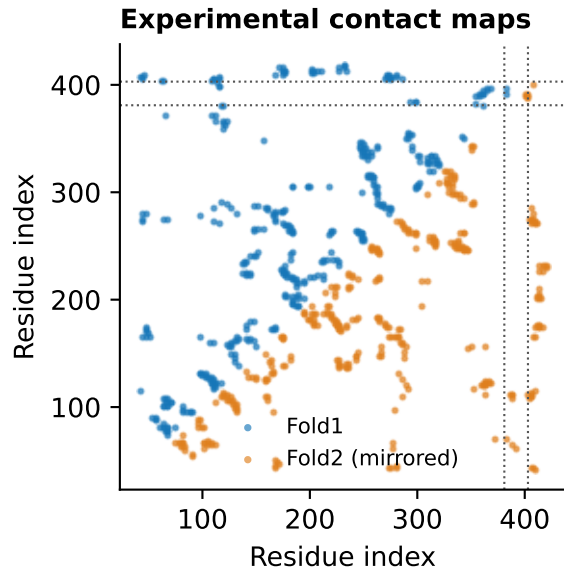
True Fold2-contact control
GDT Fold1/Fold2: 0.654 / 0.481
region: 0.213 / 0.000

No-contact baseline
GDT Fold1/Fold2: 0.169 / 0.173
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

04. 1qomb_1nocA — contact maps and structural scores

L=420 · Fold1=1qomB · Fold2=1nocA · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.875



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

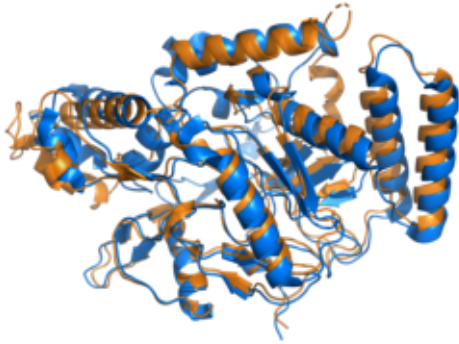
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.000 / 0.000
Fold1 TM-score F1/F2: 0.244 / 0.233
Fold1 GDT: 0.042 vs 0.026 (need 0.827)
Fold1 region: 0.102 vs 0.000
Fold1 CA RMSD: 22.71 Å
Fold1 mean pLDDT: 45.4
Fold2 contact recall F1/F2: 0.048 / 0.000
Fold2 TM-score F1/F2: 0.288 / 0.271
Fold2 GDT: 0.054 vs 0.043 (need 0.797)
Fold2 region: 0.000 vs 0.000
Fold2 CA RMSD: 22.71 Å
Fold2 mean pLDDT: 50.3

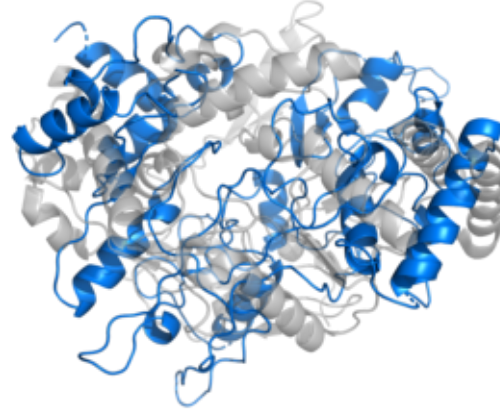
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

04. 1qomb_1nocA — experimental and Helico structures

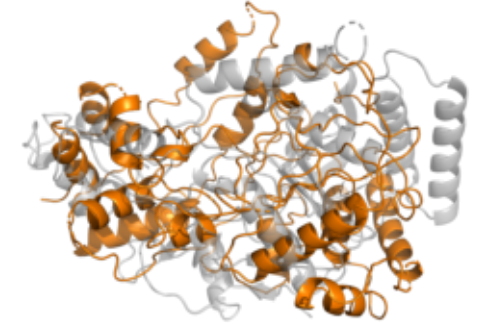
Experimental folds (aligned)



Best Miss · rollout 186 vs Fold1



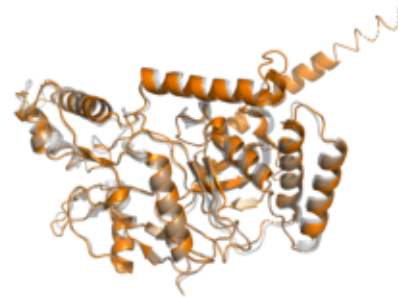
Best Miss · rollout 645 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.919 / 0.614
region: 0.636 / 0.057

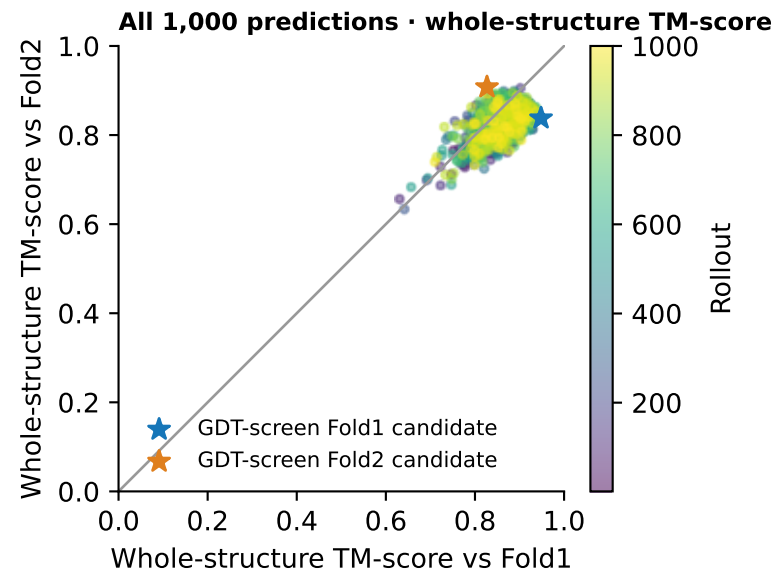
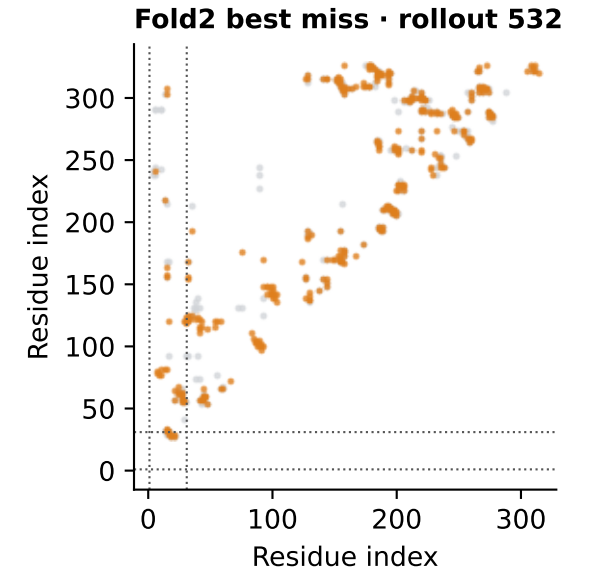
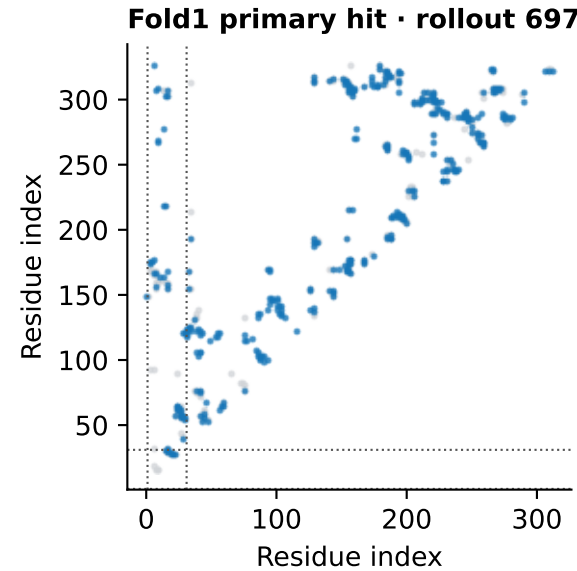
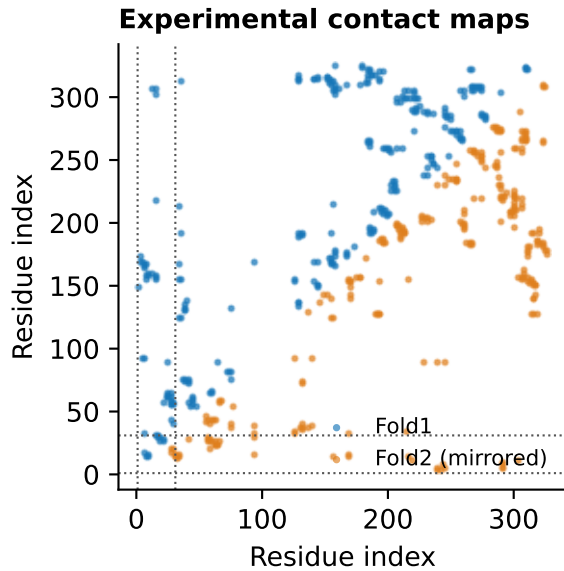
True Fold2-contact control
GDT Fold1/Fold2: 0.563 / 0.886
region: 0.068 / 0.625

No-contact baseline
GDT Fold1/Fold2: 0.026 / 0.019
region: 0.023 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

05. 1qs8b_1miqb — contact maps and structural scores

L=411 · Fold1=1qs8B · Fold2=1miqB · 1,000 rollouts · near-match sequence · state=1 modes · reference TM-score=0.843



Post-hoc structural screen
Fold1 hits: 73
Fold2 hits: 0
Contact n structure: 52 / 0
First Fold1: 7.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

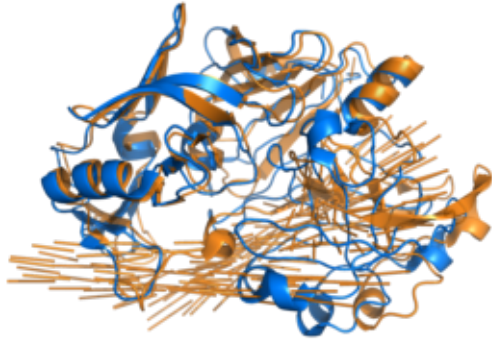
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.265 / 0.043
Fold1 TM-score F1/F2: 0.948 / 0.838
Fold1 GDT: 0.855 vs 0.592 (need 0.784)
Fold1 region: 0.833 vs 0.267
Fold1 CA RMSD: 1.73 Å
Fold1 mean pLDDT: 85.0
Fold2 contact recall F1/F2: 0.235 / 0.000
Fold2 TM-score F1/F2: 0.827 / 0.908
Fold2 GDT: 0.758 vs 0.561 (need 0.805)
Fold2 region: 0.483 vs 0.200
Fold2 CA RMSD: 2.45 Å
Fold2 mean pLDDT: 82.2

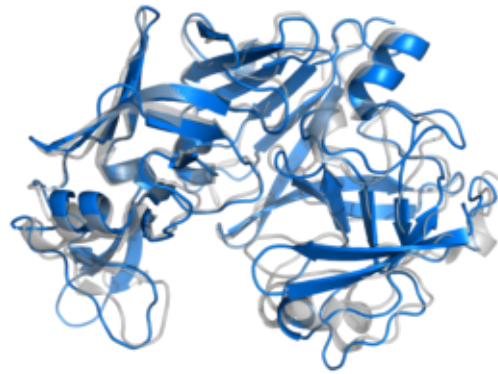
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

05. 1qs8b_1miqb — experimental and Helico structures

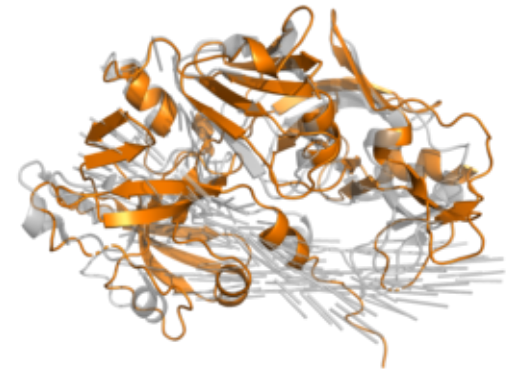
Experimental folds (aligned)



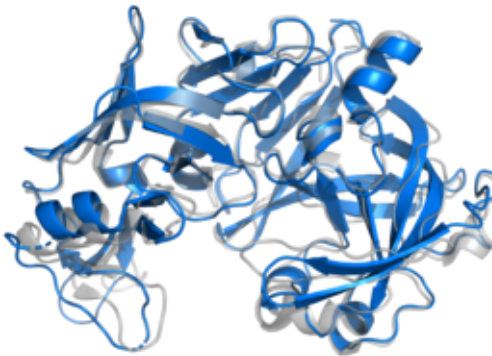
Primary Hit • rollout 697 vs Fold1



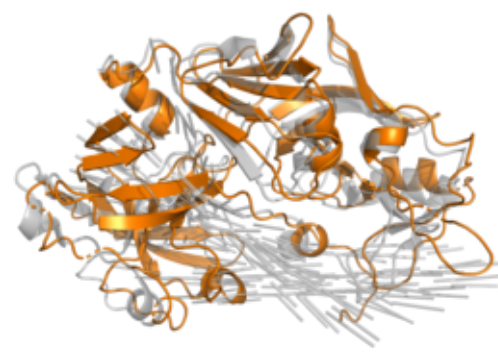
Best Miss • rollout 532 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.871 / 0.613
region: 0.833 / 0.317

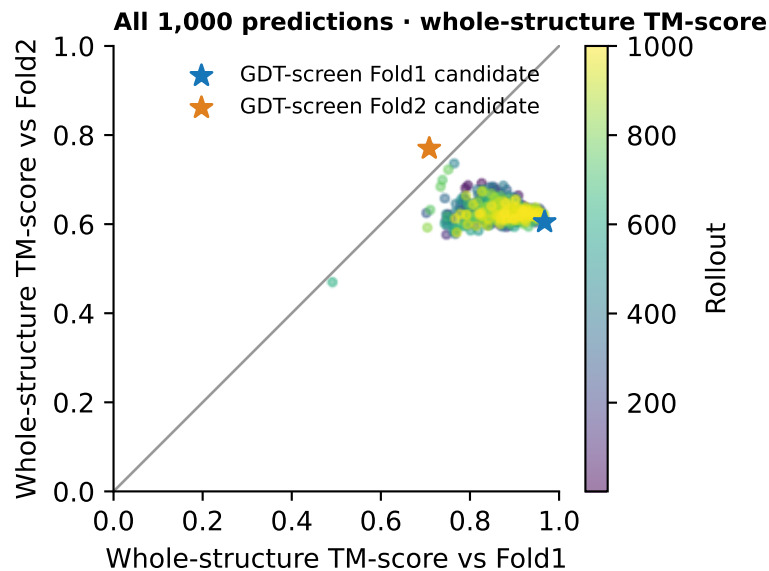
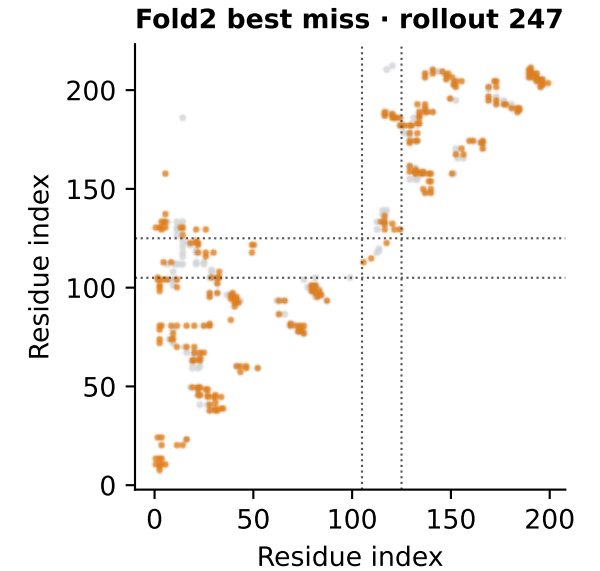
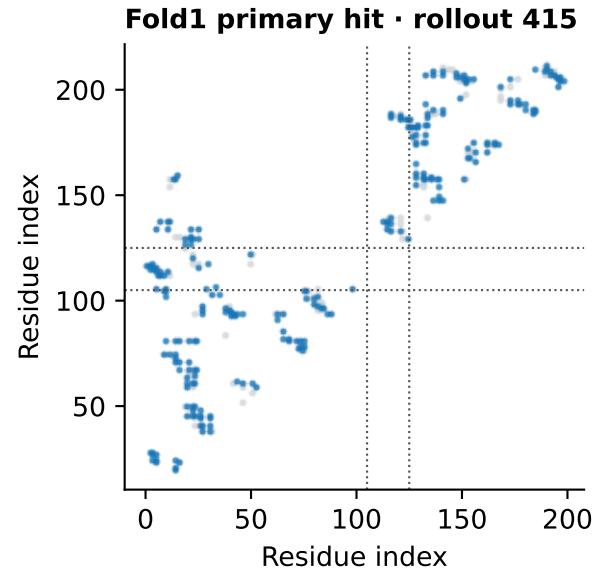
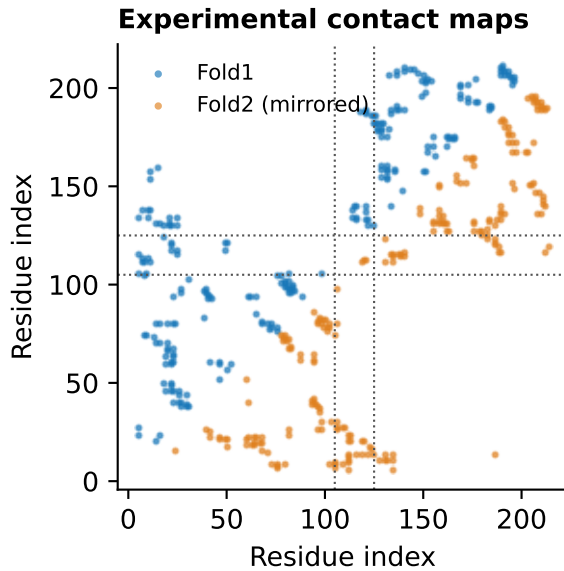
True Fold2-contact control
GDT Fold1/Fold2: 0.564 / 0.895
region: 0.225 / 0.758

No-contact baseline
GDT Fold1/Fold2: 0.019 / 0.028
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

06. 1repc_2z9ob — contact maps and structural scores

L=227 · Fold1=1repC · Fold2=2z9oB · 1,000 rollouts · near-match sequence · state=1 modes · reference TM-score=0.615



Post-hoc structural screen
Fold1 hits: 355
Fold2 hits: 0
Contact n structure: 355 / 0
First Fold1: 2.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

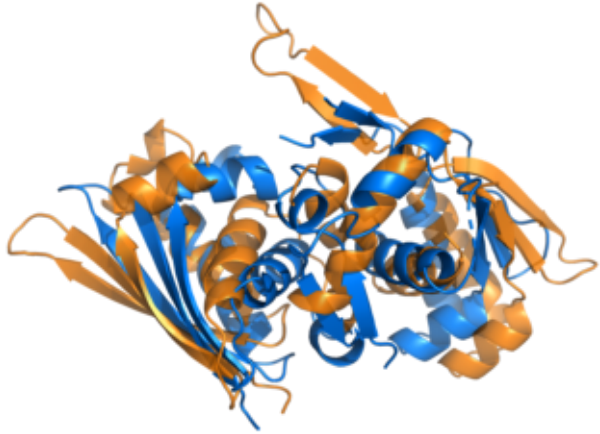
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.607 / 0.100
Fold1 TM-score F1/F2: 0.968 / 0.605
Fold1 GDT: 0.925 vs 0.276 (need 0.799)
Fold1 region: 0.863 vs 0.263
Fold1 CA RMSD: 1.02 Å
Fold1 mean pLDDT: 88.5
Fold2 contact recall F1/F2: 0.393 / 0.125
Fold2 TM-score F1/F2: 0.709 / 0.770
Fold2 GDT: 0.521 vs 0.437 (need 0.723)
Fold2 region: 0.388 vs 0.325
Fold2 CA RMSD: 3.79 Å
Fold2 mean pLDDT: 77.6

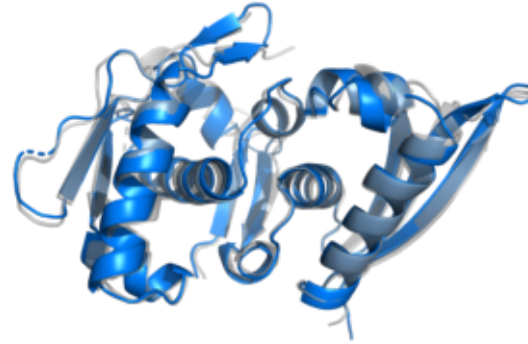
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

06. 1repc_2z9ob — experimental and Helico structures

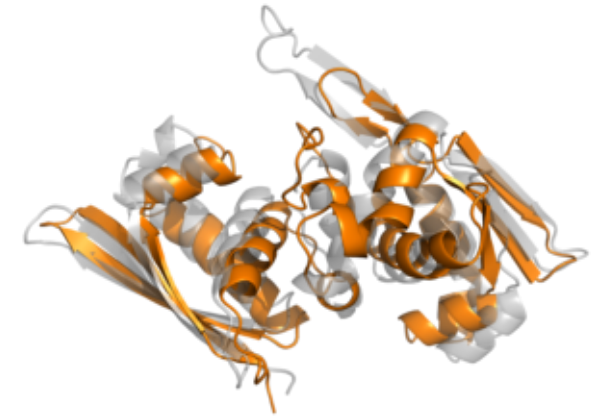
Experimental folds (aligned)



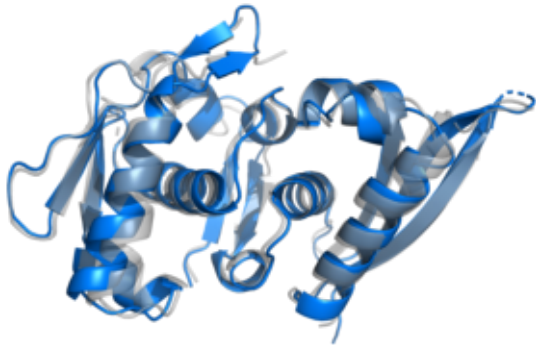
Primary Hit · rollout 415 vs Fold1



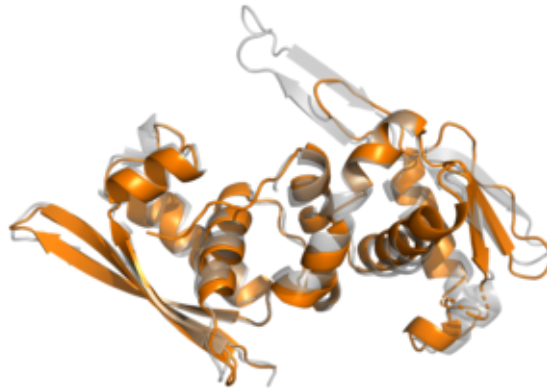
Best Miss · rollout 247 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.888 / 0.298
region: 0.850 / 0.275

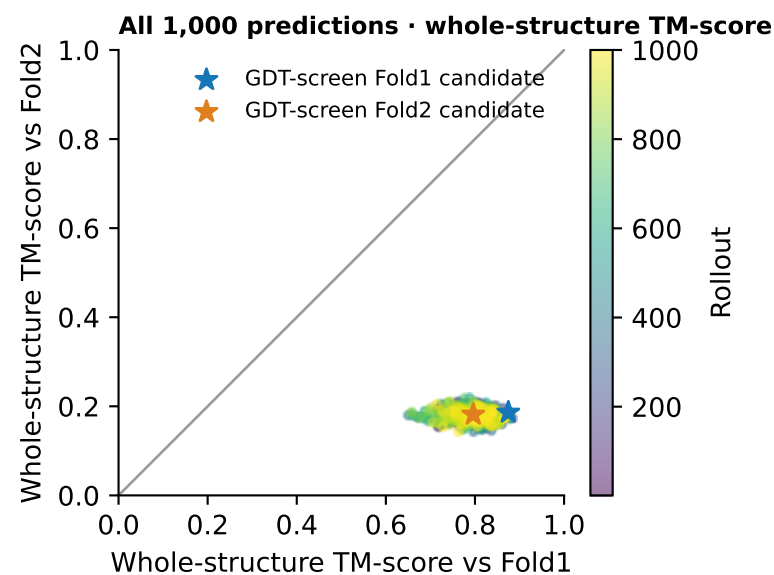
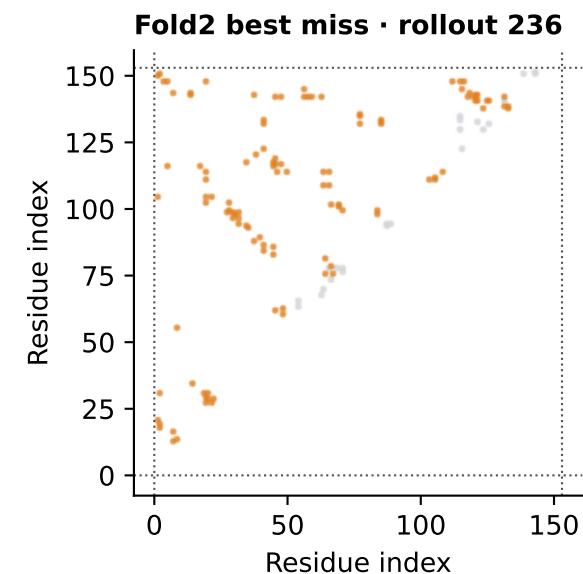
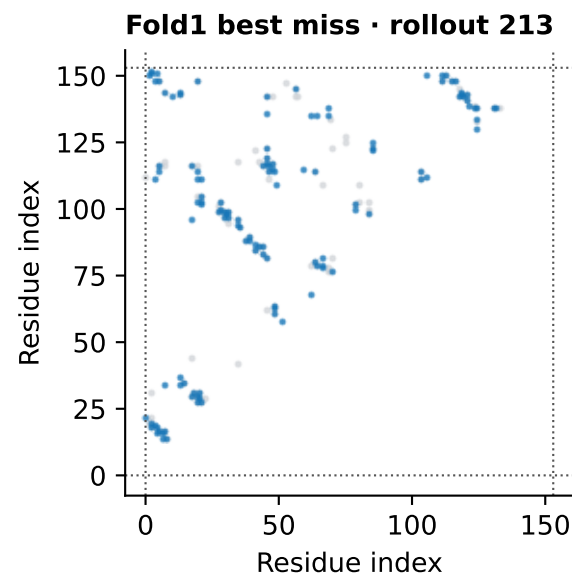
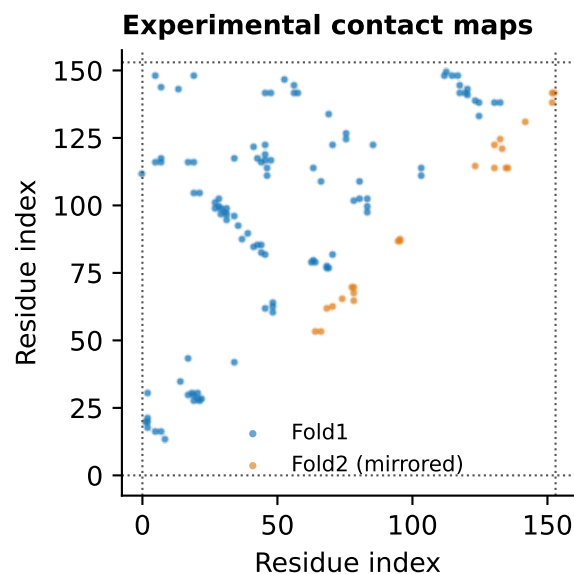
True Fold2-contact control
GDT Fold1/Fold2: 0.261 / 0.803
region: 0.225 / 0.762

No-contact baseline
GDT Fold1/Fold2: 0.039 / 0.046
region: 0.062 / 0.075

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

07. 1uxmk_2nama — contact maps and structural scores

L=153 · Fold1=1uxmk · Fold2=2nama · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.186



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

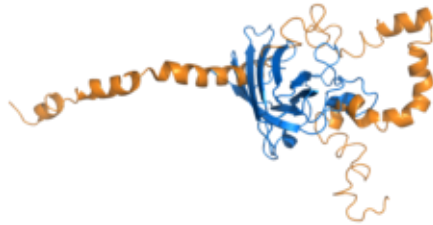
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.629 / 0.091
Fold1 TM-score F1/F2: 0.875 / 0.186
Fold1 GDT: 0.814 vs 0.005 (need 0.875)
Fold1 region: 0.814 vs 0.005
Fold1 CA RMSD: 2.39 Å
Fold1 mean pLDDT: 82.4
Fold2 contact recall F1/F2: 0.590 / 0.000
Fold2 TM-score F1/F2: 0.796 / 0.182
Fold2 GDT: 0.046 vs 0.620 (need 0.350)
Fold2 region: 0.046 vs 0.620
Fold2 CA RMSD: 30.62 Å
Fold2 mean pLDDT: 78.5

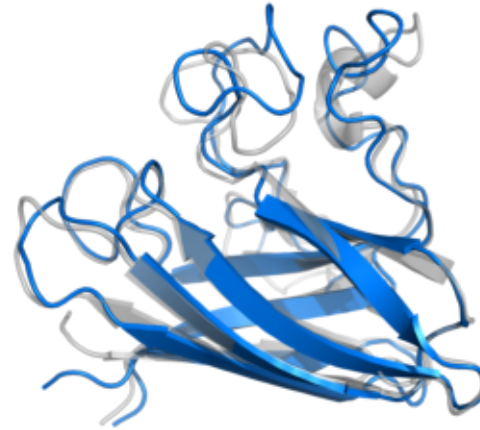
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

07. 1uxmk_2nama — experimental and Helico structures

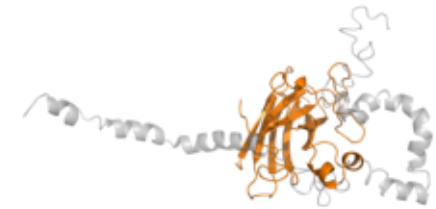
Experimental folds (aligned)



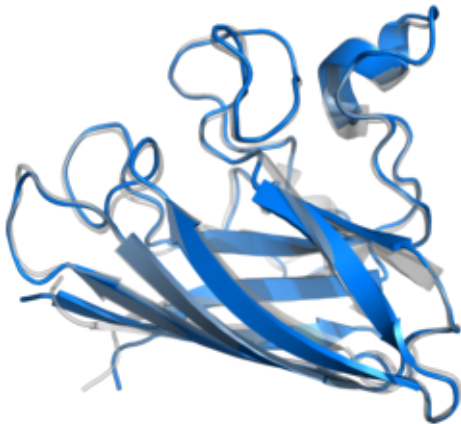
Best Miss · rollout 213 vs Fold1



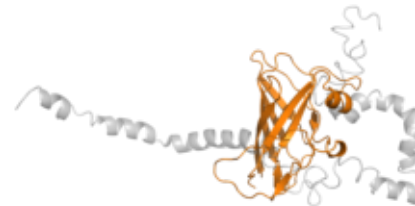
Best Miss · rollout 236 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.972 / 0.005
region: 0.972 / 0.005

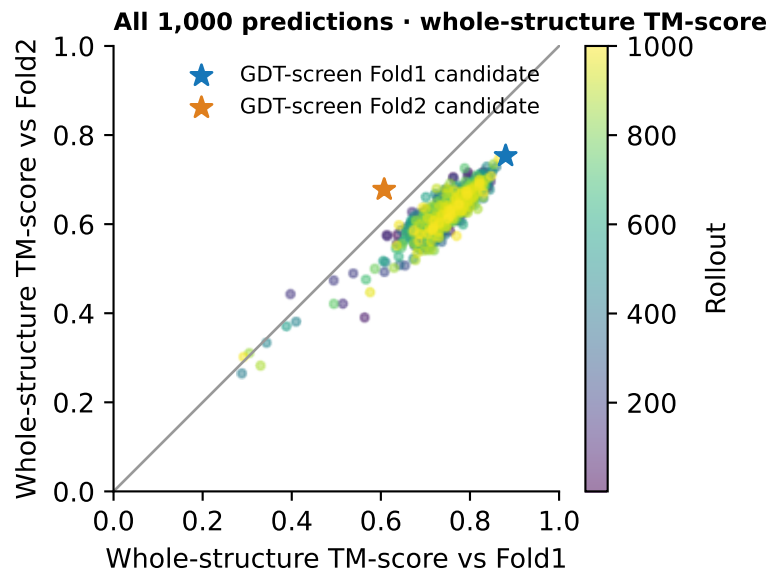
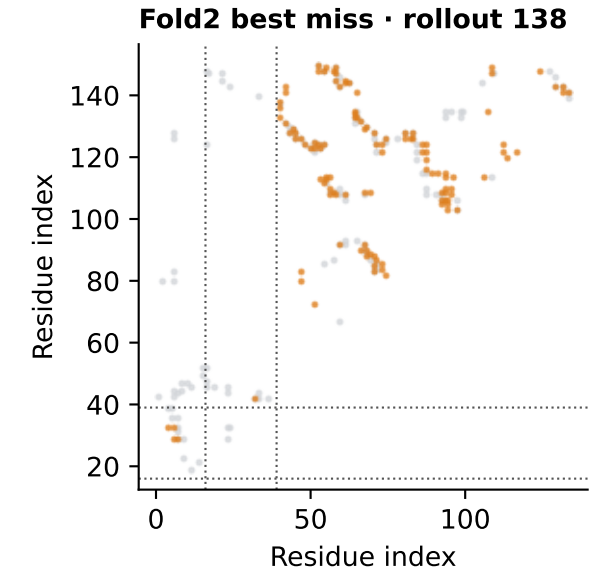
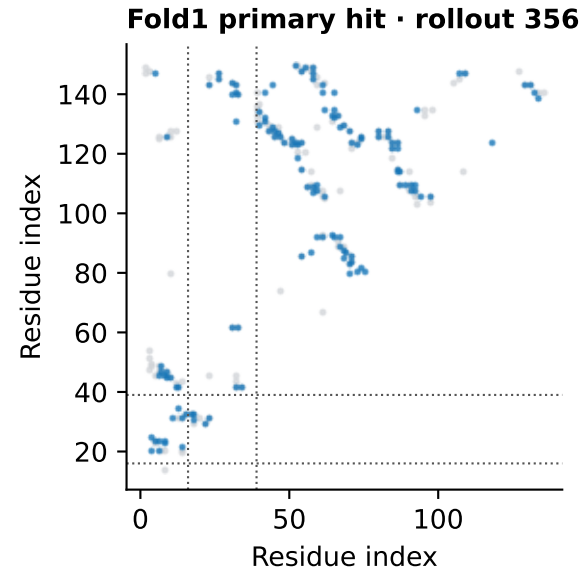
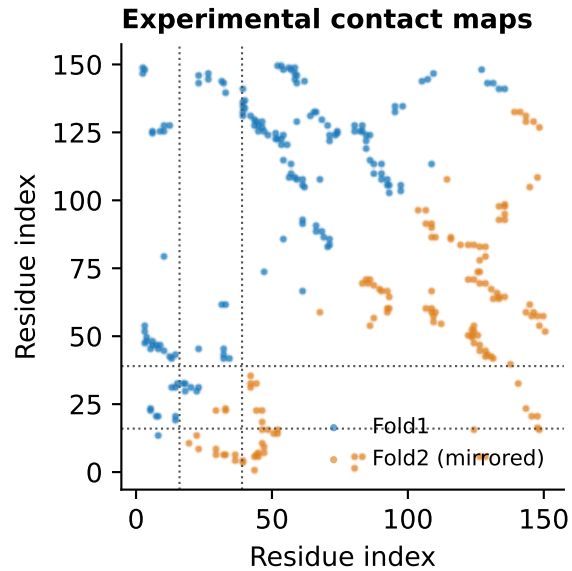
True Fold2-contact control
GDT Fold1/Fold2: 0.270 / 0.016
region: 0.270 / 0.016

No-contact baseline
GDT Fold1/Fold2: 0.367 / 0.000
region: 0.367 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

08. 1xnta_3lqca — contact maps and structural scores

L=152 · Fold1=1xntA · Fold2=3lqcA · 1,000 rollouts · exact sequence · state=1 modes · reference TM-score=0.721



Post-hoc structural screen
Fold1 hits: 16
Fold2 hits: 0
Contact n structure: 16 / 0
First Fold1: 70.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

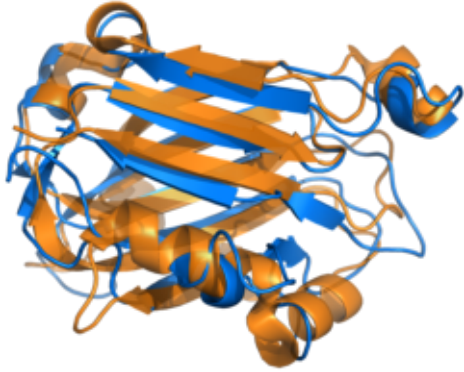
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.577 / 0.038
Fold1 TM-score F1/F2: 0.880 / 0.753
Fold1 GDT: 0.765 vs 0.567 (need 0.695)
Fold1 region: 0.739 vs 0.228
Fold1 CA RMSD: 1.93 Å
Fold1 mean pLDDT: 86.2
Fold2 contact recall F1/F2: 0.000 / 0.038
Fold2 TM-score F1/F2: 0.607 / 0.678
Fold2 GDT: 0.403 vs 0.365 (need 0.828)
Fold2 region: 0.435 vs 0.109
Fold2 CA RMSD: 5.32 Å
Fold2 mean pLDDT: 63.2

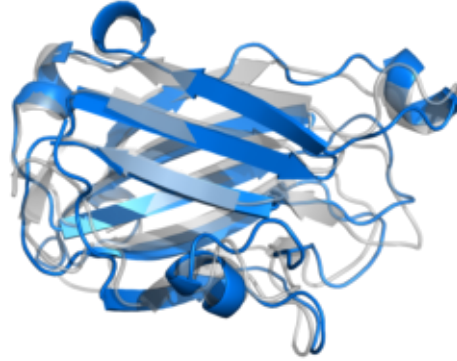
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

08. 1xnta_3lqca — experimental and Helico structures

Experimental folds (aligned)



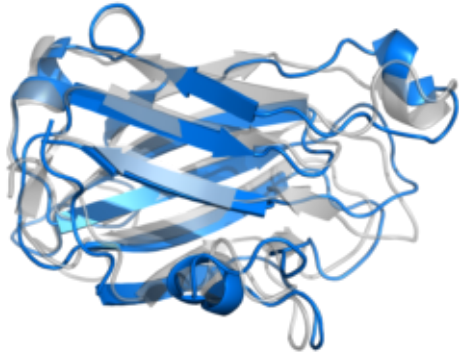
Primary Hit · rollout 356 vs Fold1



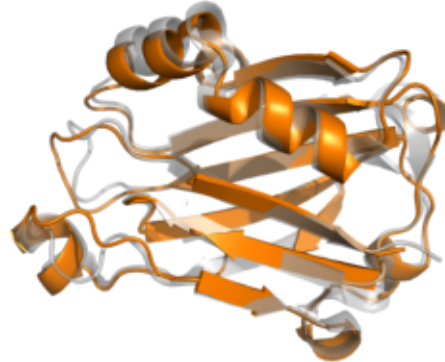
Best Miss · rollout 138 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.772 / 0.592
region: 0.739 / 0.228

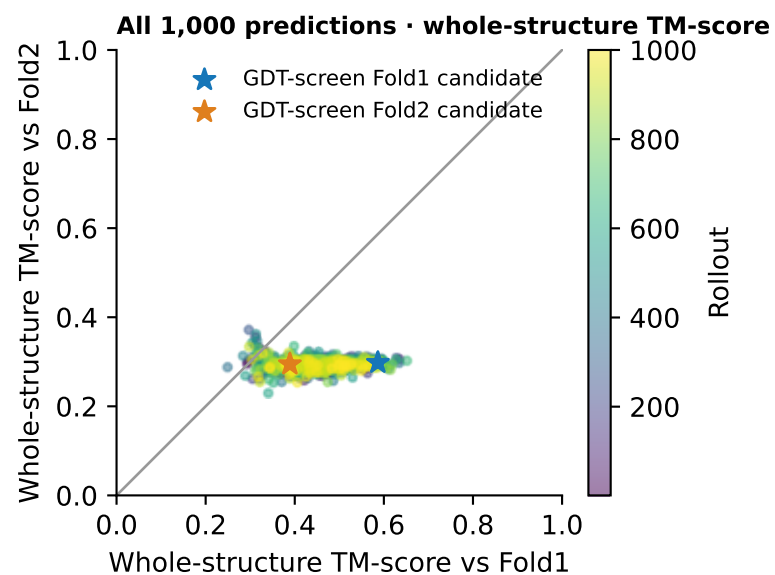
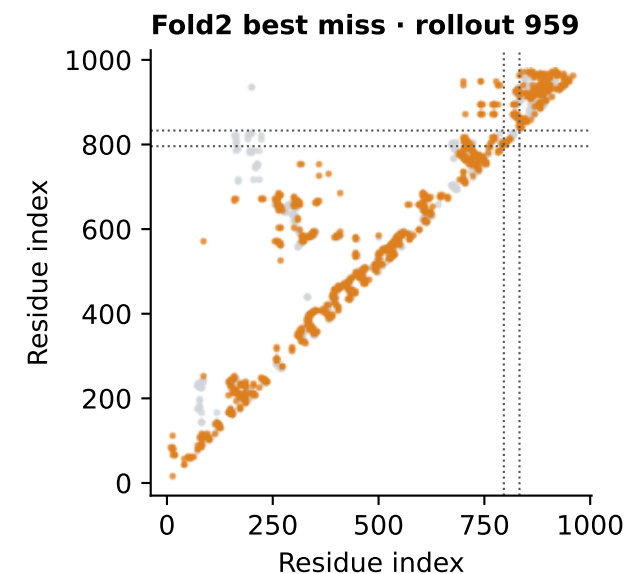
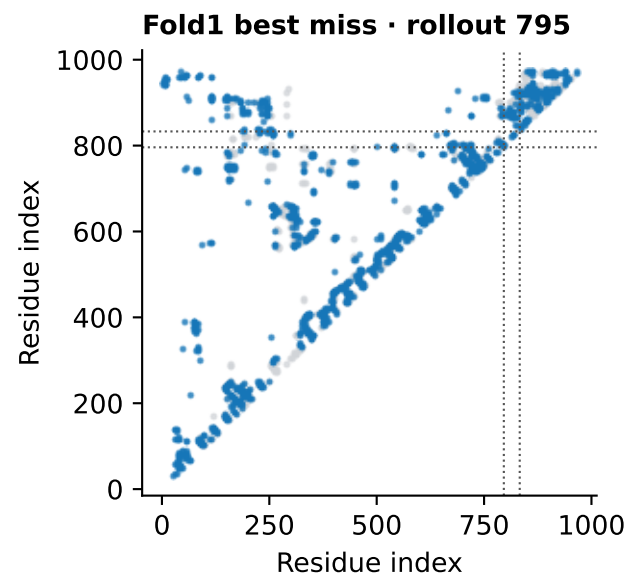
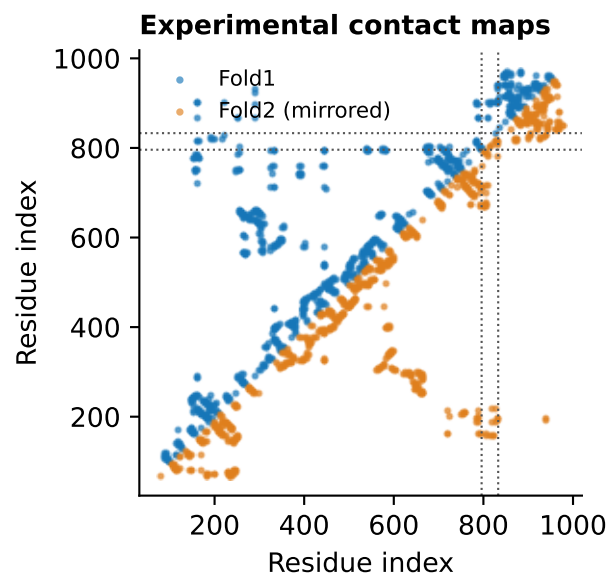
True Fold2-contact control
GDT Fold1/Fold2: 0.563 / 0.920
region: 0.239 / 0.891

No-contact baseline
GDT Fold1/Fold2: 0.068 / 0.072
region: 0.033 / 0.087

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

09. 2a73b_3l5nb — contact maps and structural scores

L=982 · Fold1=2a73B · Fold2=3l5nB · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.301



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

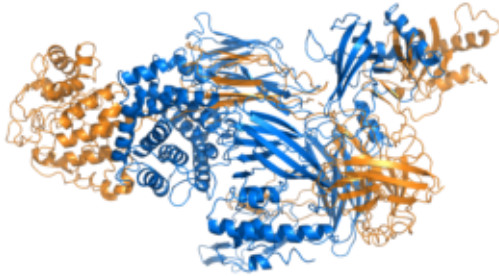
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.125 / 0.095
Fold1 TM-score F1/F2: 0.587 / 0.298
Fold1 GDT: 0.186 vs 0.009 (need 0.424)
Fold1 region: 0.101 vs 0.000
Fold1 CA RMSD: 18.68 Å
Fold1 mean pLDDT: 59.9
Fold2 contact recall F1/F2: 0.000 / 0.000
Fold2 TM-score F1/F2: 0.389 / 0.295
Fold2 GDT: 0.042 vs 0.000 (need 0.350)
Fold2 region: 0.000 vs 0.000
Fold2 CA RMSD: 60.23 Å
Fold2 mean pLDDT: 58.9

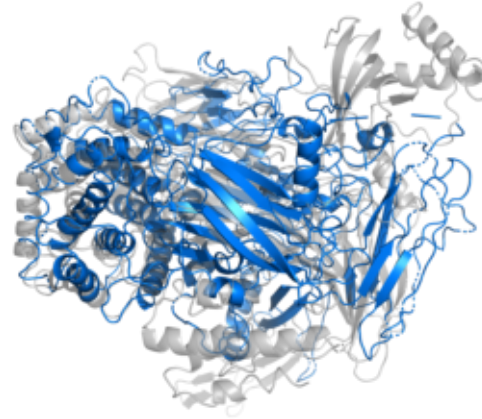
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

09. 2a73b_3l5nb — experimental and Helico structures

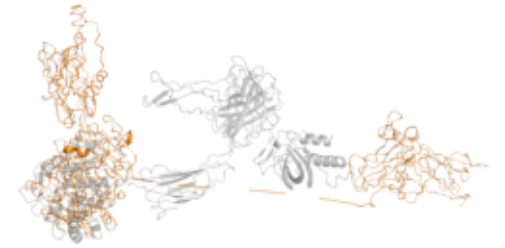
Experimental folds (aligned)



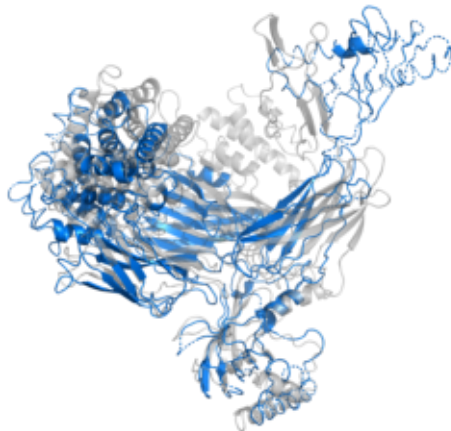
Best Miss · rollout 795 vs Fold1



Best Miss · rollout 959 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.471 / 0.007
region: 0.635 / 0.007

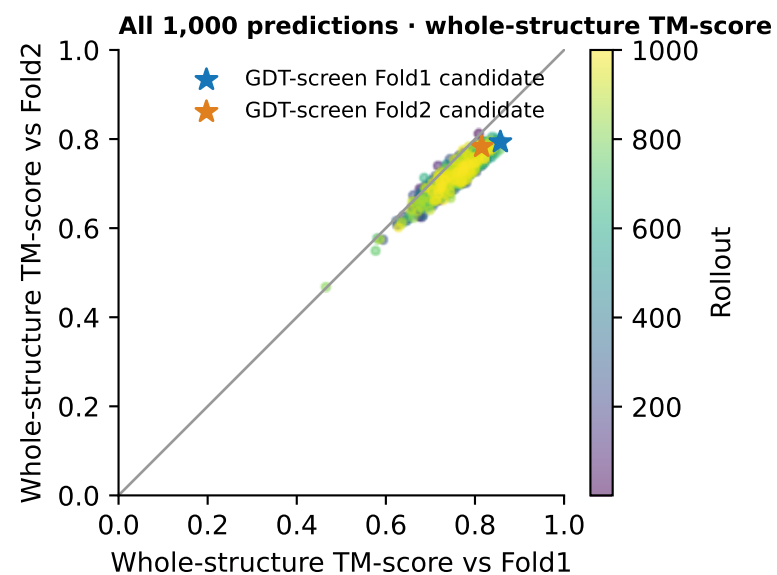
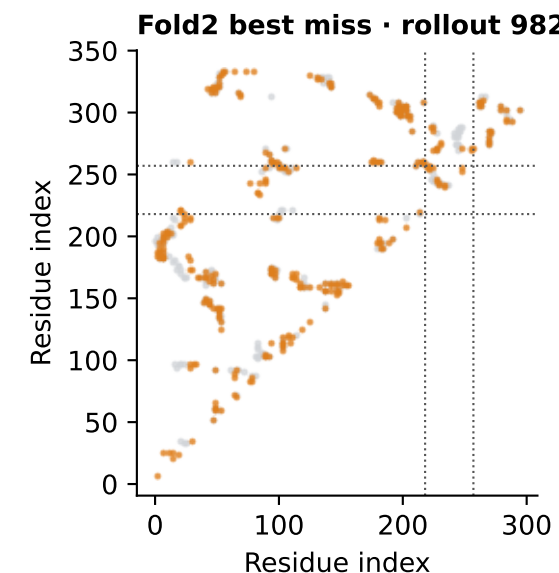
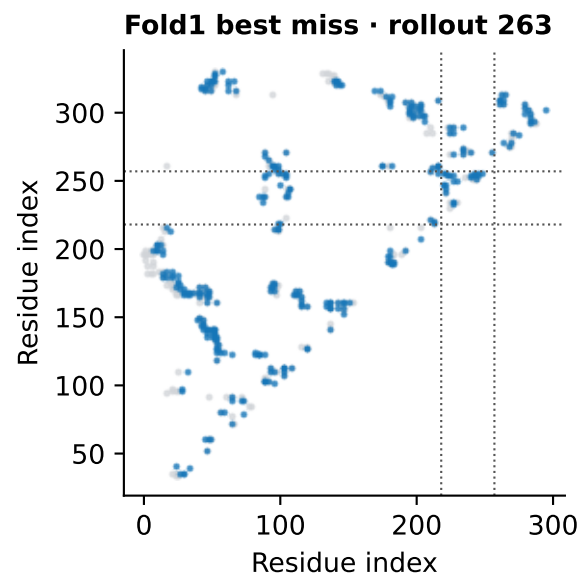
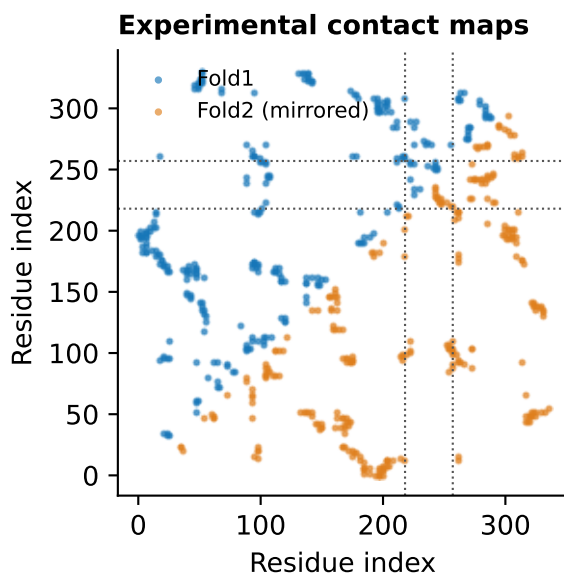
True Fold2-contact control
GDT Fold1/Fold2: 0.000 / 0.000
region: 0.000 / 0.000

No-contact baseline
GDT Fold1/Fold2: 0.004 / 0.003
region: 0.000 / 0.007

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

10. 2c1vb_2c1uc — contact maps and structural scores

L=335 · Fold1=2c1vB · Fold2=2c1uC · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.872



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

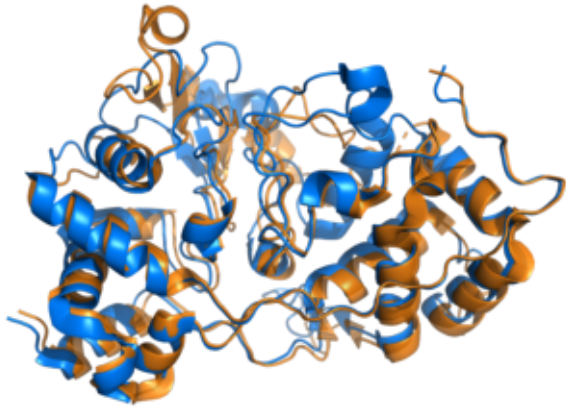
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.500 / 0.085
Fold1 TM-score F1/F2: 0.857 / 0.793
Fold1 GDT: 0.631 vs 0.474 (need 0.734)
Fold1 region: 0.474 vs 0.314
Fold1 CA RMSD: 4.07 Å
Fold1 mean pLDDT: 80.8
Fold2 contact recall F1/F2: 0.133 / 0.128
Fold2 TM-score F1/F2: 0.815 / 0.783
Fold2 GDT: 0.534 vs 0.524 (need 0.745)
Fold2 region: 0.288 vs 0.173
Fold2 CA RMSD: 6.89 Å
Fold2 mean pLDDT: 77.9

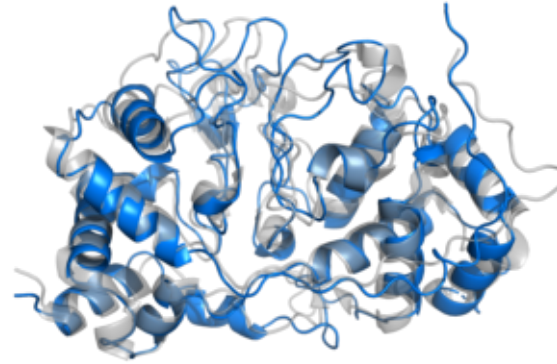
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

10. 2c1vb_2c1uc — experimental and Helico structures

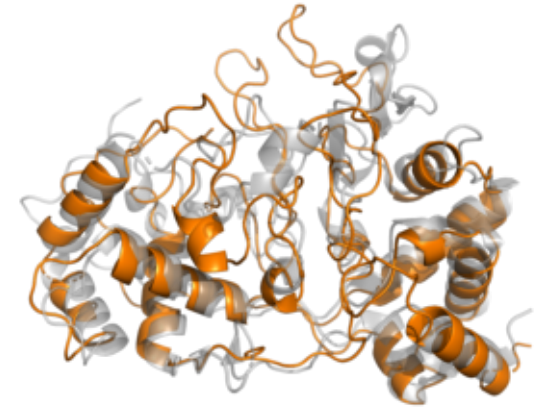
Experimental folds (aligned)



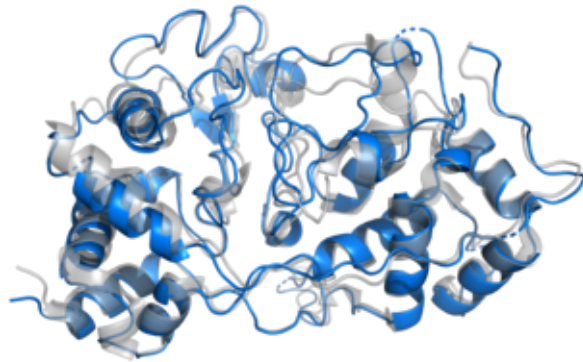
Best Miss · rollout 263 vs Fold1



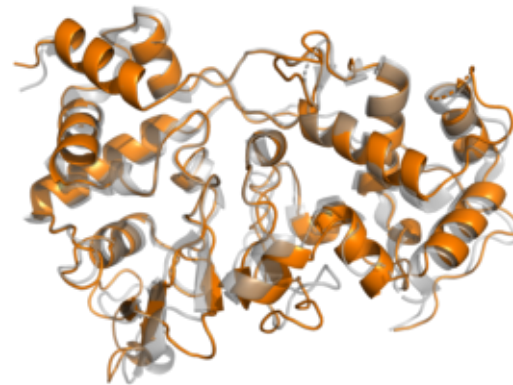
Best Miss · rollout 982 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.815 / 0.569
region: 0.814 / 0.218

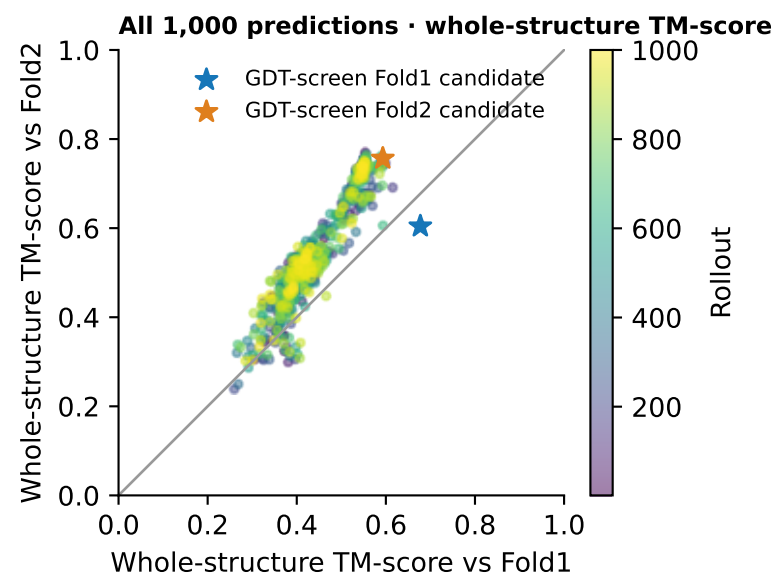
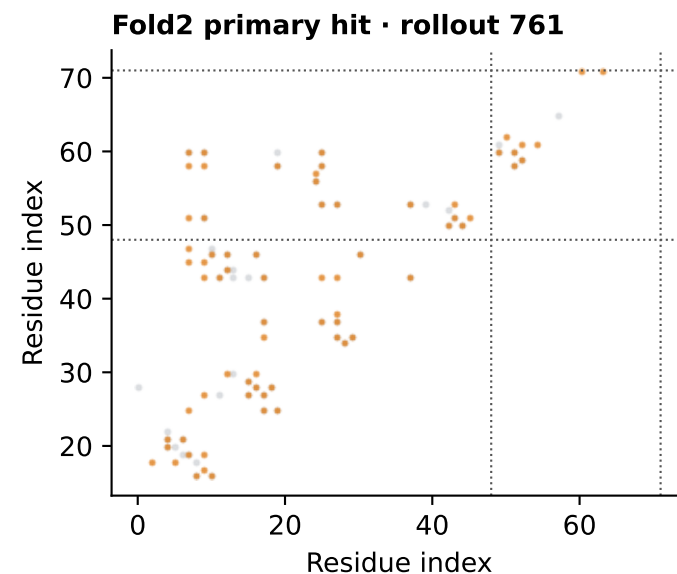
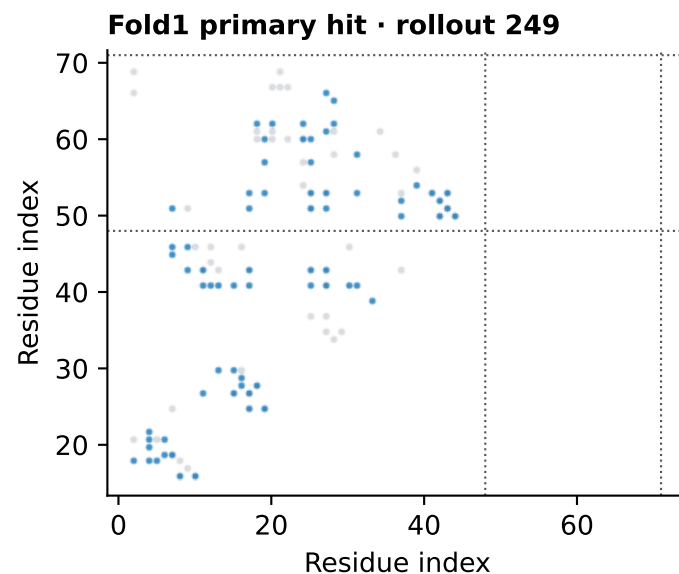
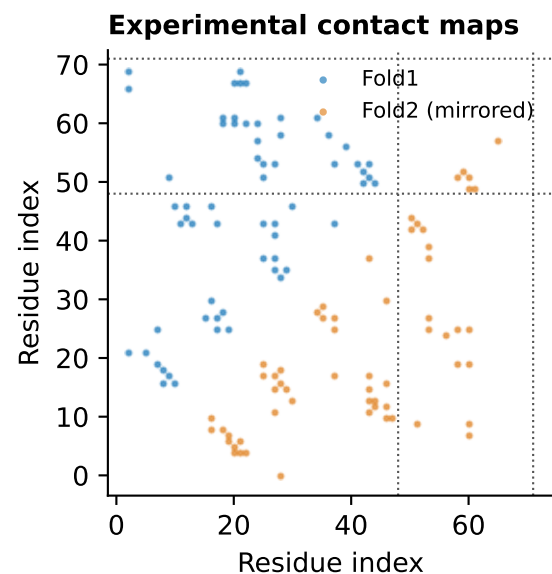
True Fold2-contact control
GDT Fold1/Fold2: 0.561 / 0.828
region: 0.186 / 0.660

No-contact baseline
GDT Fold1/Fold2: 0.041 / 0.035
region: 0.019 / 0.038

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

11. 2lela_2k0qa — contact maps and structural scores

L=74 · Fold1=2lelA · Fold2=2k0qA · 1,000 rollouts · exact sequence · state=2 modes · reference TM-score=0.542



Post-hoc structural screen
Fold1 hits: 1
Fold2 hits: 274
Contact n structure: 0 / 209
First Fold1: 249.0
First Fold2: 3.0
True-contact gates: F1 pass · F2 pass

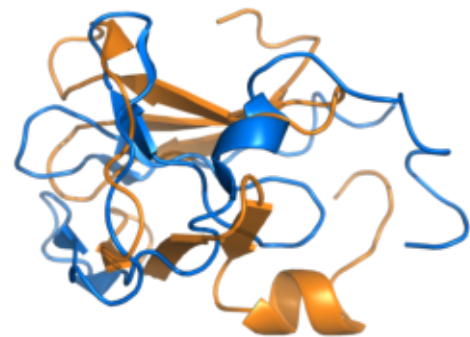
No-contact baseline hit: Fold2

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.182 / 0.143
Fold1 TM-score F1/F2: 0.677 / 0.604
Fold1 GDT: 0.449 vs 0.209 (need 0.350)
Fold1 region: 0.359 vs 0.152
Fold1 CA RMSD: 4.71 Å
Fold1 mean pLDDT: 70.5
Fold2 contact recall F1/F2: 0.091 / 0.714
Fold2 TM-score F1/F2: 0.593 / 0.756
Fold2 GDT: 0.659 vs 0.216 (need 0.417)
Fold2 region: 0.489 vs 0.087
Fold2 CA RMSD: 3.18 Å
Fold2 mean pLDDT: 86.0

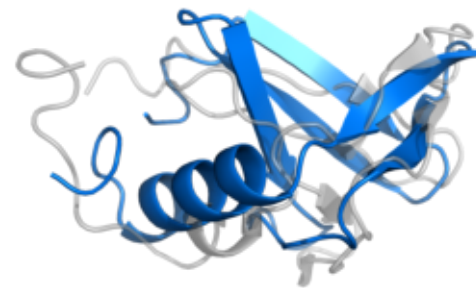
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

11. 2lela_2k0qa — experimental and Helico structures

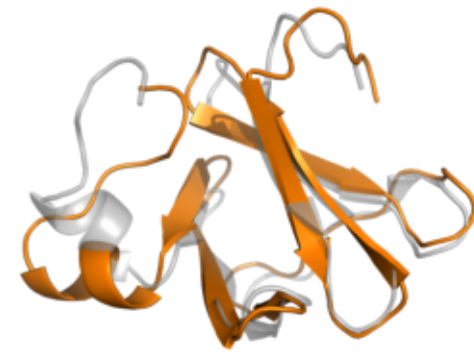
Experimental folds (aligned)



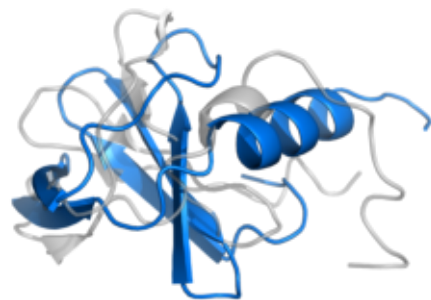
Primary Hit · rollout 249 vs Fold1



Primary Hit · rollout 761 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.355 / 0.206
region: 0.337 / 0.130

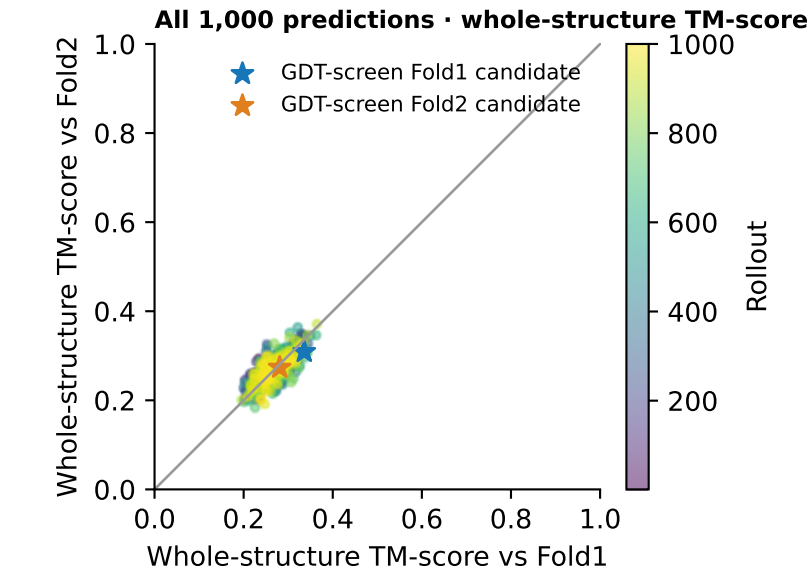
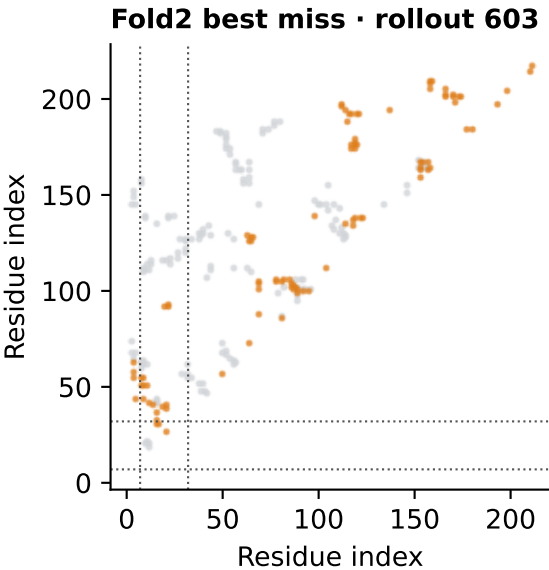
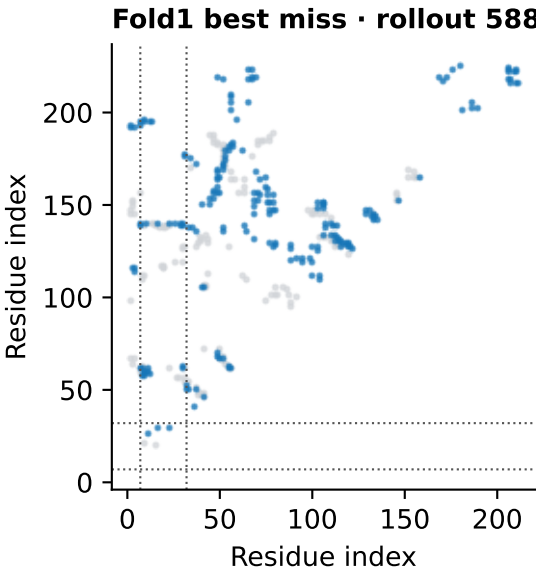
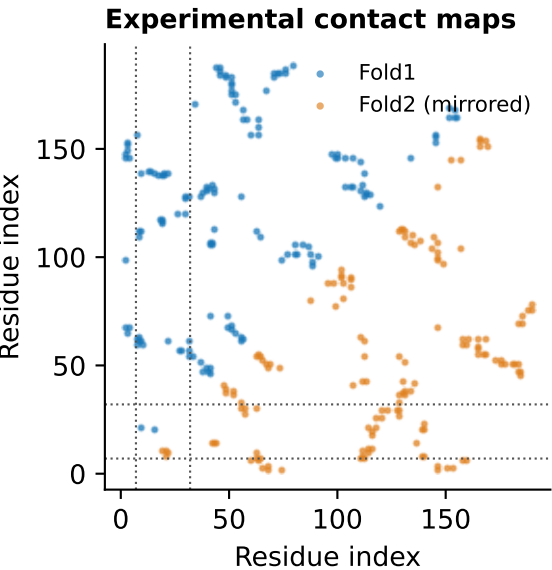
True Fold2-contact control
GDT Fold1/Fold2: 0.108 / 0.463
region: 0.022 / 0.337

No-contact baseline
GDT Fold1/Fold2: 0.155 / 0.520
region: 0.022 / 0.337

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

12. 2pbkb_3njqa — contact maps and structural scores

L=228 · Fold1=2pbkB · Fold2=3njqA · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.847



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

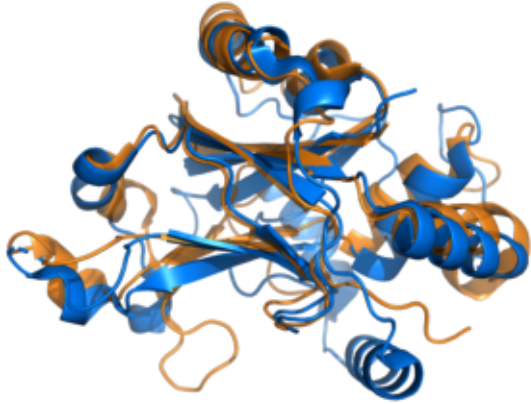
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.000 / 0.000
Fold1 TM-score F1/F2: 0.336 / 0.309
Fold1 GDT: 0.119 vs 0.094 (need 0.804)
Fold1 region: 0.360 vs 0.180
Fold1 CA RMSD: 18.11 Å
Fold1 mean pLDDT: 56.8
Fold2 contact recall F1/F2: 0.000 / 0.000
Fold2 TM-score F1/F2: 0.281 / 0.274
Fold2 GDT: 0.066 vs 0.042 (need 0.818)
Fold2 region: 0.330 vs 0.150
Fold2 CA RMSD: 19.74 Å
Fold2 mean pLDDT: 36.2

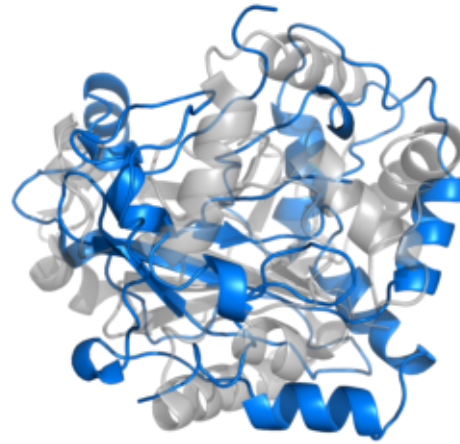
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

12. 2pbkb_3njqa — experimental and Helico structures

Experimental folds (aligned)



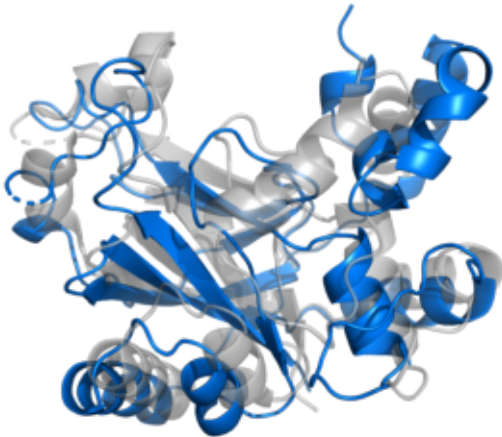
Best Miss · rollout 588 vs Fold1



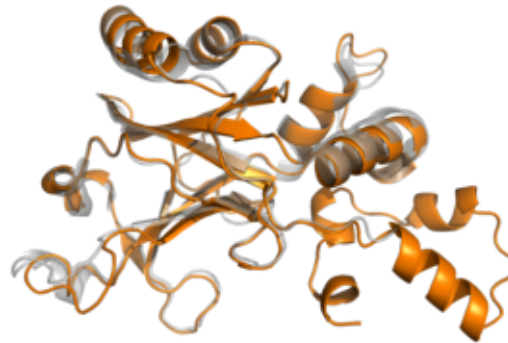
Best Miss · rollout 603 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.893 / 0.607
region: 0.860 / 0.390

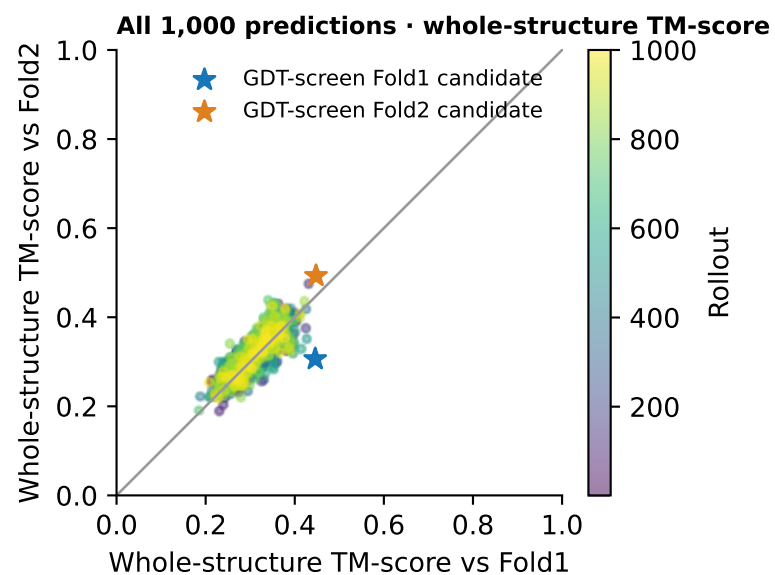
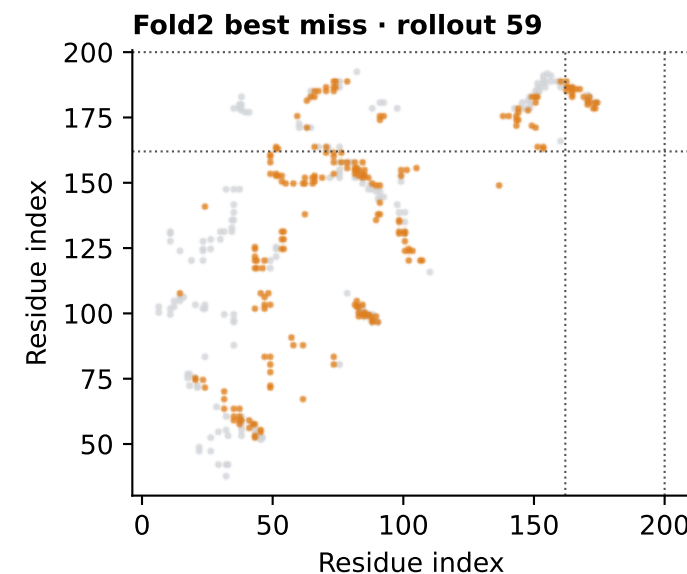
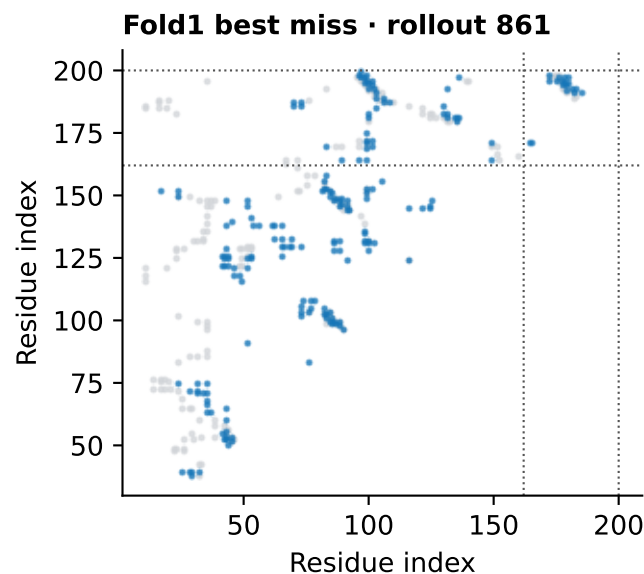
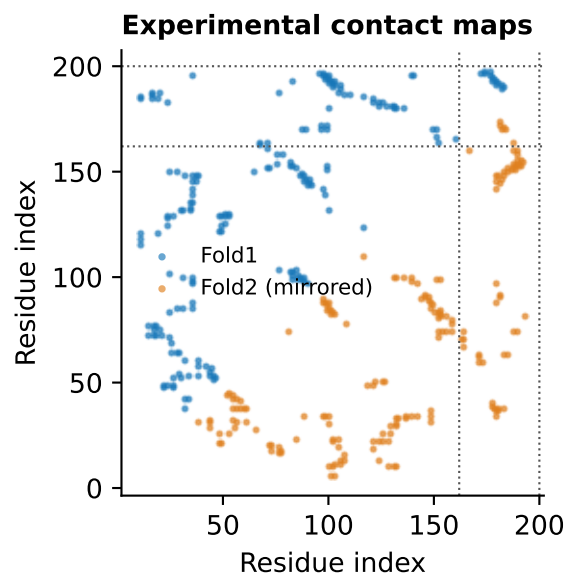
True Fold2-contact control
GDT Fold1/Fold2: 0.608 / 0.909
region: 0.380 / 0.990

No-contact baseline
GDT Fold1/Fold2: 0.046 / 0.045
region: 0.060 / 0.040

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

13. 2vfxl_3gmhl — contact maps and structural scores

L=203 · Fold1=2vfxL · Fold2=3gmhL · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.697



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

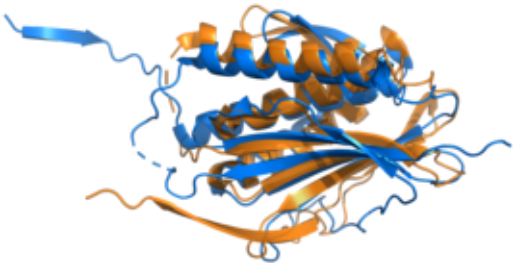
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.086 / 0.000
Fold1 TM-score F1/F2: 0.446 / 0.306
Fold1 GDT: 0.174 vs 0.098 (need 0.779)
Fold1 region: 0.270 vs 0.000
Fold1 CA RMSD: 13.60 Å
Fold1 mean pLDDT: 67.5
Fold2 contact recall F1/F2: 0.000 / 0.151
Fold2 TM-score F1/F2: 0.448 / 0.493
Fold2 GDT: 0.167 vs 0.087 (need 0.714)
Fold2 region: 0.204 vs 0.039
Fold2 CA RMSD: 12.61 Å
Fold2 mean pLDDT: 47.5

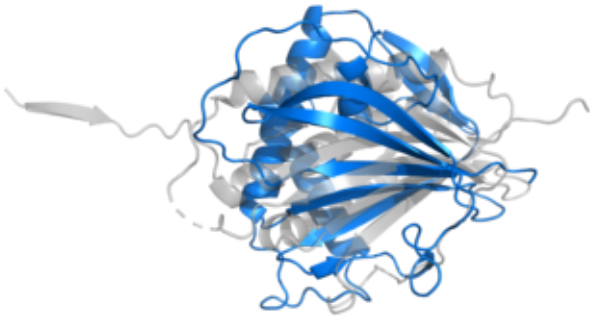
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

13. 2vfxl_3gmhl — experimental and Helico structures

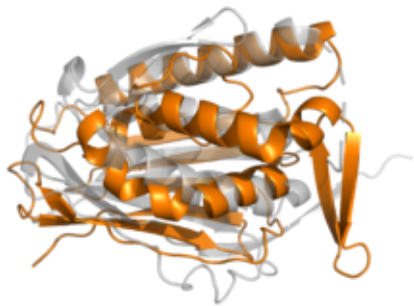
Experimental folds (aligned)



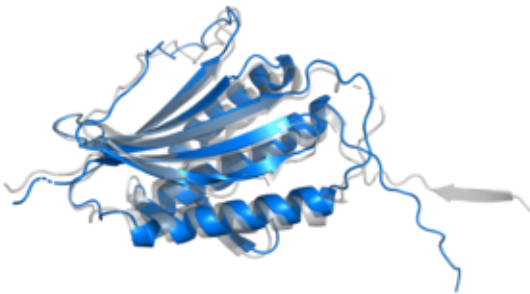
Best Miss · rollout 861 vs Fold1



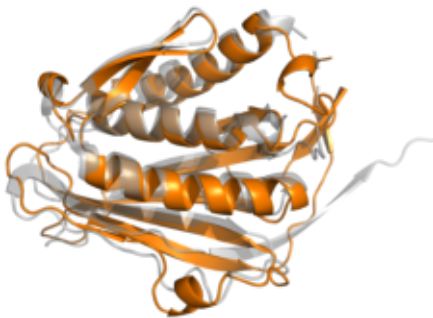
Best Miss · rollout 59 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.865 / 0.389
region: 0.829 / 0.053

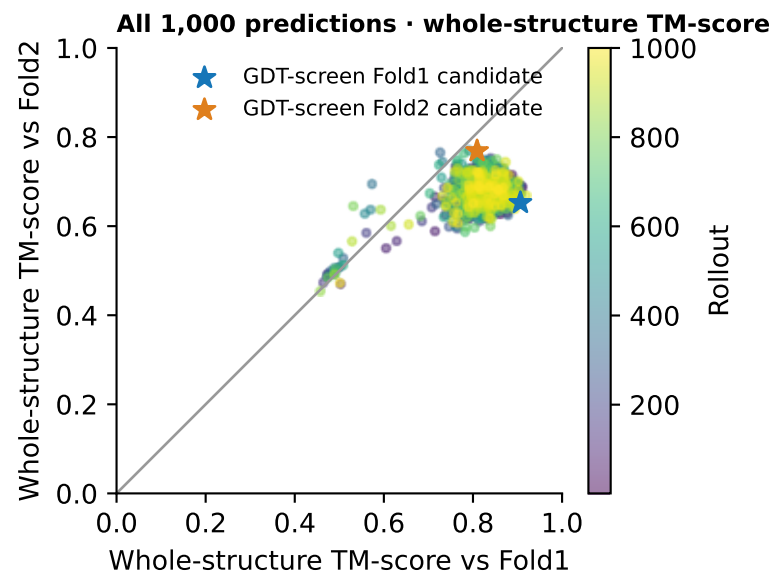
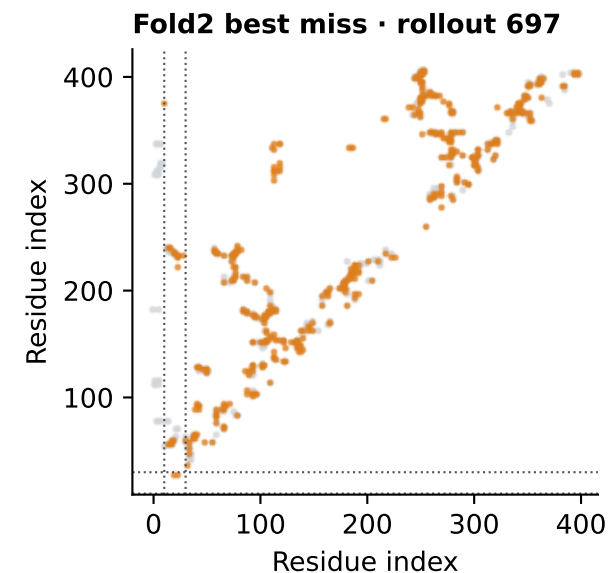
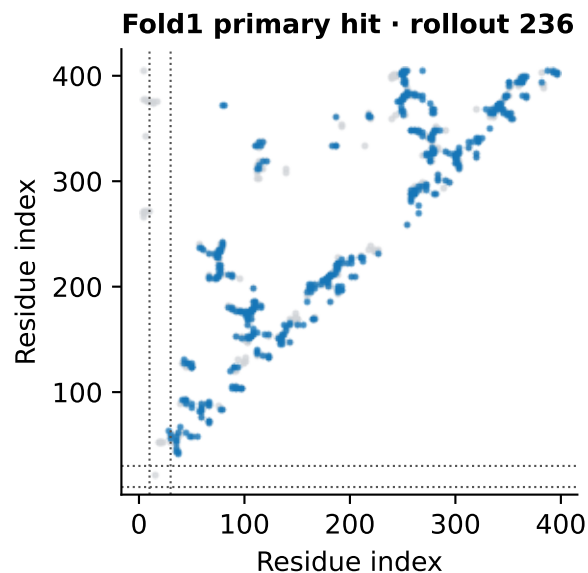
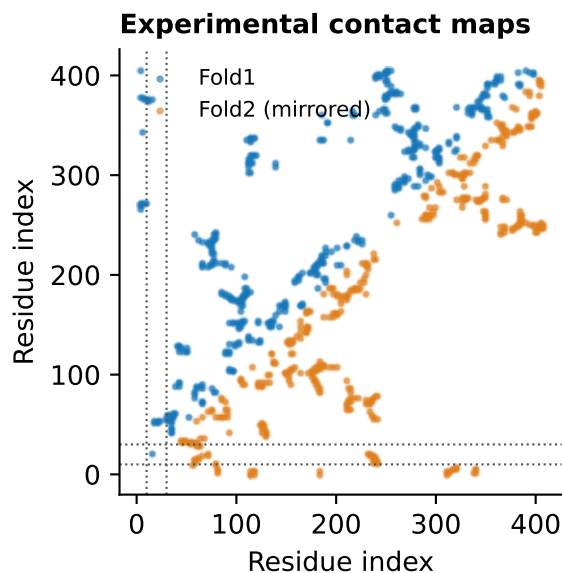
True Fold2-contact control
GDT Fold1/Fold2: 0.372 / 0.793
region: 0.046 / 0.638

No-contact baseline
GDT Fold1/Fold2: 0.073 / 0.087
region: 0.026 / 0.118

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

14. 3g0ha_3ewsb — contact maps and structural scores

L=418 · Fold1=3g0hA · Fold2=3ewsB · 1,000 rollouts · exact sequence · state=1 modes · reference TM-score=0.635



Post-hoc structural screen
Fold1 hits: 20
Fold2 hits: 0
Contact n structure: 1 / 0
First Fold1: 22.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

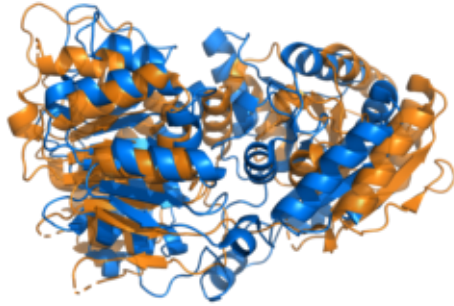
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.091 / 0.000
Fold1 TM-score F1/F2: 0.906 / 0.652
Fold1 GDT: 0.680 vs 0.291 (need 0.633)
Fold1 region: 0.283 vs 0.017
Fold1 CA RMSD: 2.72 Å
Fold1 mean pLDDT: 80.3
Fold2 contact recall F1/F2: 0.000 / 0.333
Fold2 TM-score F1/F2: 0.809 / 0.769
Fold2 GDT: 0.433 vs 0.451 (need 0.604)
Fold2 region: 0.467 vs 0.000
Fold2 CA RMSD: 6.02 Å
Fold2 mean pLDDT: 83.1

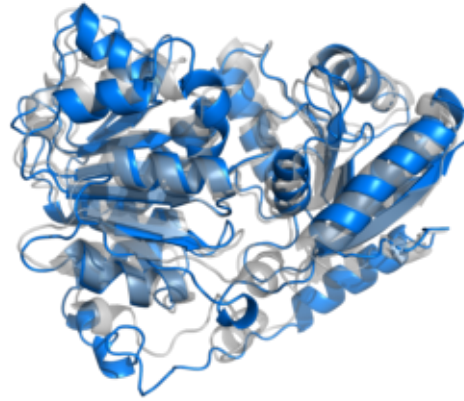
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

14. 3g0ha_3ewsb — experimental and Helico structures

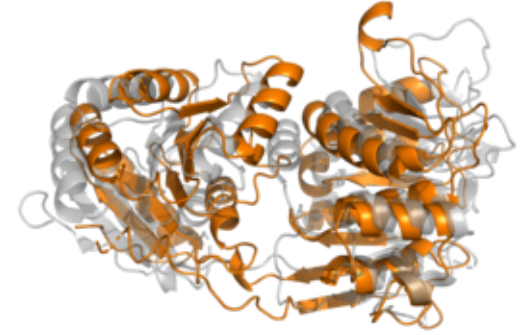
Experimental folds (aligned)



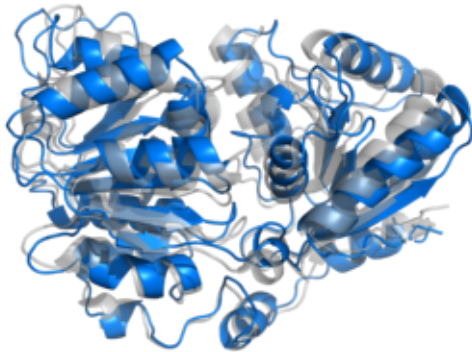
Primary Hit · rollout 236 vs Fold1



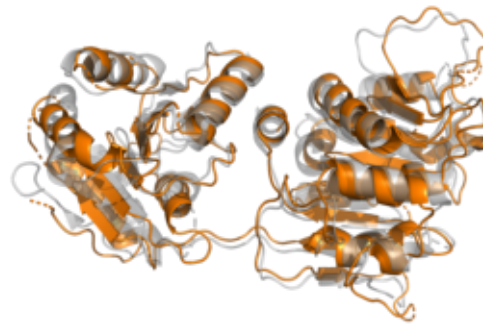
Best Miss · rollout 697 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.703 / 0.293
region: 0.817 / 0.000

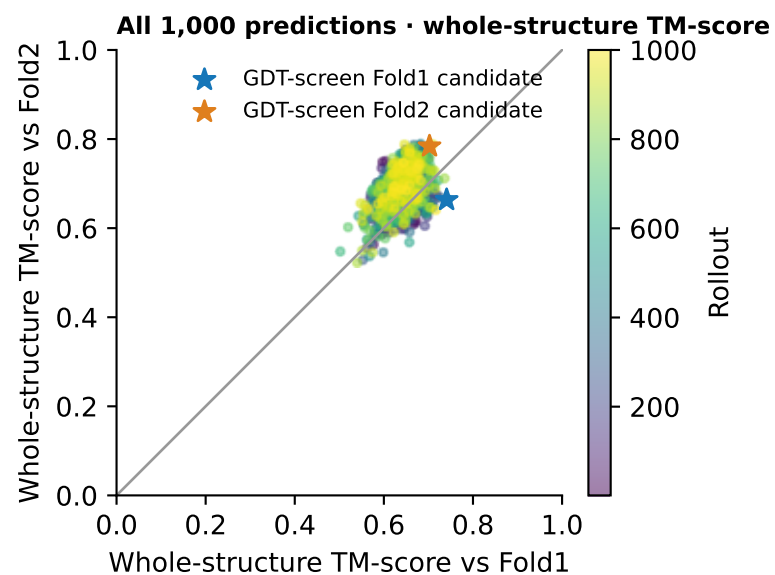
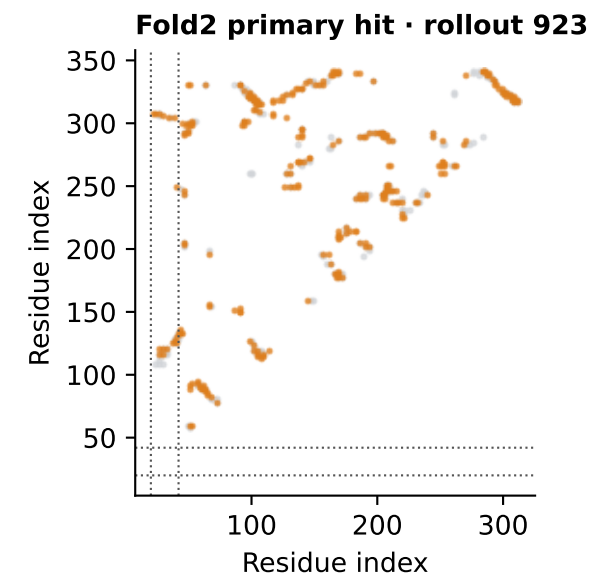
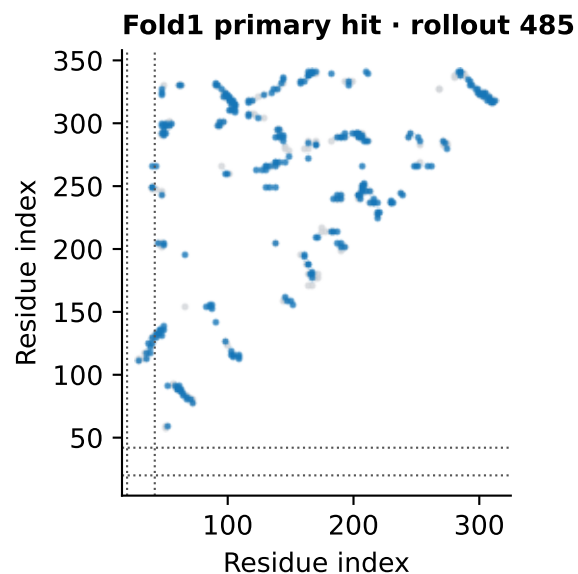
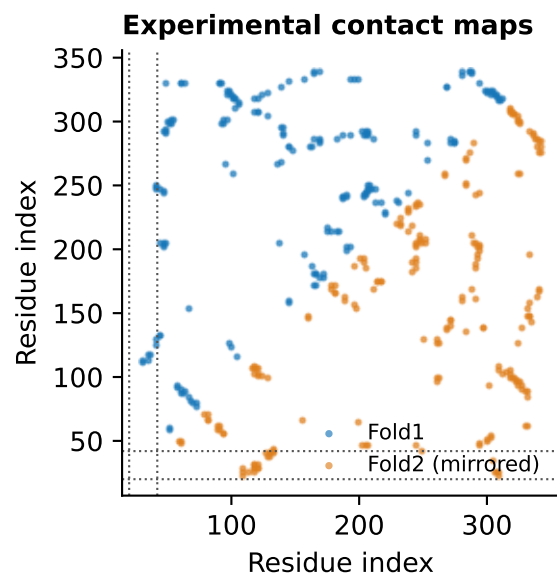
True Fold2-contact control
GDT Fold1/Fold2: 0.199 / 0.672
region: 0.000 / 0.600

No-contact baseline
GDT Fold1/Fold2: 0.008 / 0.002
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

15. 3j7wb_3j7vg — contact maps and structural scores

L=343 · Fold1=3j7wb · Fold2=3j7vg · 1,000 rollouts · exact sequence · state=2 modes · reference TM-score=0.690



Post-hoc structural screen
Fold1 hits: 2
Fold2 hits: 94
Contact n structure: 1 / 82
First Fold1: 144.0
First Fold2: 2.0
True-contact gates: F1 pass · F2 pass

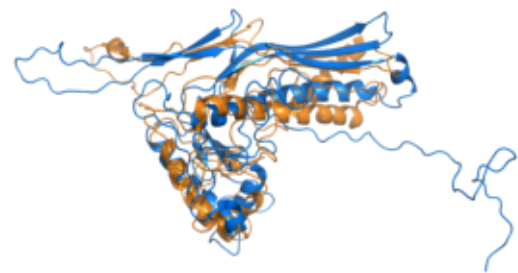
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.400 / 0.000
Fold1 TM-score F1/F2: 0.741 / 0.663
Fold1 GDT: 0.420 vs 0.283 (need 0.386)
Fold1 region: 0.238 vs 0.036
Fold1 CA RMSD: 5.69 Å
Fold1 mean pLDDT: 73.0
Fold2 contact recall F1/F2: 0.100 / 0.286
Fold2 TM-score F1/F2: 0.702 / 0.784
Fold2 GDT: 0.517 vs 0.317 (need 0.434)
Fold2 region: 0.798 vs 0.036
Fold2 CA RMSD: 5.77 Å
Fold2 mean pLDDT: 79.8

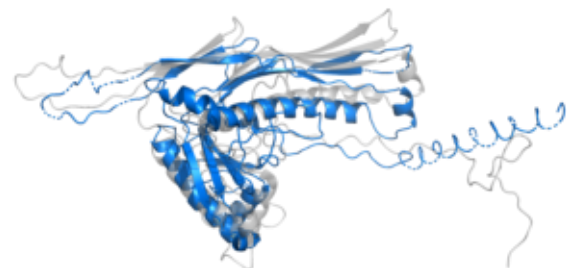
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

15. 3j7wb_3j7vg — experimental and Helico structures

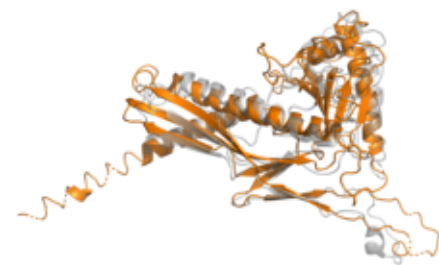
Experimental folds (aligned)



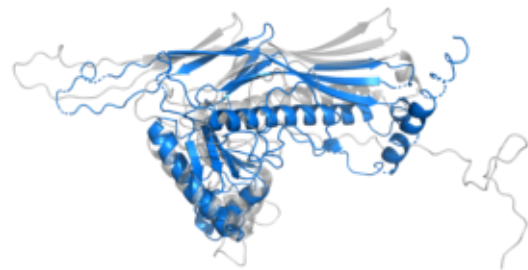
Primary Hit · rollout 485 vs Fold1



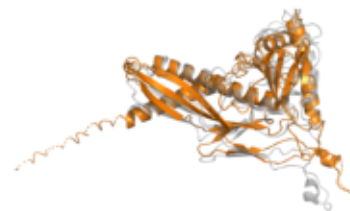
Primary Hit · rollout 923 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.429 / 0.403
region: 0.190 / 0.060

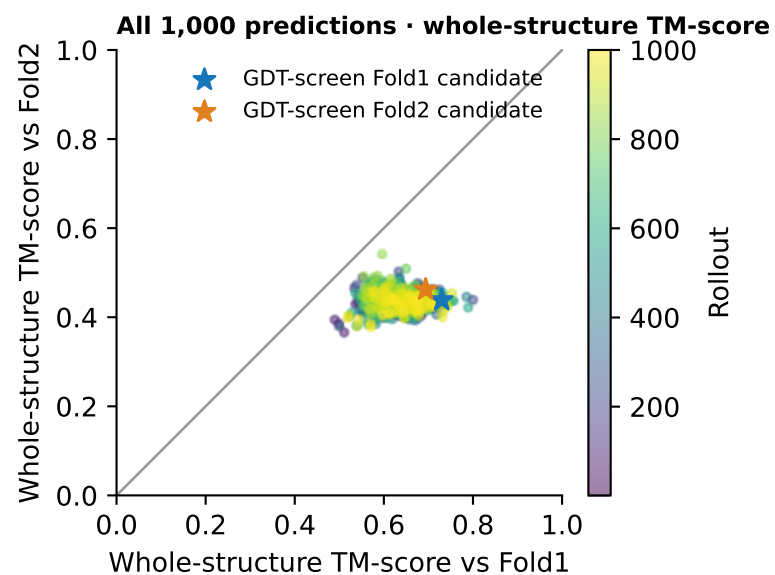
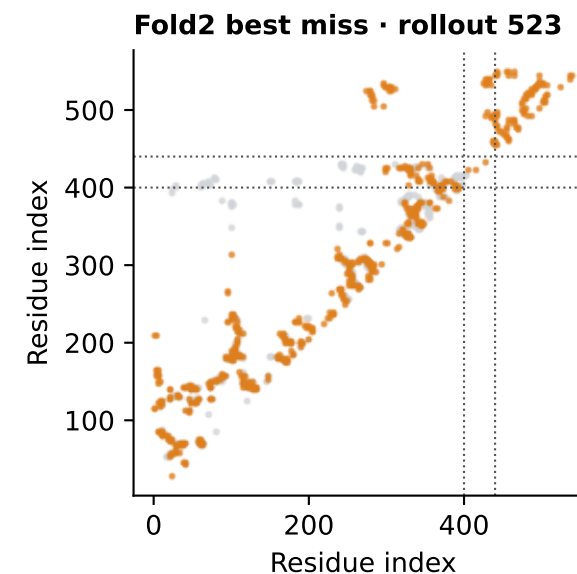
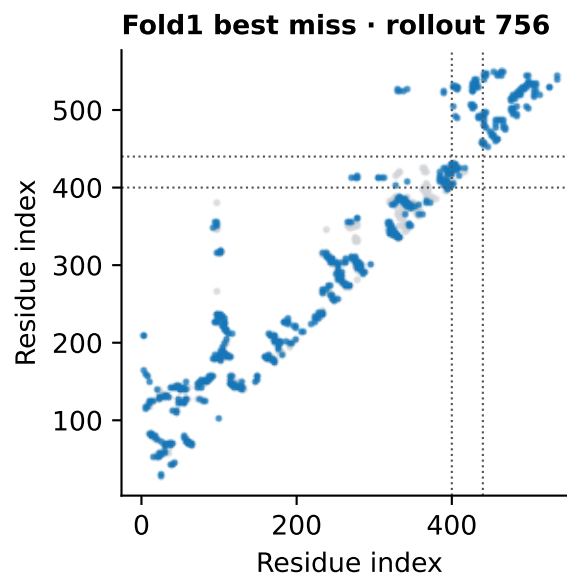
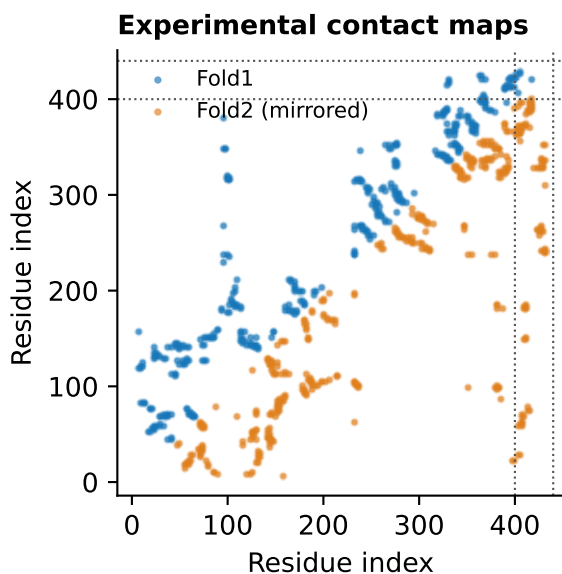
True Fold2-contact control
GDT Fold1/Fold2: 0.323 / 0.483
region: 0.036 / 0.607

No-contact baseline
GDT Fold1/Fold2: 0.013 / 0.022
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

16. 3meeA_4b3ob — contact maps and structural scores

L=552 · Fold1=3meeA · Fold2=4b3ob · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.443



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

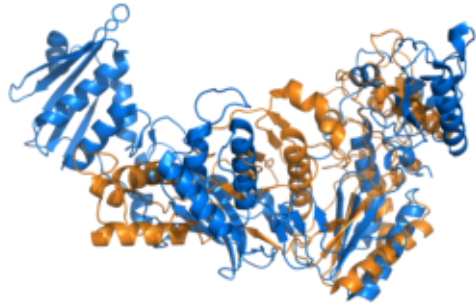
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.273 / 0.025
Fold1 TM-score F1/F2: 0.730 / 0.439
Fold1 GDT: 0.416 vs 0.101 (need 0.609)
Fold1 region: 0.117 vs 0.000
Fold1 CA RMSD: 8.50 Å
Fold1 mean pLDDT: 76.2
Fold2 contact recall F1/F2: 0.045 / 0.050
Fold2 TM-score F1/F2: 0.694 / 0.462
Fold2 GDT: 0.082 vs 0.308 (need 0.777)
Fold2 region: 0.133 vs 0.033
Fold2 CA RMSD: 19.22 Å
Fold2 mean pLDDT: 72.6

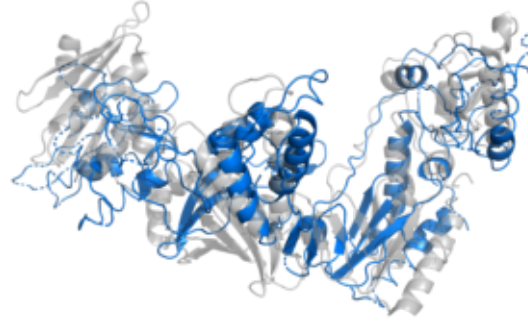
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

16. 3meea_4b3ob — experimental and Helico structures

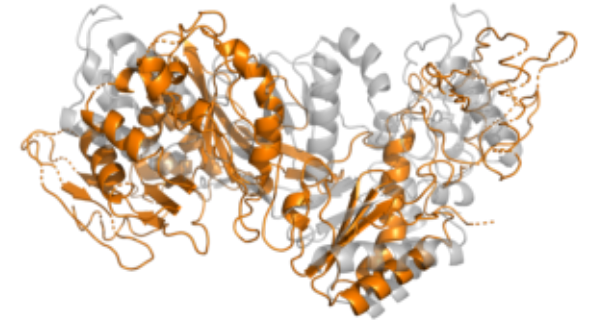
Experimental folds (aligned)



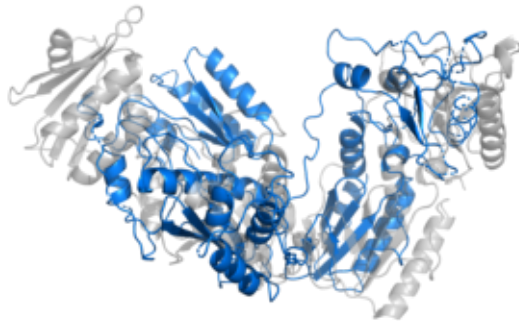
Best Miss · rollout 756 vs Fold1



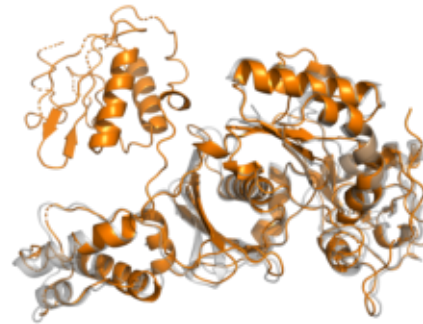
Best Miss · rollout 523 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.677 / 0.072
region: 0.433 / 0.000

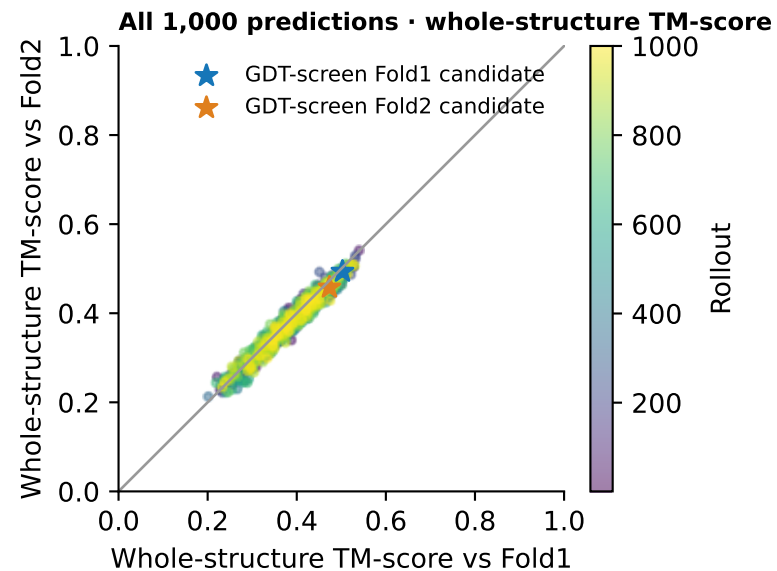
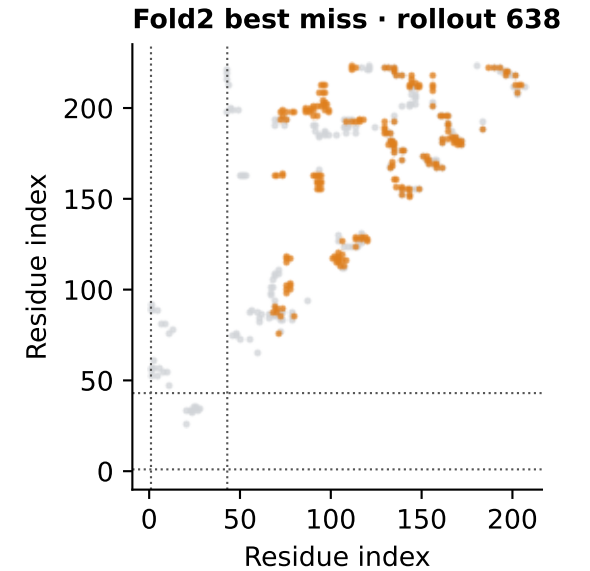
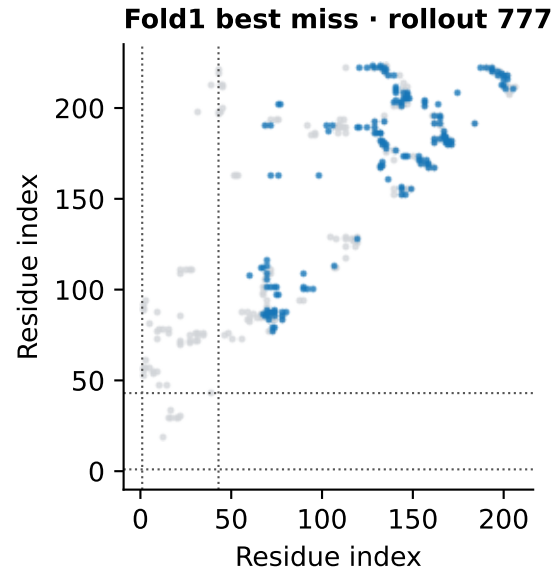
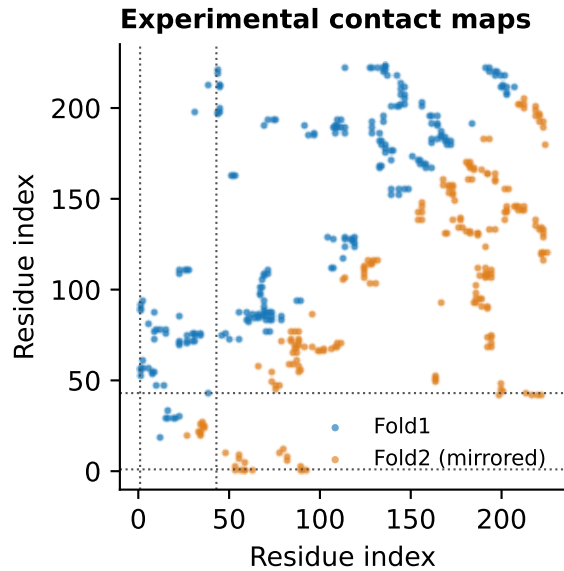
True Fold2-contact control
GDT Fold1/Fold2: 0.080 / 0.863
region: 0.000 / 0.808

No-contact baseline
GDT Fold1/Fold2: 0.018 / 0.004
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

17. 3tp2a_5lj3m — contact maps and structural scores

L=252 · Fold1=3tp2A · Fold2=5lj3M · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.817



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

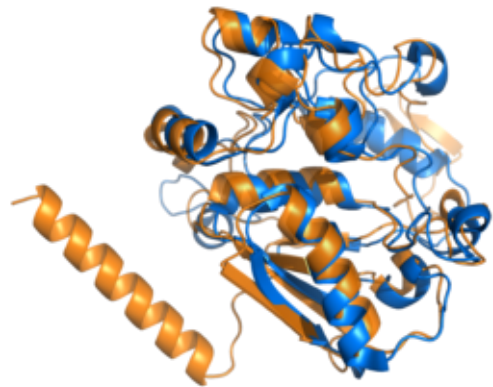
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.000 / 0.000
Fold1 TM-score F1/F2: 0.503 / 0.494
Fold1 GDT: 0.192 vs 0.157 (need 0.806)
Fold1 region: 0.161 vs 0.060
Fold1 CA RMSD: 14.38 Å
Fold1 mean pLDDT: 52.3
Fold2 contact recall F1/F2: 0.000 / 0.000
Fold2 TM-score F1/F2: 0.473 / 0.458
Fold2 GDT: 0.246 vs 0.206 (need 0.425)
Fold2 region: 0.060 vs 0.036
Fold2 CA RMSD: 16.80 Å
Fold2 mean pLDDT: 49.1

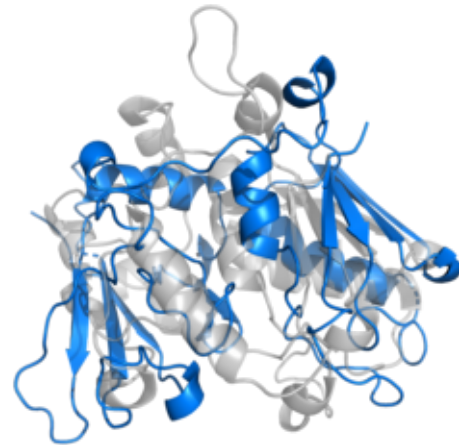
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

17. 3tp2a_5lj3m — experimental and Helico structures

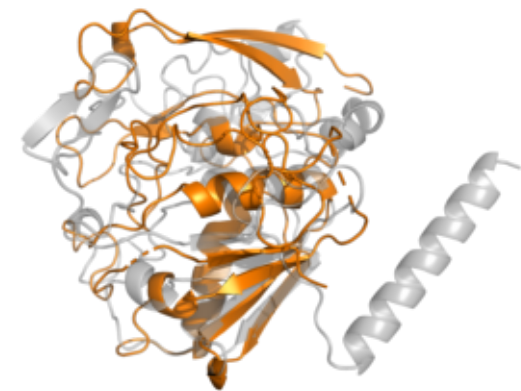
Experimental folds (aligned)



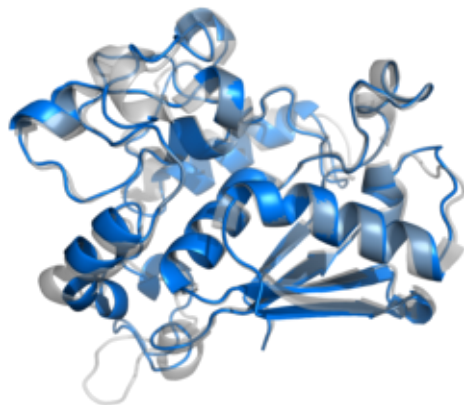
Best Miss · rollout 777 vs Fold1



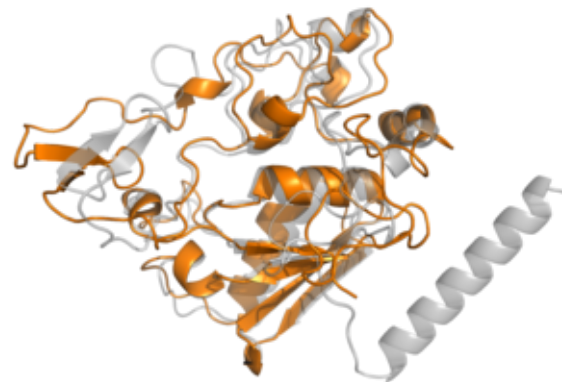
Best Miss · rollout 638 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.895 / 0.557
region: 0.810 / 0.161

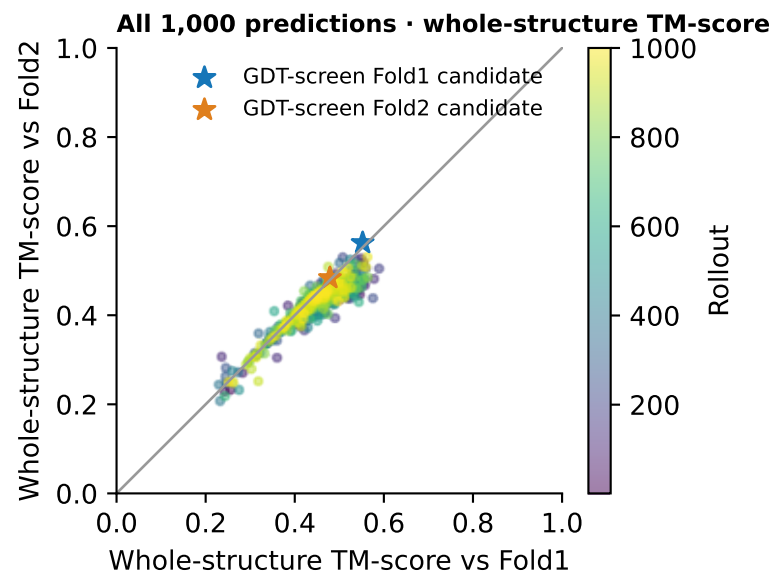
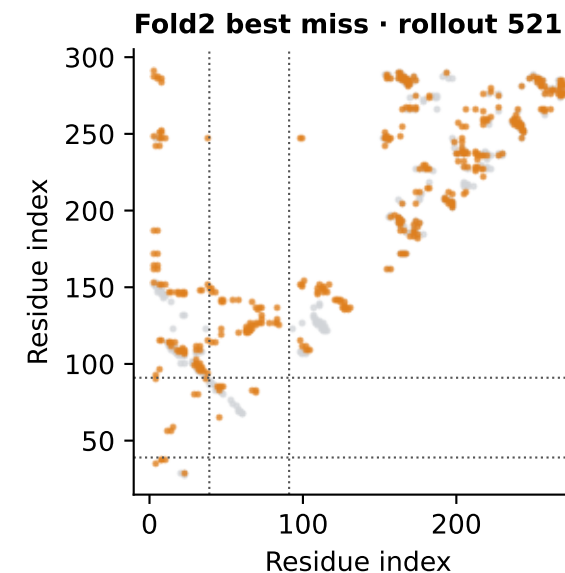
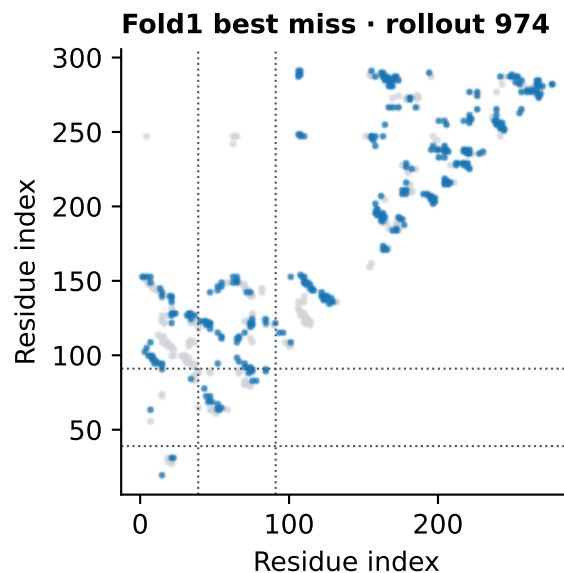
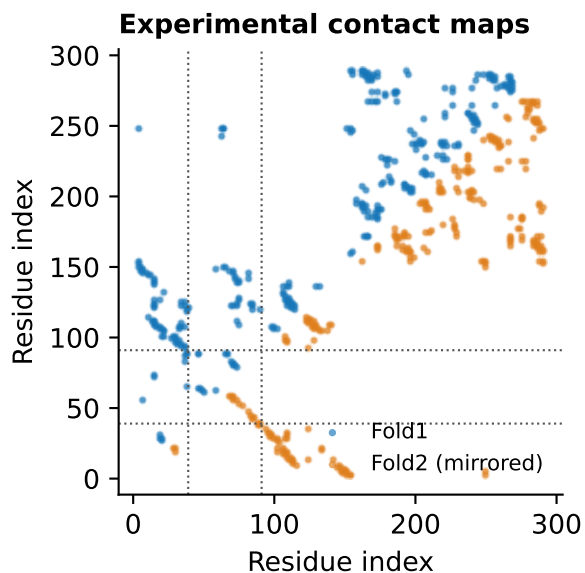
True Fold2-contact control
GDT Fold1/Fold2: 0.363 / 0.472
region: 0.149 / 0.179

No-contact baseline
GDT Fold1/Fold2: 0.011 / 0.007
region: 0.018 / 0.006

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

18. 3zrgb_5ec5p — contact maps and structural scores

L=293 · Fold1=3zrgb · Fold2=5ec5P · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.587



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

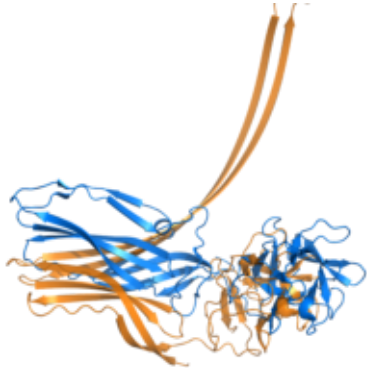
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.036 / 0.000
Fold1 TM-score F1/F2: 0.552 / 0.562
Fold1 GDT: 0.297 vs 0.011 (need 0.481)
Fold1 region: 0.000 vs 0.000
Fold1 CA RMSD: 17.03 Å
Fold1 mean pLDDT: 72.5
Fold2 contact recall F1/F2: 0.018 / 0.059
Fold2 TM-score F1/F2: 0.479 / 0.484
Fold2 GDT: 0.114 vs 0.072 (need 0.350)
Fold2 region: 0.026 vs 0.031
Fold2 CA RMSD: 23.69 Å
Fold2 mean pLDDT: 64.9

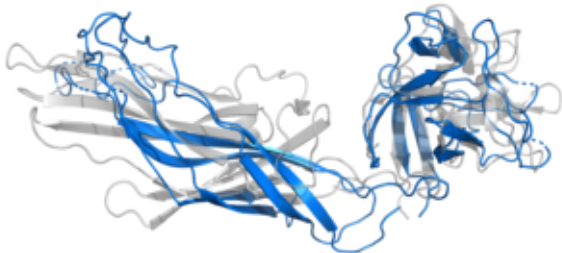
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

18. 3zxgb_5ec5p — experimental and Helico structures

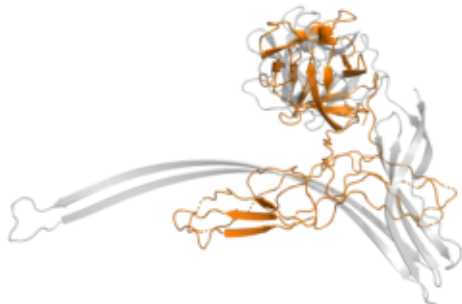
Experimental folds (aligned)



Best Miss · rollout 974 vs Fold1



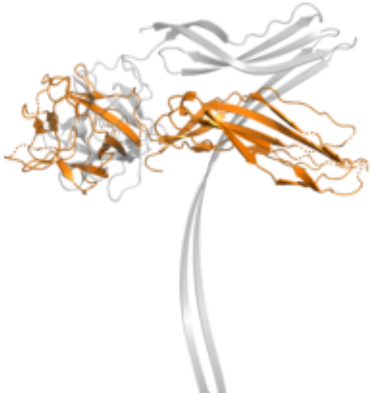
Best Miss · rollout 521 vs Fold2



True Fold1 contacts → Helico



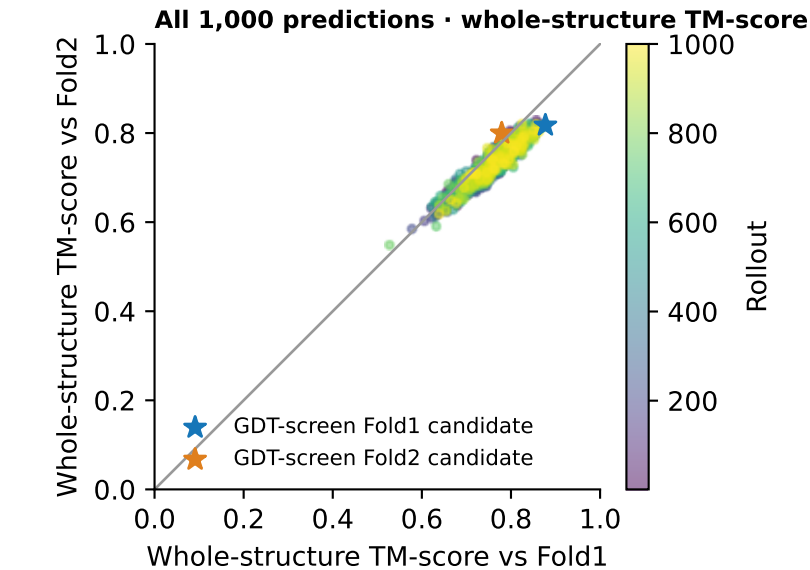
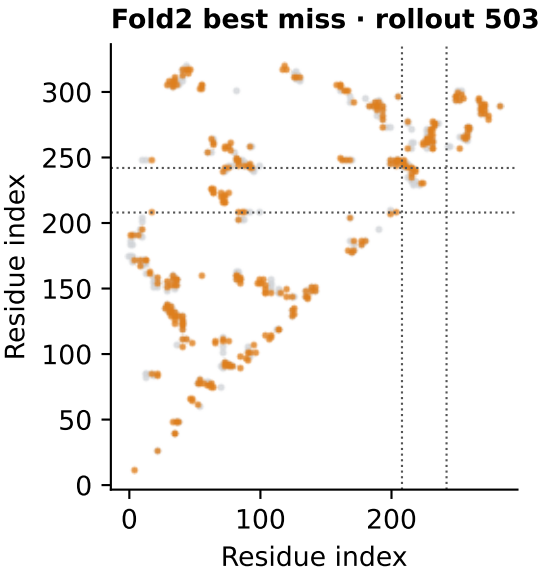
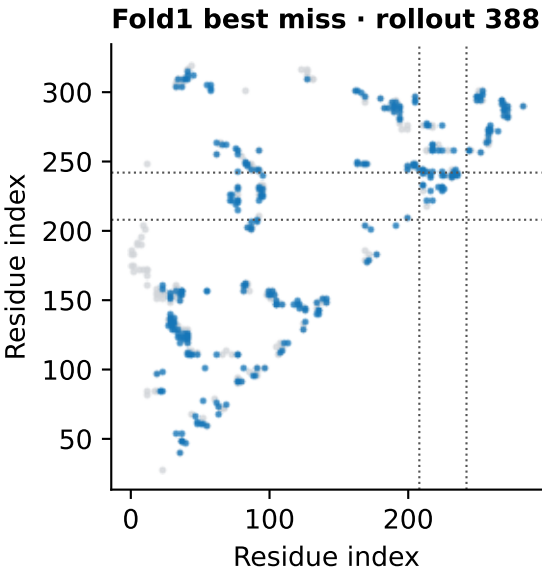
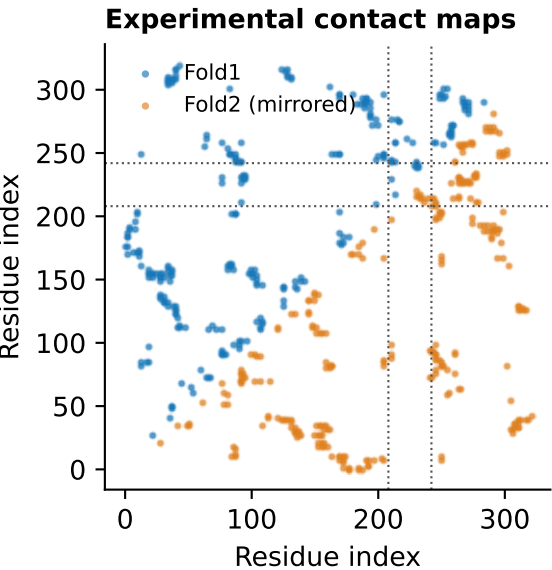
True Fold2 contacts → Helico



Cross-reference metrics	
True Fold1-contact control	
GDT Fold1/Fold2:	0.534 / 0.032
region:	0.454 / 0.031
True Fold2-contact control	
GDT Fold1/Fold2:	0.336 / 0.011
region:	0.051 / 0.000
No-contact baseline	
GDT Fold1/Fold2:	0.011 / 0.005
region:	0.020 / 0.000
Blue/orange: prediction or corresponding reference	
Gray: target experimental reference	
Cartoons are visual aids; tables use exact common-position scoring.	

19. 4aana_4aala — contact maps and structural scores

L=324 · Fold1=4aanA · Fold2=4aala · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.876



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

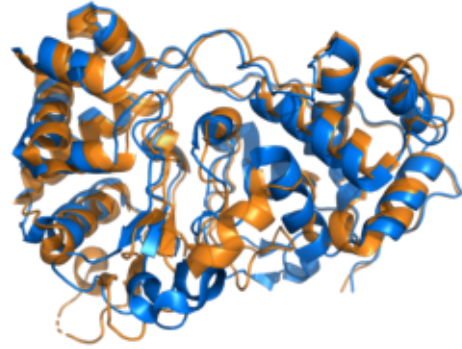
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.241 / 0.159
Fold1 TM-score F1/F2: 0.877 / 0.817
Fold1 GDT: 0.684 vs 0.524 (need 0.777)
Fold1 region: 0.325 vs 0.167
Fold1 CA RMSD: 3.54 Å
Fold1 mean pLDDT: 80.0
Fold2 contact recall F1/F2: 0.069 / 0.205
Fold2 TM-score F1/F2: 0.779 / 0.800
Fold2 GDT: 0.523 vs 0.518 (need 0.811)
Fold2 region: 0.283 vs 0.125
Fold2 CA RMSD: 5.24 Å
Fold2 mean pLDDT: 70.4

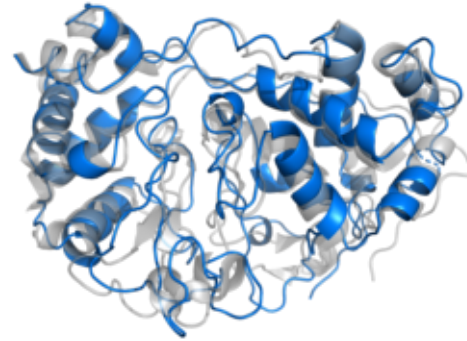
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

19. 4aana_4aala — experimental and Helico structures

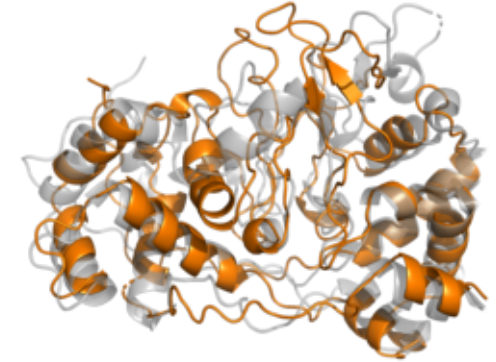
Experimental folds (aligned)



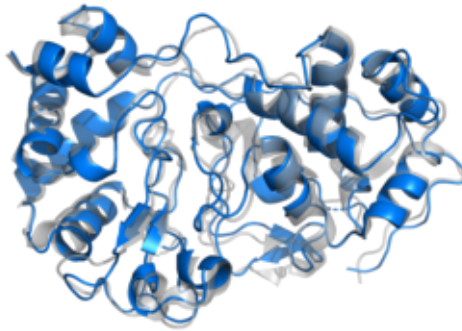
Best Miss · rollout 388 vs Fold1



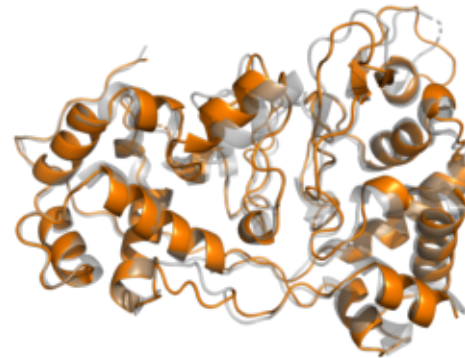
Best Miss · rollout 503 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.864 / 0.592
region: 0.783 / 0.142

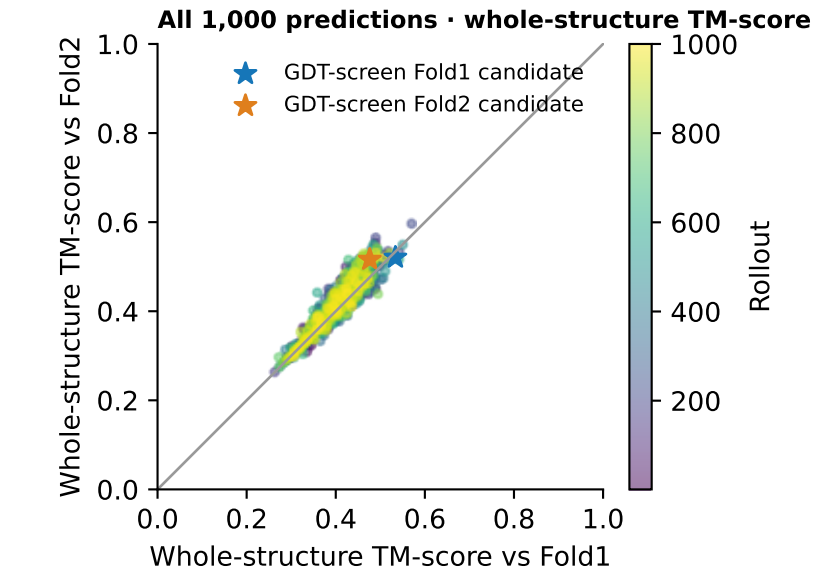
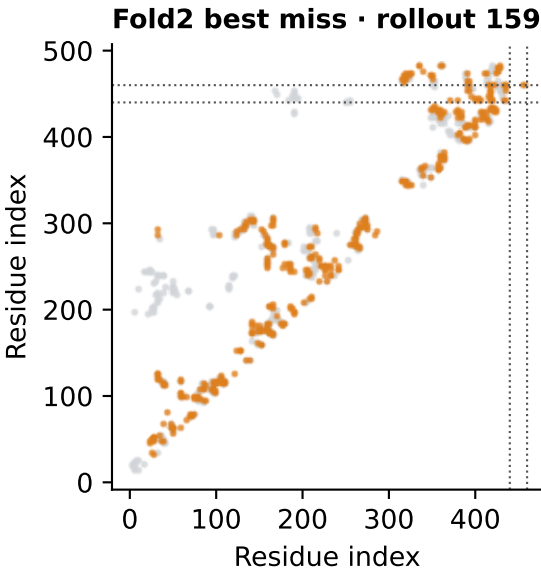
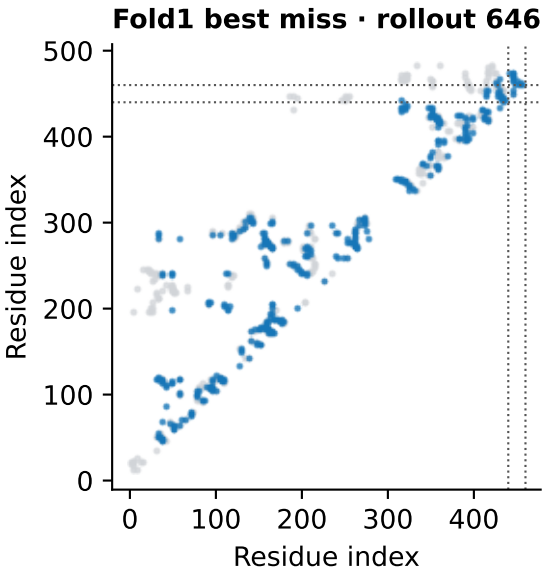
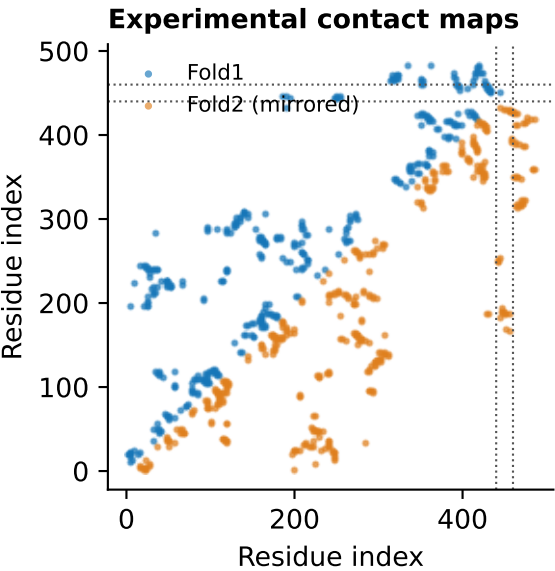
True Fold2-contact control
GDT Fold1/Fold2: 0.661 / 0.901
region: 0.142 / 0.758

No-contact baseline
GDT Fold1/Fold2: 0.015 / 0.027
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

20. 4o0pa_4o01d — contact maps and structural scores

L=489 · Fold1=4o0pA · Fold2=4o01D · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.855



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 FAIL · F2 FAIL

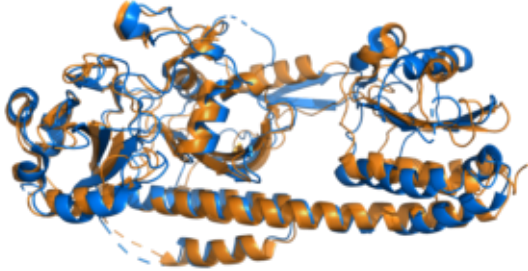
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.053 / 0.000
Fold1 TM-score F1/F2: 0.534 / 0.521
Fold1 GDT: 0.151 vs 0.104 (need 0.350)
Fold1 region: 0.000 vs 0.000
Fold1 CA RMSD: 17.23 Å
Fold1 mean pLDDT: 61.9
Fold2 contact recall F1/F2: 0.000 / 0.000
Fold2 TM-score F1/F2: 0.476 / 0.516
Fold2 GDT: 0.094 vs 0.070 (need 0.350)
Fold2 region: 0.158 vs 0.026
Fold2 CA RMSD: 17.41 Å
Fold2 mean pLDDT: 72.0

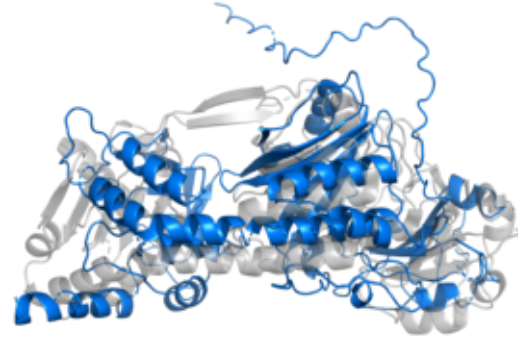
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

20. 4o0pa_4o01d — experimental and Helico structures

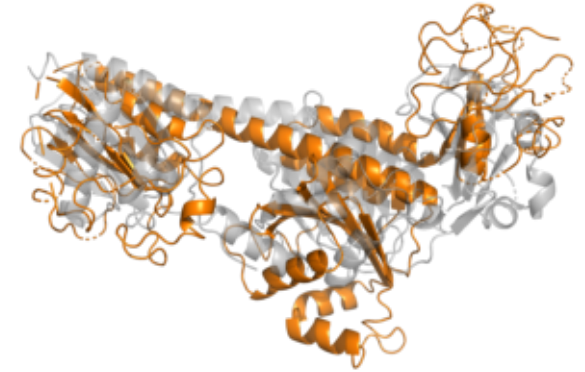
Experimental folds (aligned)



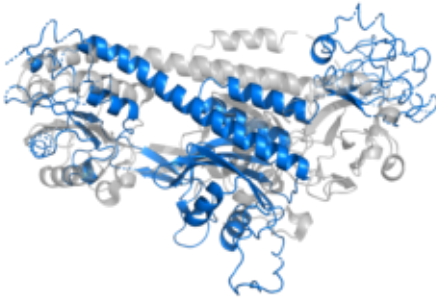
Best Miss · rollout 646 vs Fold1



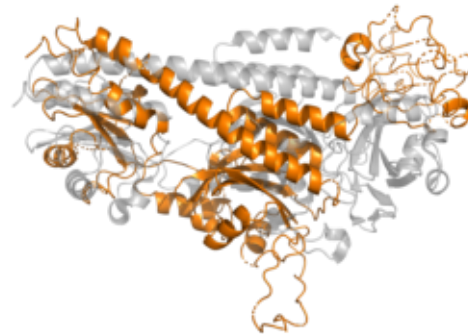
Best Miss · rollout 159 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.068 / 0.070
region: 0.250 / 0.079

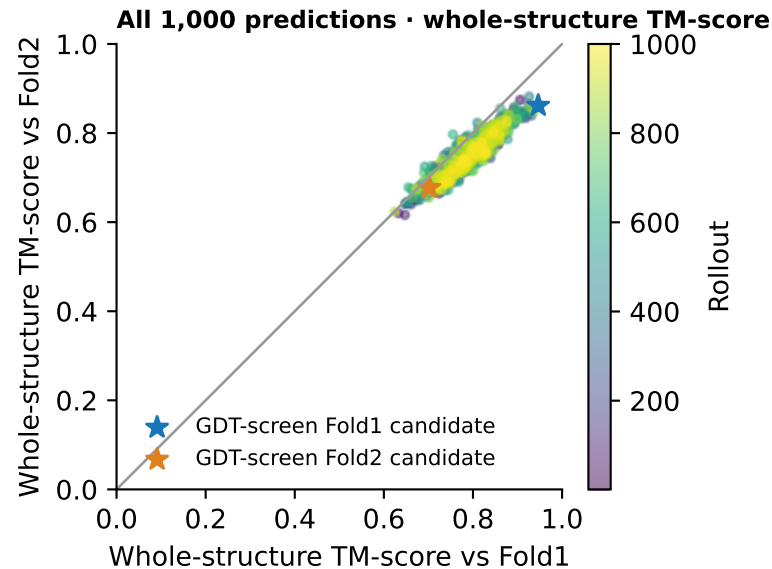
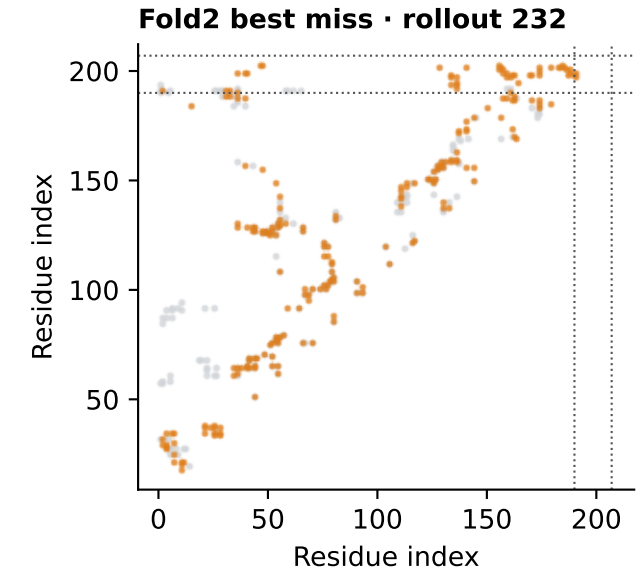
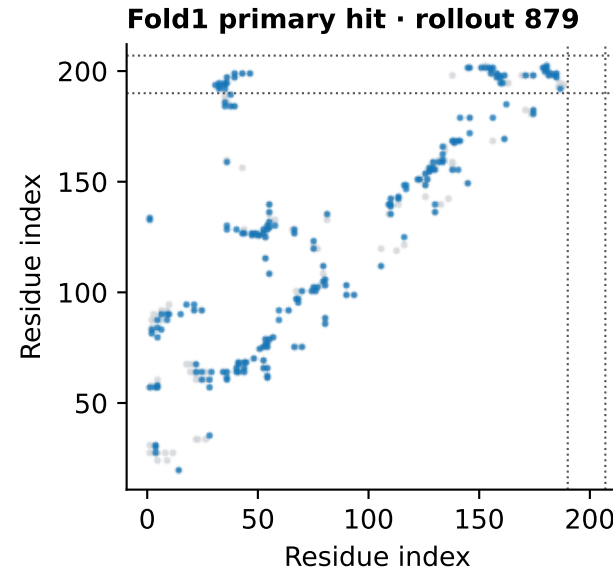
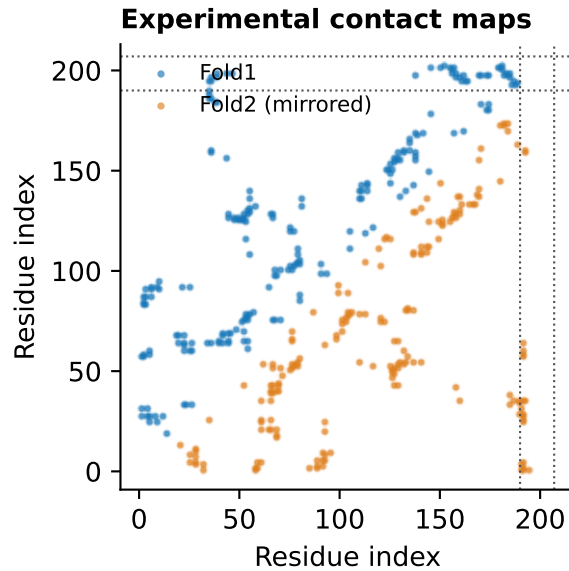
True Fold2-contact control
GDT Fold1/Fold2: 0.057 / 0.086
region: 0.118 / 0.132

No-contact baseline
GDT Fold1/Fold2: 0.010 / 0.009
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

21. 4pyia_4pyja — contact maps and structural scores

L=208 · Fold1=4pyiA · Fold2=4pyjA · 1,000 rollouts · exact sequence · state=1 modes · reference TM-score=0.882



Post-hoc structural screen
Fold1 hits: 5
Fold2 hits: 0
Contact n structure: 5 / 0
First Fold1: 286.0
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

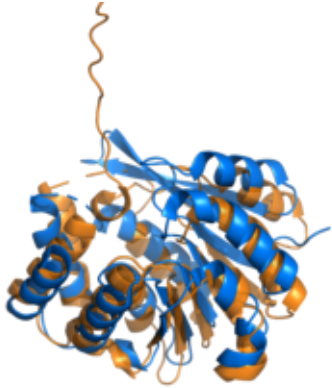
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.725 / 0.000
Fold1 TM-score F1/F2: 0.946 / 0.862
Fold1 GDT: 0.896 vs 0.400 (need 0.850)
Fold1 region: 0.809 vs 0.015
Fold1 CA RMSD: 1.70 Å
Fold1 mean pLDDT: 86.0
Fold2 contact recall F1/F2: 0.250 / 0.053
Fold2 TM-score F1/F2: 0.702 / 0.677
Fold2 GDT: 0.208 vs 0.219 (need 0.399)
Fold2 region: 0.103 vs 0.000
Fold2 CA RMSD: 13.72 Å
Fold2 mean pLDDT: 78.9

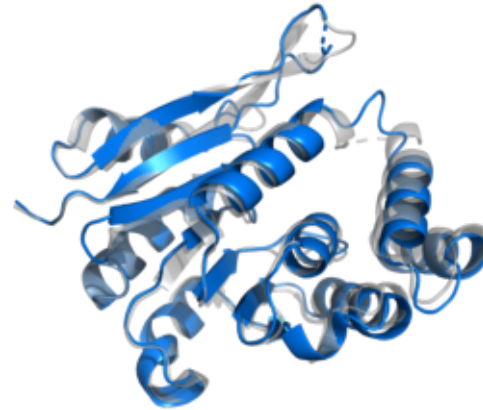
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

21. 4pyia_4pyja — experimental and Helico structures

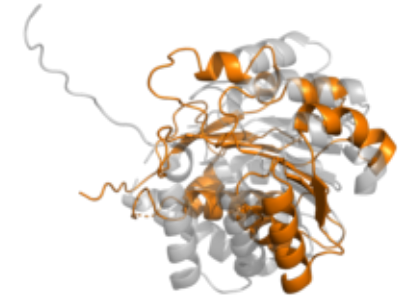
Experimental folds (aligned)



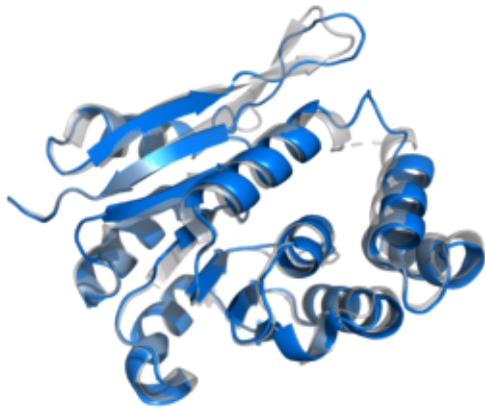
Primary Hit · rollout 879 vs Fold1



Best Miss · rollout 232 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.944 / 0.400
region: 0.897 / 0.000

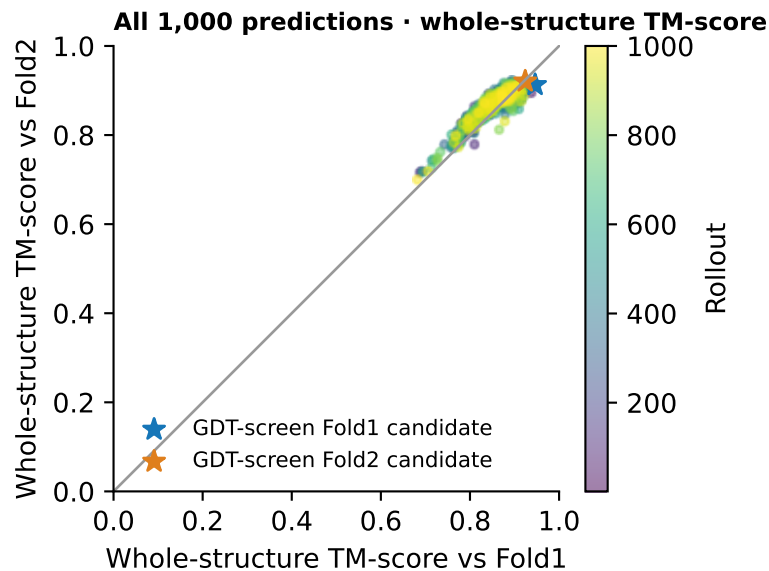
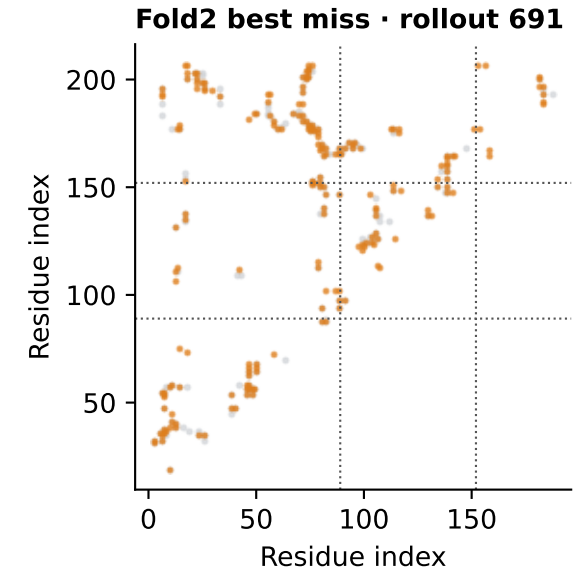
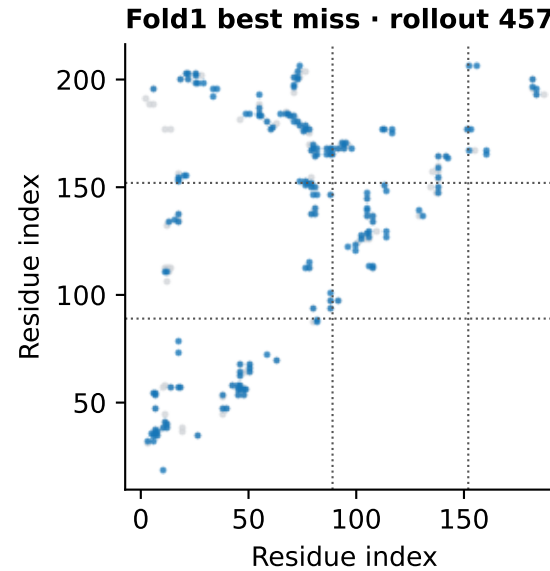
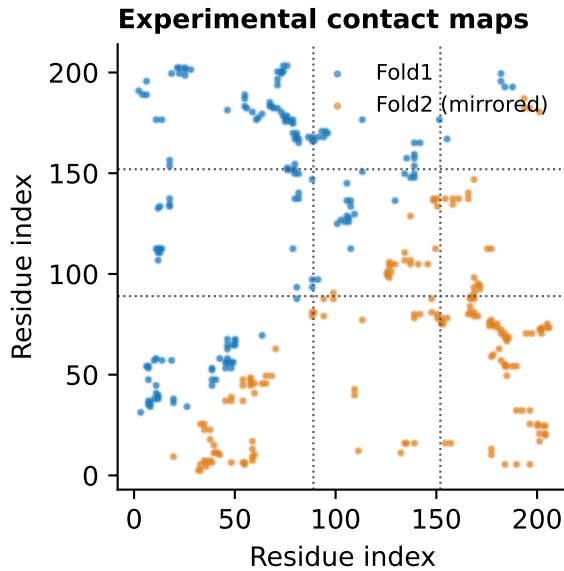
True Fold2-contact control
GDT Fold1/Fold2: 0.880 / 0.443
region: 0.647 / 0.118

No-contact baseline
GDT Fold1/Fold2: 0.165 / 0.116
region: 0.029 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

22. 4y0mj_4xwsd — contact maps and structural scores

L=209 · Fold1=4y0mj · Fold2=4xwsd · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.942



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 FAIL · F2 FAIL

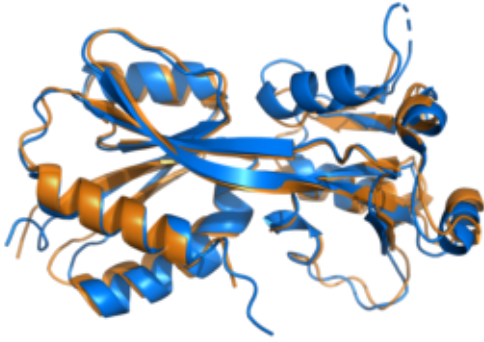
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.385 / 0.250
Fold1 TM-score F1/F2: 0.946 / 0.913
Fold1 GDT: 0.882 vs 0.799 (need 0.724)
Fold1 region: 0.825 vs 0.680
Fold1 CA RMSD: 1.55 Å
Fold1 mean pLDDT: 87.4
Fold2 contact recall F1/F2: 0.462 / 0.312
Fold2 TM-score F1/F2: 0.924 / 0.921
Fold2 GDT: 0.835 vs 0.836 (need 0.770)
Fold2 region: 0.765 vs 0.785
Fold2 CA RMSD: 1.79 Å
Fold2 mean pLDDT: 86.2

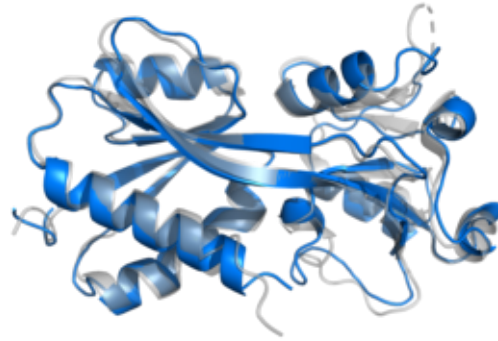
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

22. 4y0mj_4xwsd — experimental and Helico structures

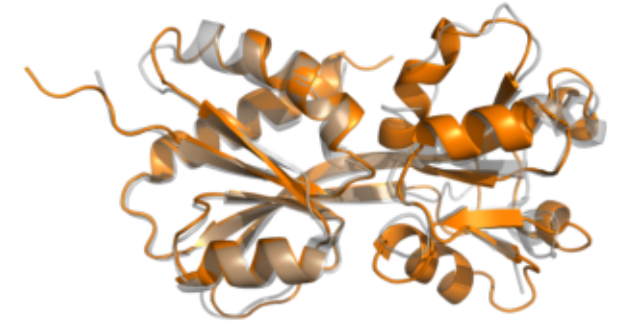
Experimental folds (aligned)



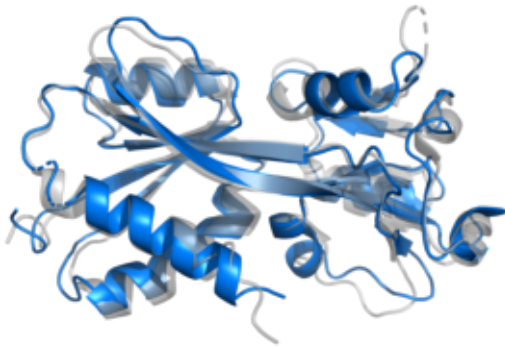
Best Miss · rollout 457 vs Fold1



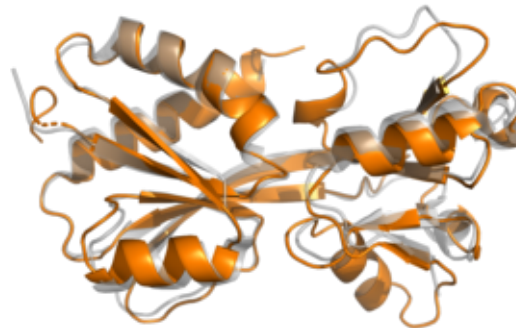
Best Miss · rollout 691 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.805 / 0.833
region: 0.770 / 0.725

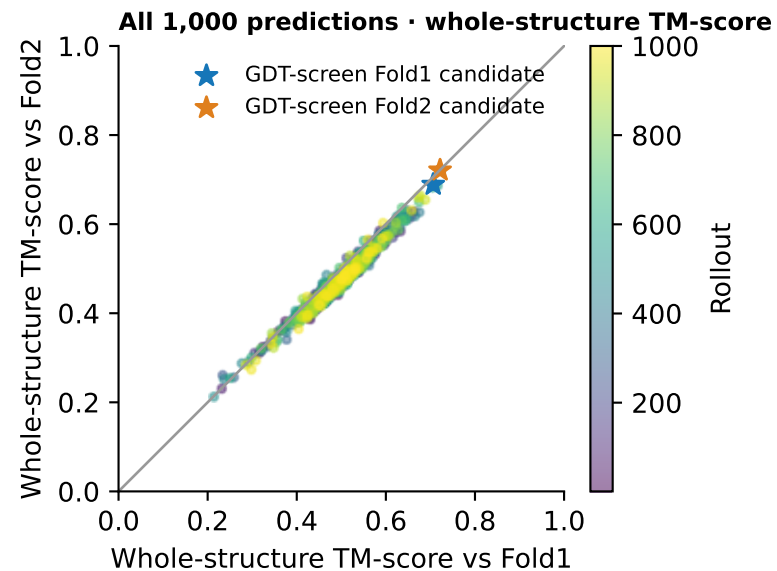
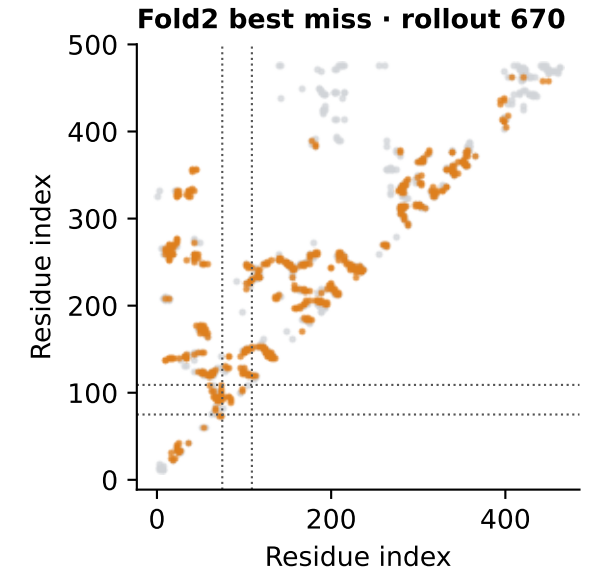
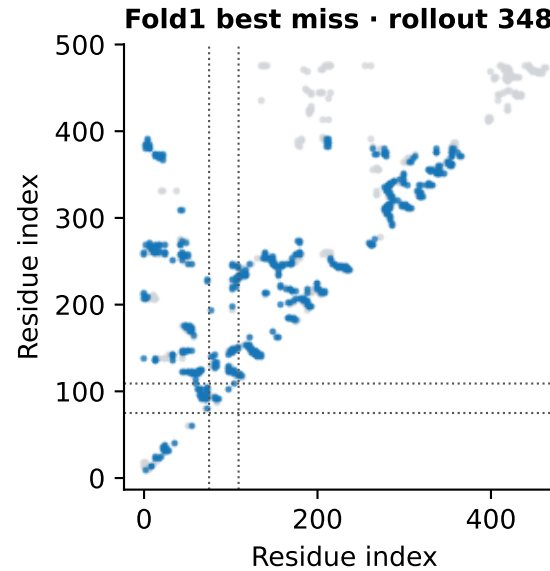
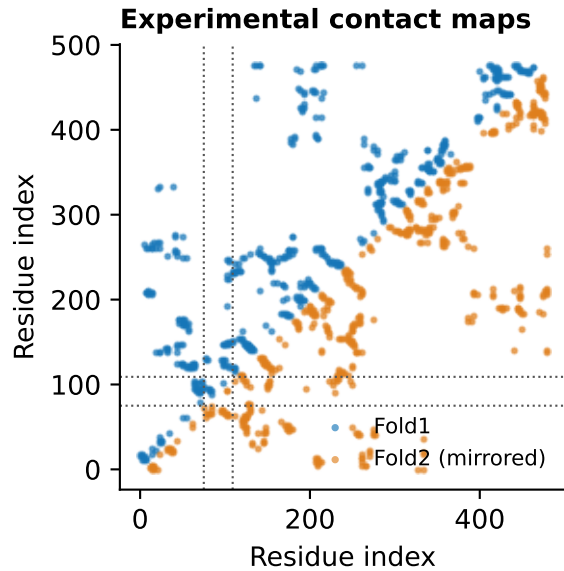
True Fold2-contact control
GDT Fold1/Fold2: 0.767 / 0.855
region: 0.710 / 0.765

No-contact baseline
GDT Fold1/Fold2: 0.052 / 0.047
region: 0.010 / 0.015

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

23. 4ydqb_4twaa — contact maps and structural scores

L=488 · Fold1=4ydqb · Fold2=4twaA · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.940



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

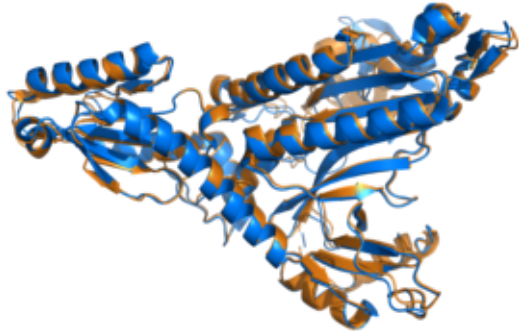
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.425 / 0.148
Fold1 TM-score F1/F2: 0.706 / 0.689
Fold1 GDT: 0.323 vs 0.313 (need 0.763)
Fold1 region: 0.279 vs 0.147
Fold1 CA RMSD: 9.24 Å
Fold1 mean pLDDT: 66.3
Fold2 contact recall F1/F2: 0.525 / 0.148
Fold2 TM-score F1/F2: 0.722 / 0.721
Fold2 GDT: 0.366 vs 0.361 (need 0.820)
Fold2 region: 0.199 vs 0.294
Fold2 CA RMSD: 9.83 Å
Fold2 mean pLDDT: 68.7

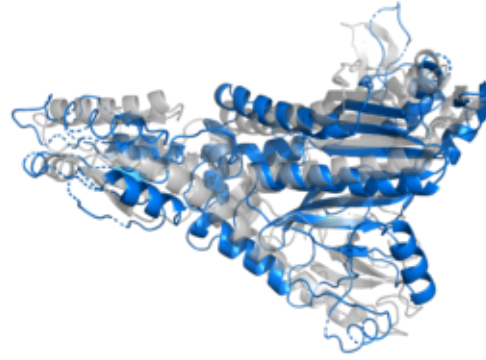
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

23. 4ydqb_4twaa — experimental and Helico structures

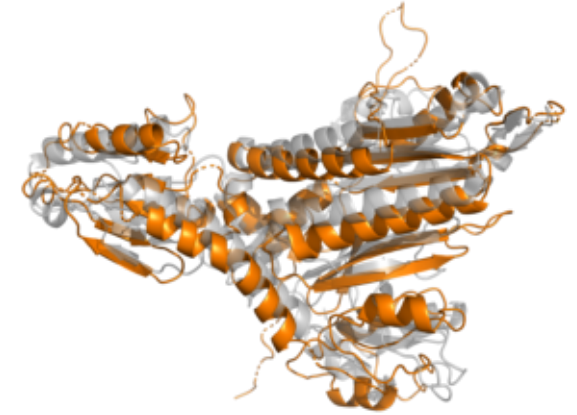
Experimental folds (aligned)



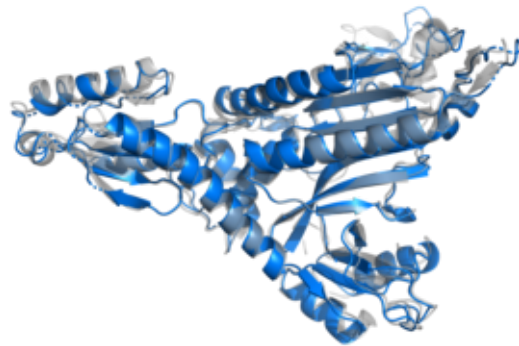
Best Miss · rollout 348 vs Fold1



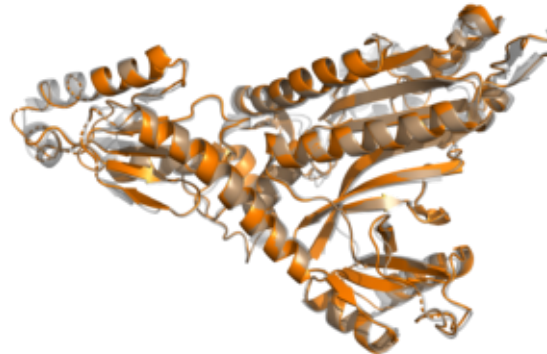
Best Miss · rollout 670 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.848 / 0.848
region: 0.537 / 0.265

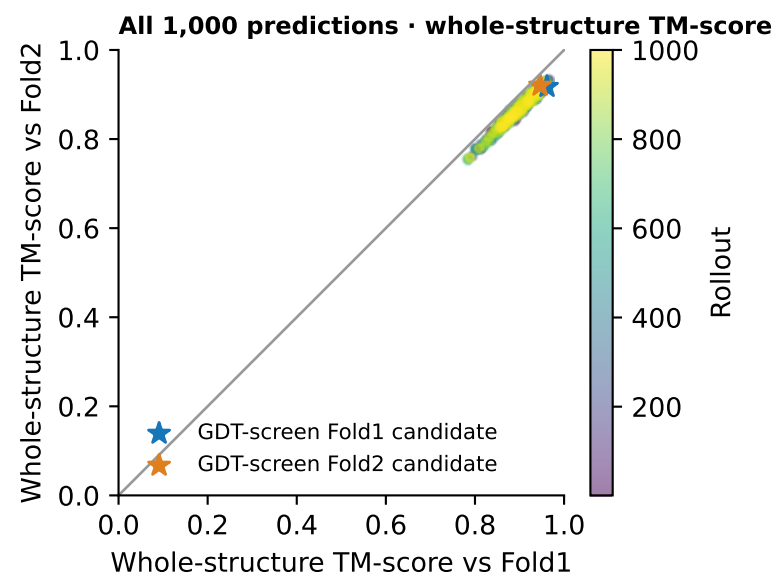
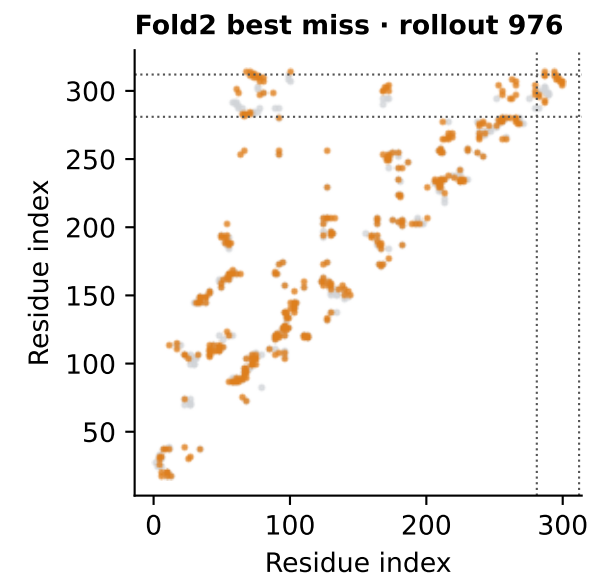
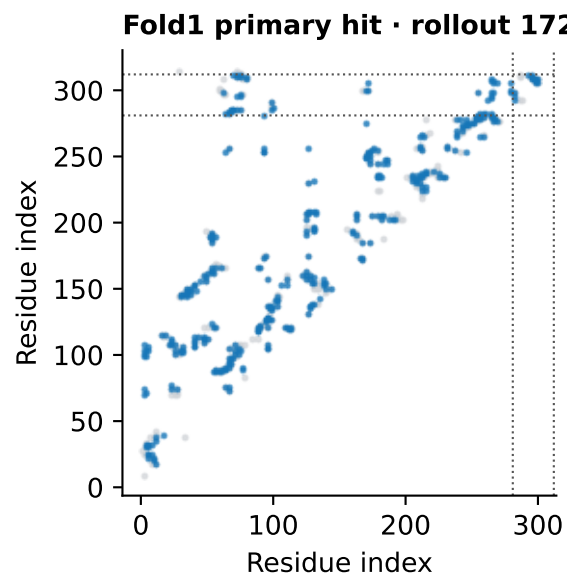
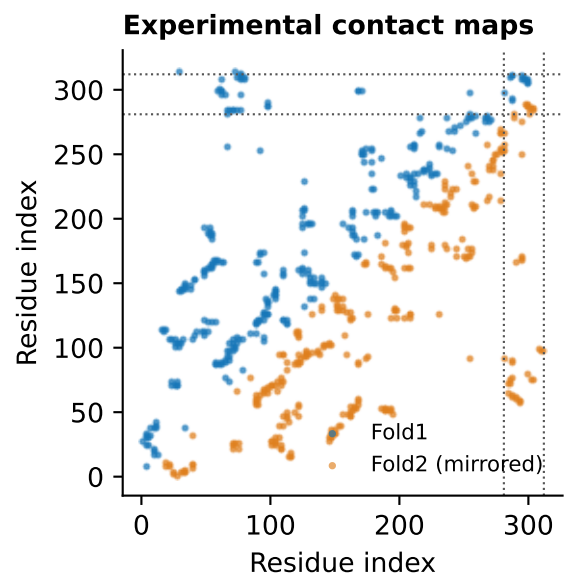
True Fold2-contact control
GDT Fold1/Fold2: 0.811 / 0.912
region: 0.250 / 0.640

No-contact baseline
GDT Fold1/Fold2: 0.003 / 0.003
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

24. 5c1va_5c1vb — contact maps and structural scores

L=318 · Fold1=5c1vA · Fold2=5c1vB · 1,000 rollouts · exact sequence · state=1 modes · reference TM-score=0.944



Post-hoc structural screen
Fold1 hits: 85
Fold2 hits: 0
Contact n structure: 85 / 0
First Fold1: 10.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

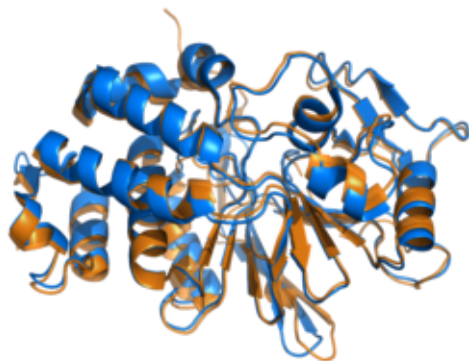
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.500 / 0.028
Fold1 TM-score F1/F2: 0.962 / 0.918
Fold1 GDT: 0.913 vs 0.772 (need 0.872)
Fold1 region: 0.774 vs 0.097
Fold1 CA RMSD: 1.52 Å
Fold1 mean pLDDT: 88.8
Fold2 contact recall F1/F2: 0.714 / 0.056
Fold2 TM-score F1/F2: 0.946 / 0.920
Fold2 GDT: 0.793 vs 0.872 (need 0.833)
Fold2 region: 0.105 vs 0.774
Fold2 CA RMSD: 7.23 Å
Fold2 mean pLDDT: 88.9

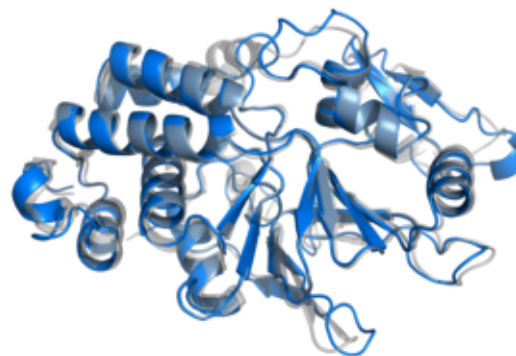
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

24. 5c1va_5c1vb — experimental and Helico structures

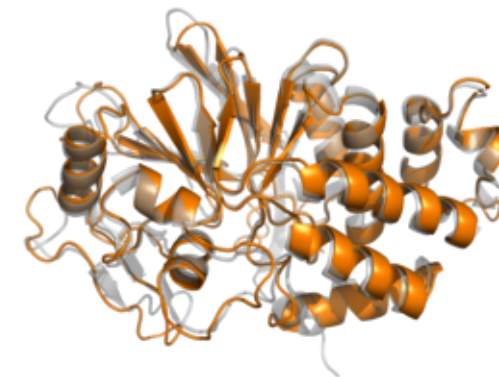
Experimental folds (aligned)



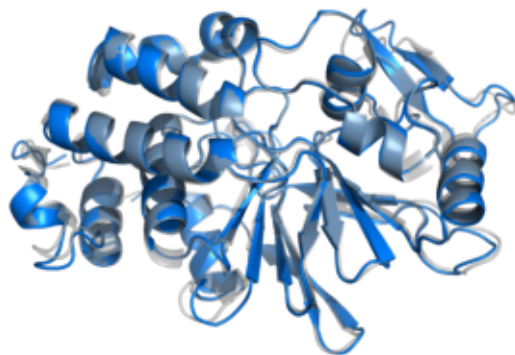
Primary Hit · rollout 172 vs Fold1



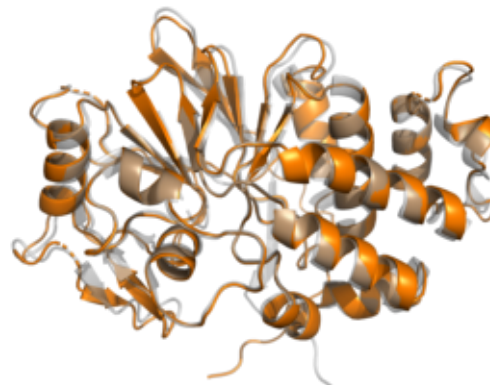
Best Miss · rollout 976 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.969 / 0.773
region: 1.000 / 0.105

True Fold2-contact control
GDT Fold1/Fold2: 0.752 / 0.926
region: 0.105 / 0.669

No-contact baseline
GDT Fold1/Fold2: 0.034 / 0.031
region: 0.000 / 0.016

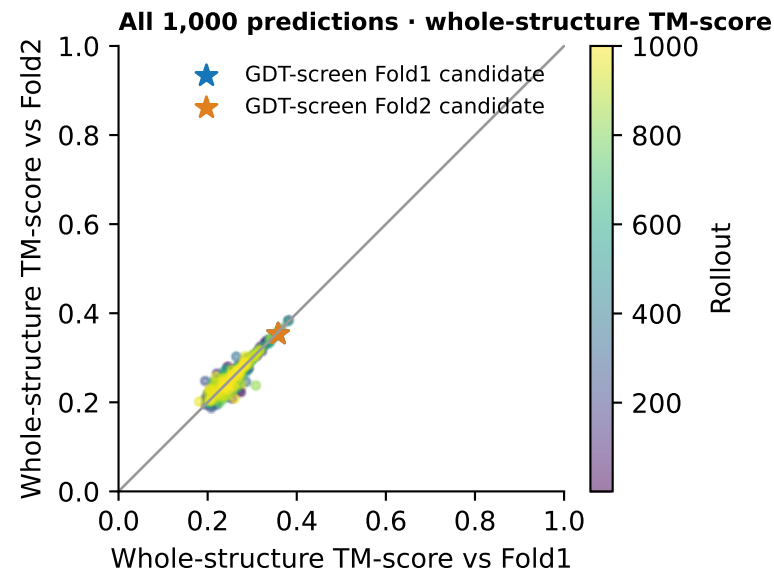
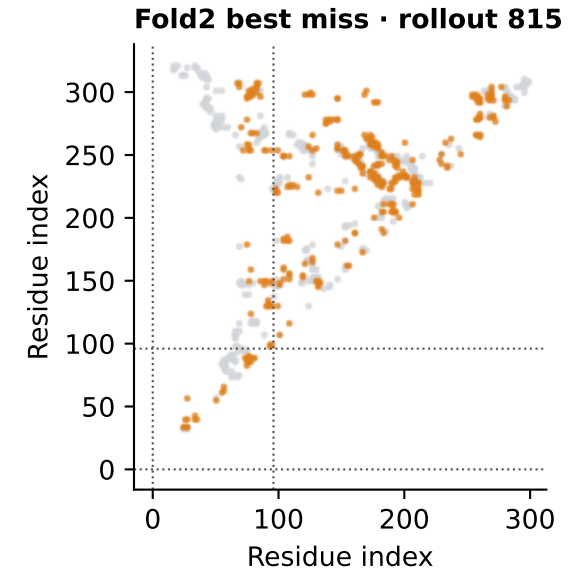
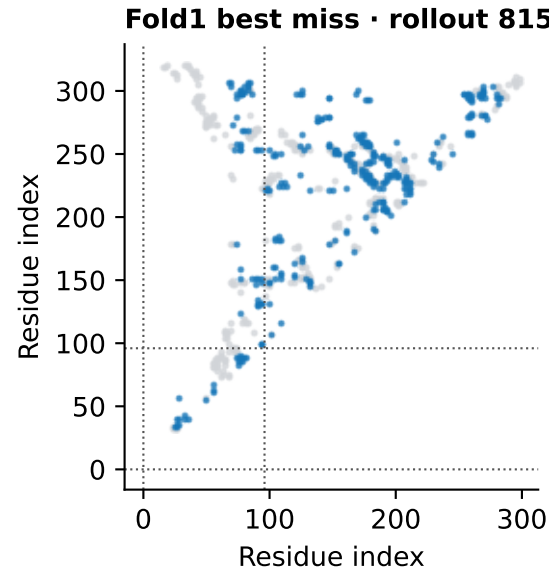
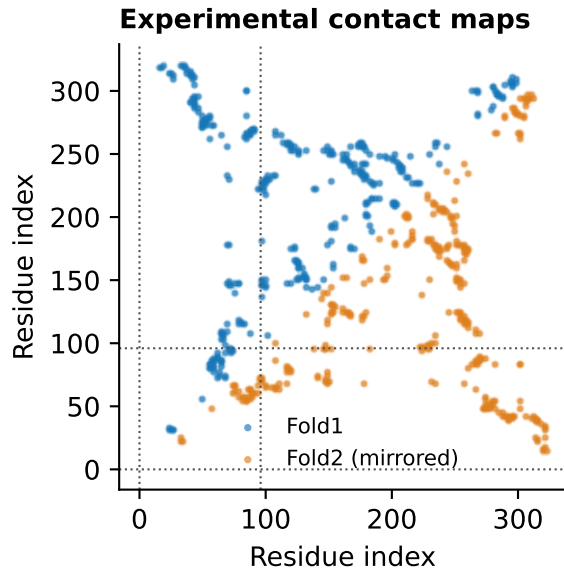
Blue/orange: prediction or corresponding reference

Gray: target experimental reference

Cartoons are visual aids; tables use exact common-position scoring.

25. 5hmgA_3ztjE — contact maps and structural scores

L=328 · Fold1=5hmgA · Fold2=3ztjE · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.994



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 FAIL · F2 FAIL

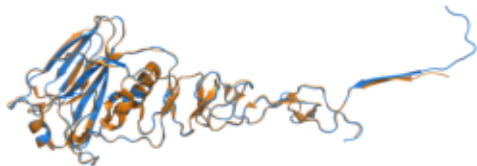
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.000 / 0.000
Fold1 TM-score F1/F2: 0.358 / 0.353
Fold1 GDT: 0.055 vs 0.057 (need 0.633)
Fold1 region: 0.077 vs 0.077
Fold1 CA RMSD: 19.42 Å
Fold1 mean pLDDT: 52.5
Fold2 contact recall F1/F2: 0.000 / 0.000
Fold2 TM-score F1/F2: 0.358 / 0.353
Fold2 GDT: 0.057 vs 0.055 (need 0.703)
Fold2 region: 0.077 vs 0.077
Fold2 CA RMSD: 19.54 Å
Fold2 mean pLDDT: 52.5

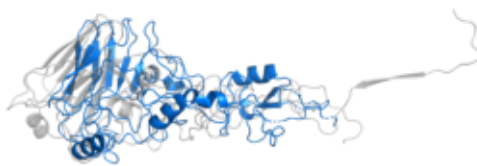
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

25. 5hmga_3ztje — experimental and Helico structures

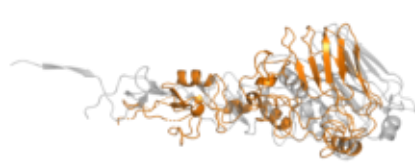
Experimental folds (aligned)



Best Miss · rollout 815 vs Fold1



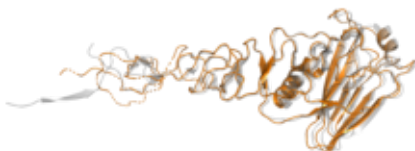
Best Miss · rollout 815 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.703 / 0.701
region: 0.665 / 0.673

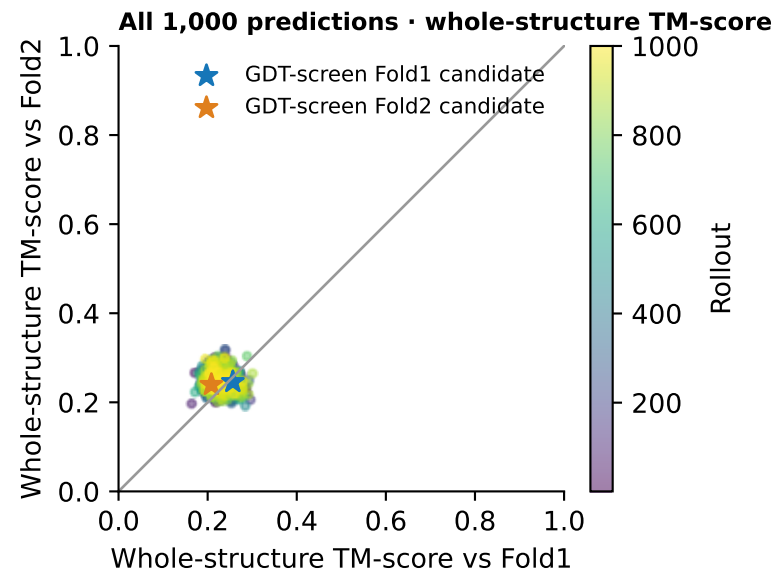
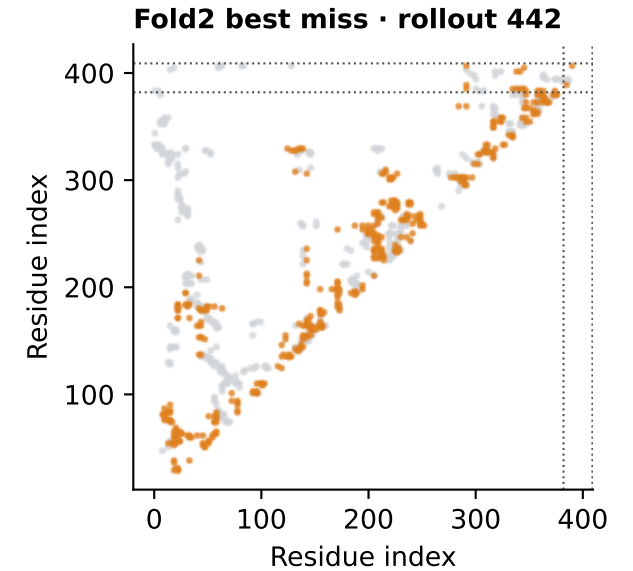
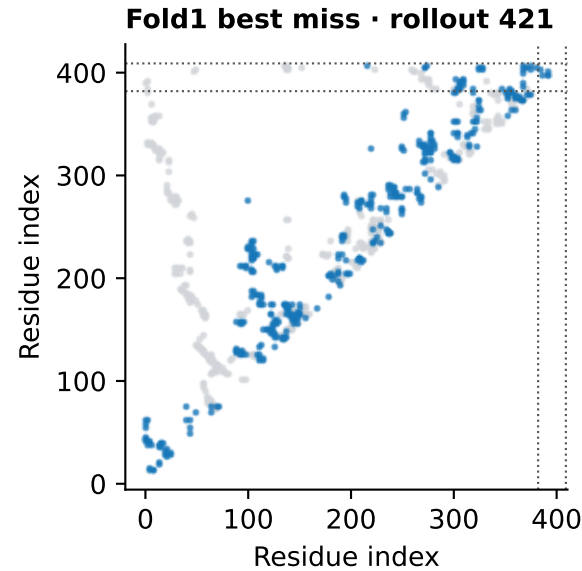
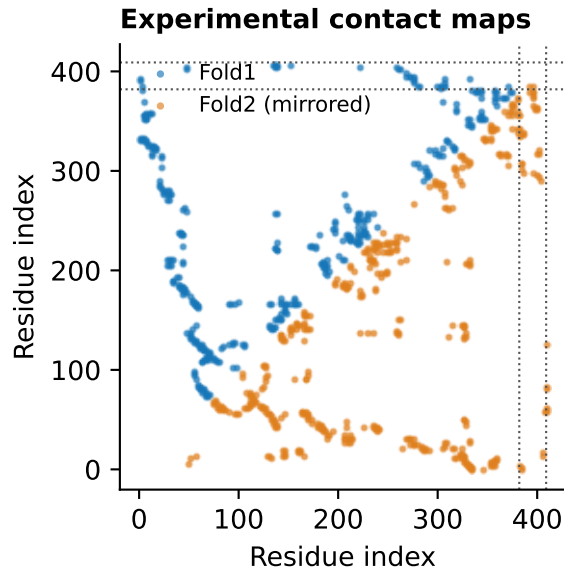
True Fold2-contact control
GDT Fold1/Fold2: 0.785 / 0.781
region: 0.724 / 0.722

No-contact baseline
GDT Fold1/Fold2: 0.013 / 0.013
region: 0.023 / 0.023

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

26. 5i2ma_5i2sa — contact maps and structural scores

L=413 · Fold1=5i2mA · Fold2=5i2sA · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.373



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 pass

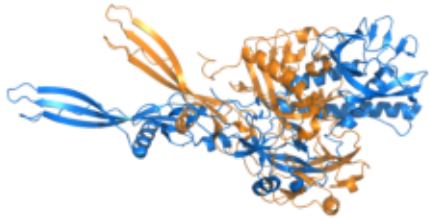
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.057 / 0.000
Fold1 TM-score F1/F2: 0.257 / 0.246
Fold1 GDT: 0.021 vs 0.012 (need 0.410)
Fold1 region: 0.120 vs 0.000
Fold1 CA RMSD: 35.03 Å
Fold1 mean pLDDT: 44.2
Fold2 contact recall F1/F2: 0.000 / 0.028
Fold2 TM-score F1/F2: 0.208 / 0.240
Fold2 GDT: 0.035 vs 0.008 (need 0.574)
Fold2 region: 0.130 vs 0.000
Fold2 CA RMSD: 24.98 Å
Fold2 mean pLDDT: 45.4

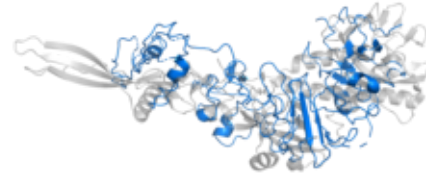
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

26. 5i2ma_5i2sa — experimental and Helico structures

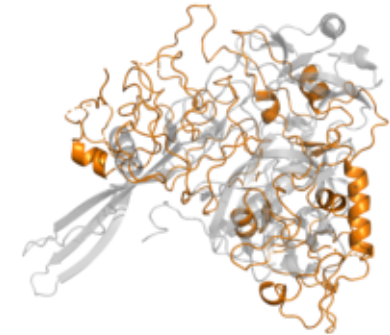
Experimental folds (aligned)



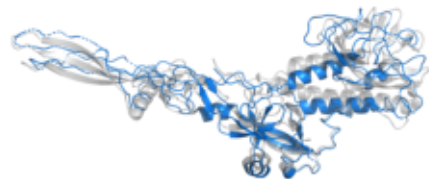
Best Miss · rollout 421 vs Fold1



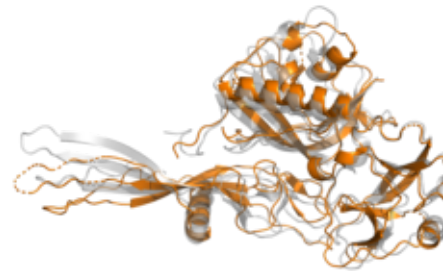
Best Miss · rollout 442 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.455 / 0.010
region: 0.444 / 0.000

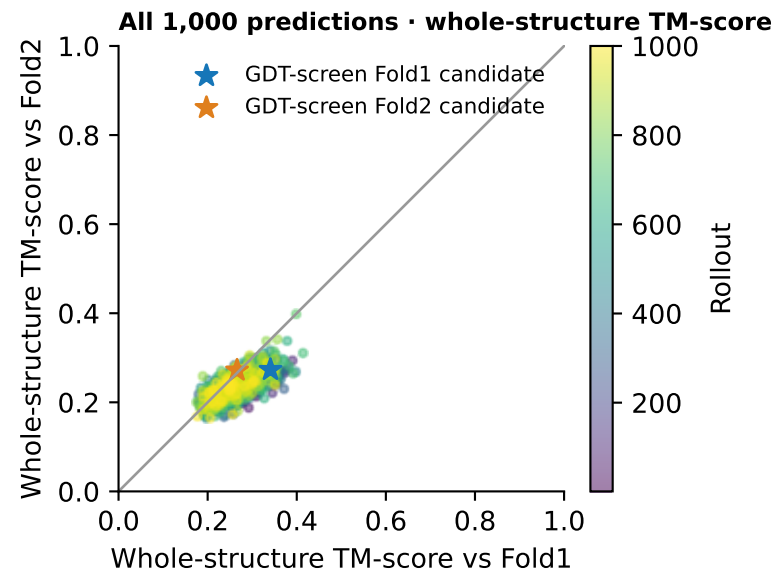
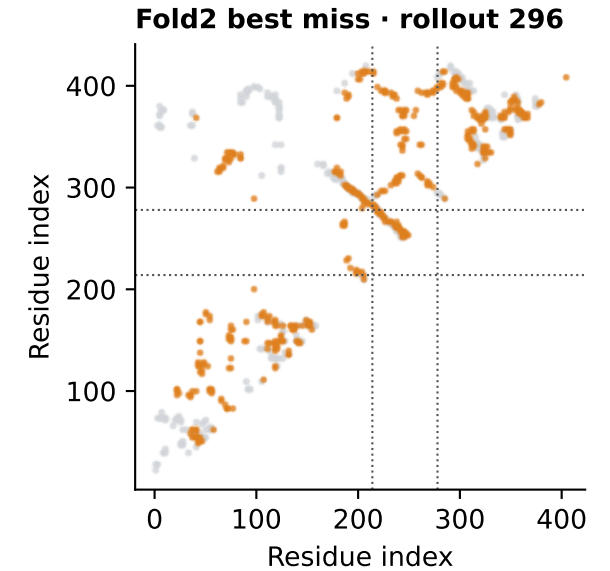
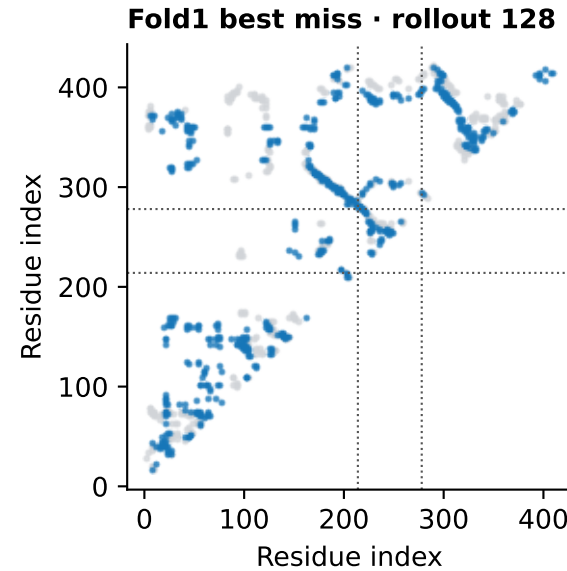
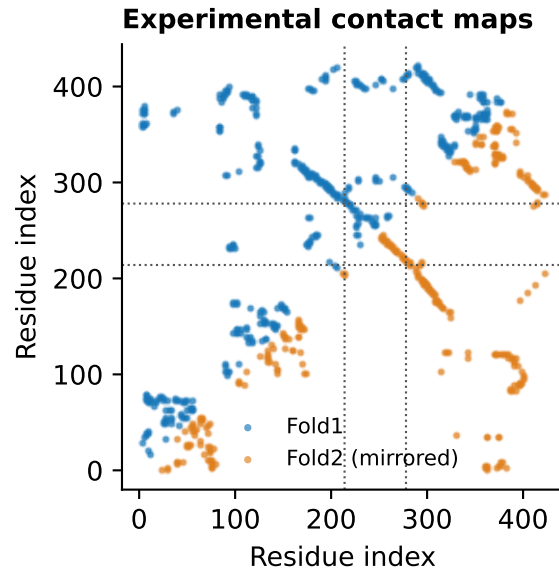
True Fold2-contact control
GDT Fold1/Fold2: 0.001 / 0.638
region: 0.009 / 0.454

No-contact baseline
GDT Fold1/Fold2: 0.006 / 0.015
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

27. 5jzha_5jztg — contact maps and structural scores

L=424 · Fold1=5jzhA · Fold2=5jztG · 1,000 rollouts · near-match sequence · state=0 modes · reference TM-score=0.575



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 pass · F2 FAIL

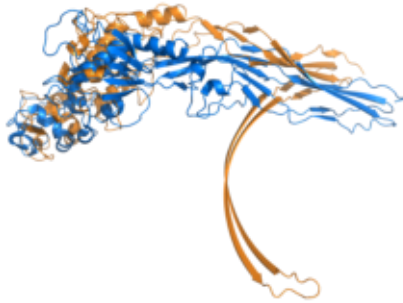
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.216 / 0.129
Fold1 TM-score F1/F2: 0.341 / 0.273
Fold1 GDT: 0.049 vs 0.005 (need 0.523)
Fold1 region: 0.127 vs 0.012
Fold1 CA RMSD: 23.12 Å
Fold1 mean pLDDT: 63.7
Fold2 contact recall F1/F2: 0.095 / 0.194
Fold2 TM-score F1/F2: 0.266 / 0.272
Fold2 GDT: 0.042 vs 0.008 (need 0.350)
Fold2 region: 0.091 vs 0.024
Fold2 CA RMSD: 29.57 Å
Fold2 mean pLDDT: 48.5

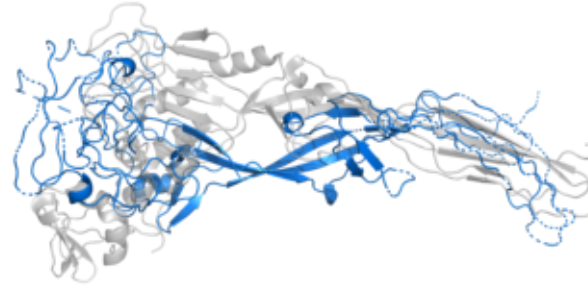
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

27. 5jzha_5jztg — experimental and Helico structures

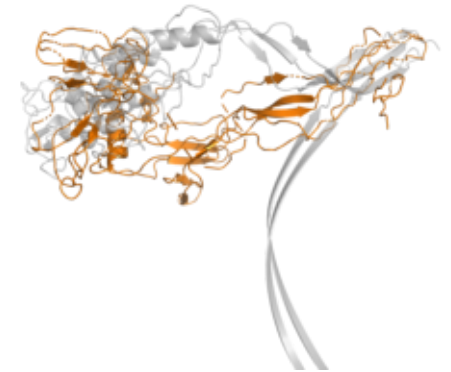
Experimental folds (aligned)



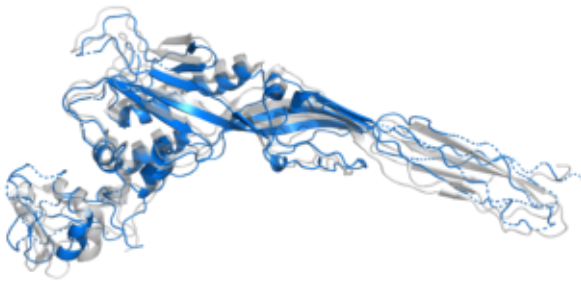
Best Miss · rollout 128 vs Fold1



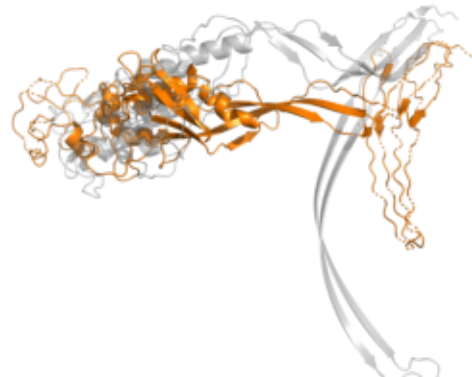
Best Miss · rollout 296 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.581 / 0.094
region: 0.575 / 0.000

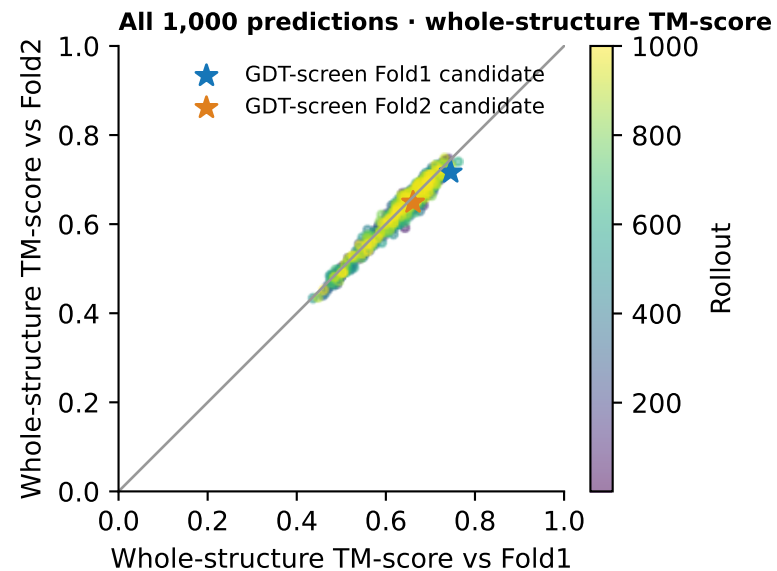
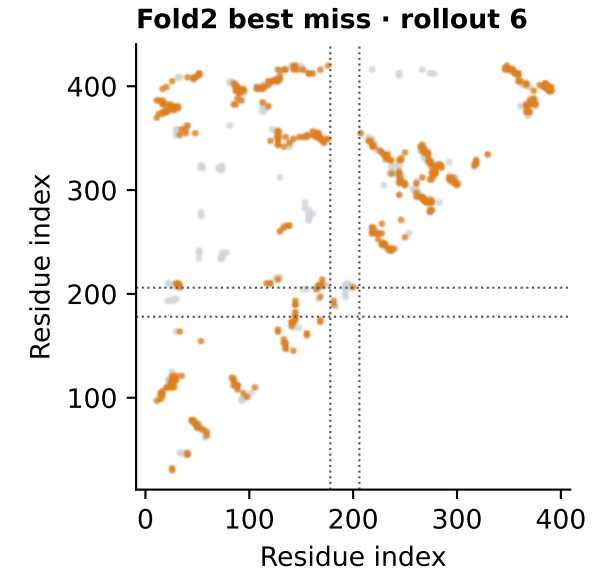
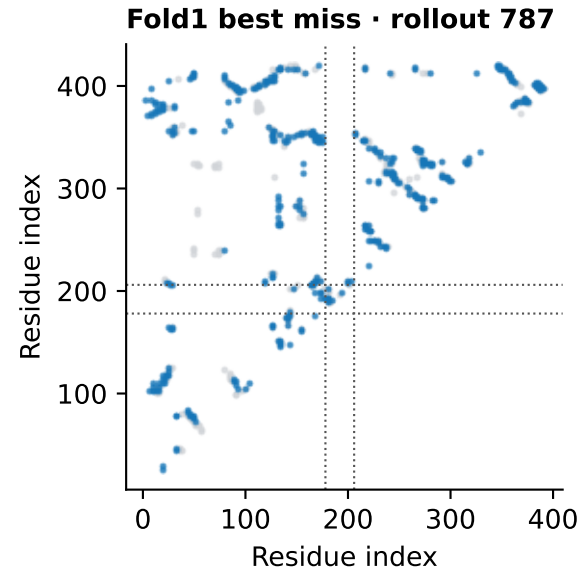
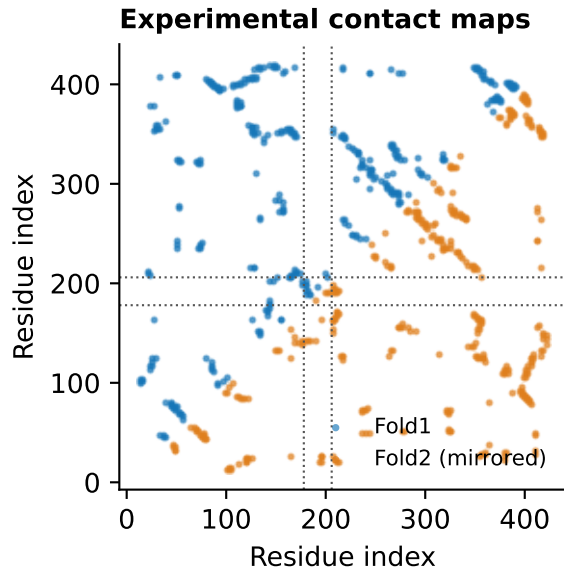
True Fold2-contact control
GDT Fold1/Fold2: 0.105 / 0.053
region: 0.012 / 0.000

No-contact baseline
GDT Fold1/Fold2: 0.004 / 0.002
region: 0.000 / 0.000

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

28. 5l35d_5l35g — contact maps and structural scores

L=422 · Fold1=5l35D · Fold2=5l35G · 1,000 rollouts · exact sequence · state=0 modes · reference TM-score=0.895



Post-hoc structural screen
Fold1 hits: 0
Fold2 hits: 0
Contact n structure: 0 / 0
First Fold1: nan
First Fold2: nan
True-contact gates: F1 FAIL · F2 pass

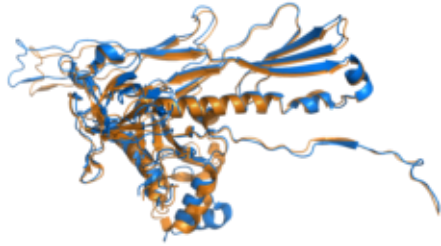
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.333 / 0.000
Fold1 TM-score F1/F2: 0.746 / 0.716
Fold1 GDT: 0.390 vs 0.370 (need 0.527)
Fold1 region: 0.348 vs 0.080
Fold1 CA RMSD: 9.99 Å
Fold1 mean pLDDT: 76.2
Fold2 contact recall F1/F2: 0.067 / 0.053
Fold2 TM-score F1/F2: 0.661 / 0.649
Fold2 GDT: 0.245 vs 0.267 (need 0.564)
Fold2 region: 0.241 vs 0.116
Fold2 CA RMSD: 9.83 Å
Fold2 mean pLDDT: 75.1

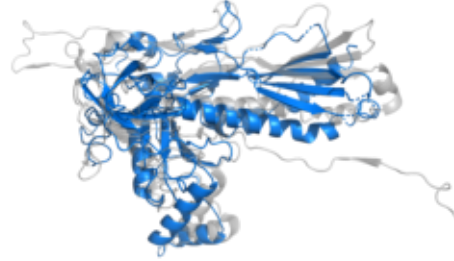
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

28. 5l35d_5l35g — experimental and Helico structures

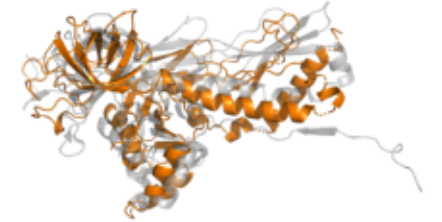
Experimental folds (aligned)



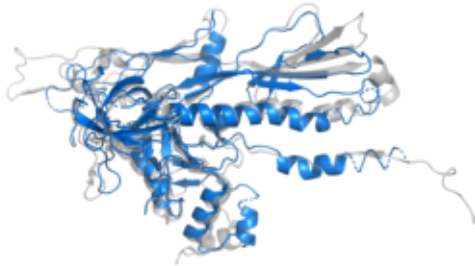
Best Miss · rollout 787 vs Fold1



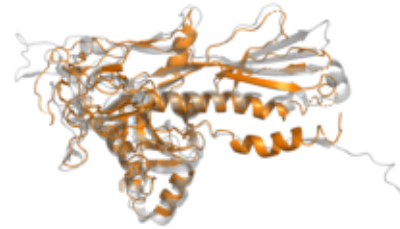
Best Miss · rollout 6 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.585 / 0.562
region: 0.223 / 0.134

True Fold2-contact control
GDT Fold1/Fold2: 0.579 / 0.626
region: 0.116 / 0.554

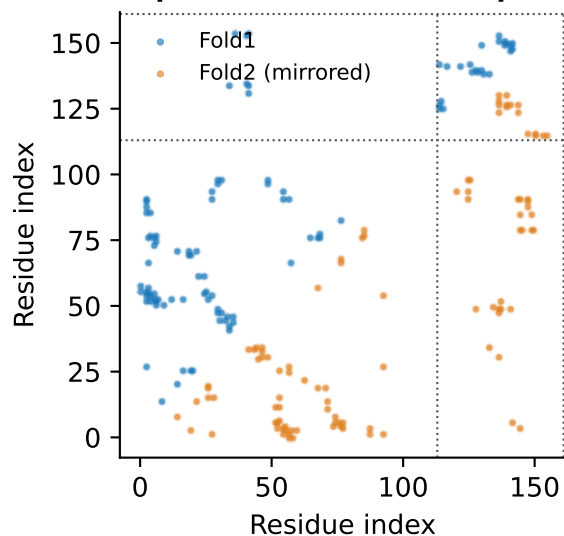
No-contact baseline
GDT Fold1/Fold2: 0.016 / 0.012
region: 0.080 / 0.027

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.

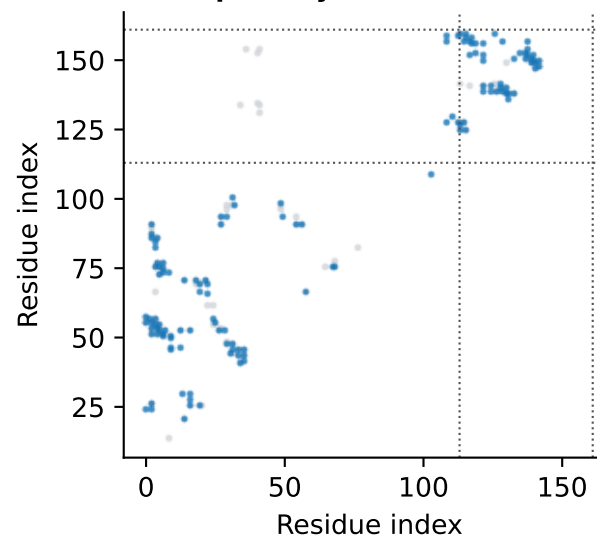
29. 6c6sd_2ougc — contact maps and structural scores

L=161 · Fold1=6c6sD · Fold2=2ougC · 1,000 rollouts · exact sequence · state=1 modes · reference TM-score=0.581

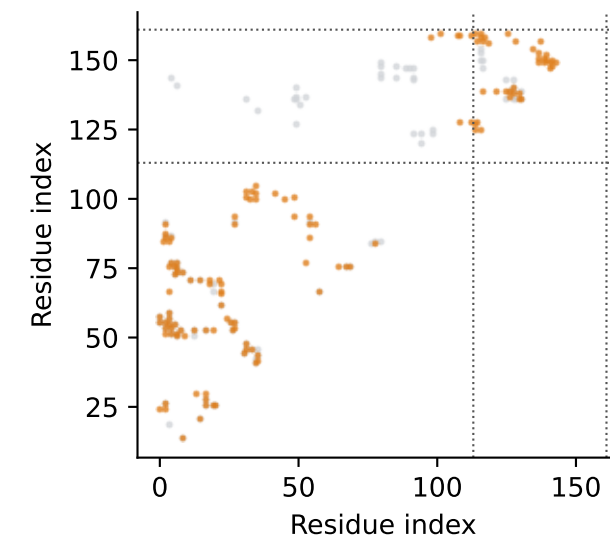
Experimental contact maps



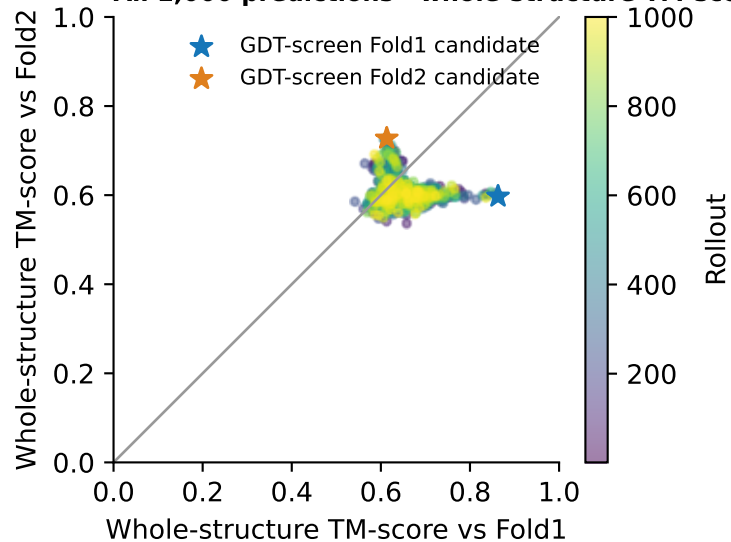
Fold1 primary hit · rollout 705



Fold2 best miss · rollout 812



All 1,000 predictions · whole-structure TM-score



Post-hoc structural screen
Fold1 hits: 31
Fold2 hits: 0
Contact n structure: 25 / 0
First Fold1: 28.0
First Fold2: nan
True-contact gates: F1 pass · F2 pass

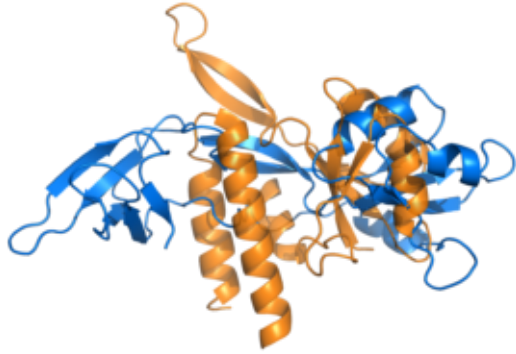
No-contact baseline hit: none

GDT-screen-selected structures
Fold1 contact recall F1/F2: 0.621 / 0.025
Fold1 TM-score F1/F2: 0.863 / 0.597
Fold1 GDT: 0.777 vs 0.035 (need 0.530)
Fold1 region: 0.655 vs 0.000
Fold1 CA RMSD: 2.34 Å
Fold1 mean pLDDT: 85.7
Fold2 contact recall F1/F2: 0.448 / 0.025
Fold2 TM-score F1/F2: 0.613 / 0.727
Fold2 GDT: 0.480 vs 0.067 (need 0.739)
Fold2 region: 0.036 vs 0.018
Fold2 CA RMSD: 8.05 Å
Fold2 mean pLDDT: 75.5

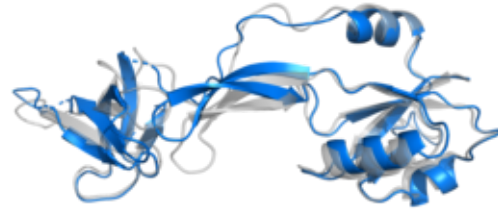
Scatter definition
TM-align each prediction to each reference using the full set of common Cα positions. The score is normalized by the reference length (tm_norm_chain2). No switching-region restriction is used. Higher is closer; 1 is a perfect match. The stars remain candidates chosen by the separately labeled GDT screen.

29. 6c6sd_2ougc — experimental and Helico structures

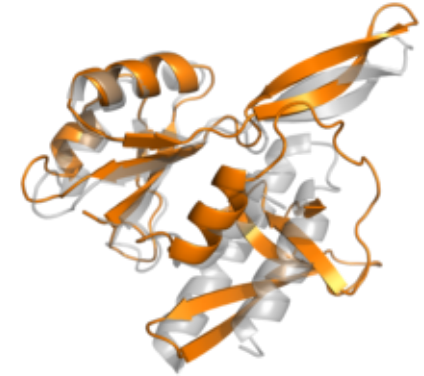
Experimental folds (aligned)



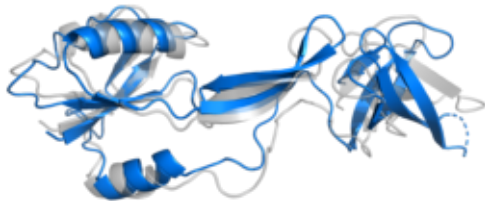
Primary Hit · rollout 705 vs Fold1



Best Miss · rollout 812 vs Fold2



True Fold1 contacts → Helico



True Fold2 contacts → Helico



Cross-reference metrics

True Fold1-contact control
GDT Fold1/Fold2: 0.589 / 0.048
region: 0.423 / 0.000

True Fold2-contact control
GDT Fold1/Fold2: 0.051 / 0.821
region: 0.000 / 0.845

No-contact baseline
GDT Fold1/Fold2: 0.018 / 0.053
region: 0.000 / 0.060

Blue/orange: prediction or corresponding reference
Gray: target experimental reference
Cartoons are visual aids; tables use exact common-position scoring.